

TOPICAL REVIEW

Application of Spatial Data and Analysis for Bus Transportation Optimization and Planning: A State-of-the-Art Review

RAFAEL L. SANTOS¹, HENRIQUE C. OLIVEIRA¹, AND DÉBORA P. SIMÕES¹

School of Civil Engineering, Architecture and Urban Design, State University of Campinas, Campinas 13083-889, Brazil

Corresponding author: Rafael L. Santos (r228749@dac.unicamp.br)

This work was supported by the Coordination for the Improvement of Higher Education Personnel (CAPES) under Grant 88887.703933/2022-00.

ABSTRACT Spatial data and spatial analysis play a crucial role in enhancing the efficiency and planning of bus transportation systems. Despite recent advances in geospatial technologies, including the widespread availability of Global Navigation Satellite System (GNSS) data and its growing use in Public Transportation (PT), there is still a lack of comprehensive reviews focused exclusively on bus services. This state-of-the-art review investigates recent studies from 2020 to 2025 that apply spatial data and spatial analysis approaches to improve bus service performance. A bibliometric analysis was conducted to identify research applications and trends in the transportation field. The review identifies five key application areas: bus route optimization, bus stop placement, passenger counting systems, bus operations and service reliability, and electric bus and sustainable mobility. The results highlight the growing integration of spatial data into bus service planning and management, which is responsible for enhancing service efficiency, passenger experience, and sustainability. While significant progress has been made, certain areas such as bus stop placement remain underexplored compared to other applications. This review offers a comprehensive reference to support future research and the development of more intelligent, reliable, and sustainable bus transportation systems.

INDEX TERMS Bus service, GNSS, transportation planning, passenger experience, sustainability.

I. INTRODUCTION

As cities and urban populations continue to grow, the demand for sustainable, accessible, and efficient public transportation (PT) systems becomes more essential [1]. PT offers a suitable alternative to private car use, contributing significantly to the development of sustainable cities [2]. Passengers' preference for PT in urban areas is primarily influenced by the quality of bus services [3], including factors such as route design [4], service coverage and availability [5], fast and reliable service [6], [7], short walking distance to bus stops and accessibility [8], [9], [10], travel conditions [11], [12], affordable service [13], and safe drivers behavior [14], [15].

Recent debates on urban mobility transitions highlight the need to reduce private car usage toward more sustainable, inclusive, and efficient public transport services [16], [17].

The associate editor coordinating the review of this manuscript and approving it for publication was Xin Sun².

These transitions focus on improving service reliability, accessibility, and user experience while using data-driven technologies to support planning and operational decision-making [18], [19]. In this context, spatial-data-driven approaches have a critical role in strengthening bus operation and planning, enhancing bus network performance, and promoting a sustainable mobility system [20].

Although bus services are widely used and it is one of the most popular PT systems [21], they often face challenges, including overcrowded vehicles, waiting times at bus stops, extended travel durations, and frequent delays [22]. However, the advancement of positioning technologies such as the Global Navigation Satellite System (GNSS) offers new opportunities to monitor and address these issues by enhancing the availability and quality of information for the planning and operation of PT systems [23]. By using GNSS technology integrated with spatial analysis, it is possible to understand the actual behavior of the transit

network and implement real-time control strategies [24], [25], [26], [27]. Moreover, Spatial data from GNSS devices technology have been used in the transportation field for decades [28].

GNSS technology is responsible for determining positions on the Earth's surface by using a constellation of satellites as references. The most well-known example of GNSS is the Global Positioning System (GPS) operated by the United States. However, several other GNSS systems have been developed by different countries, including Galileo (European Union), BeiDou (China), GLONASS (Russia), IRNSS (India), and QZSS (Japan) [29], [30]. GNSS technology and spatial data have been widely employed in the transportation sector, particularly for the optimization and planning of bus service operations [28].

Different transportation modes, including buses, rely on GNSS technology to collect real-time data - such as location, travel time, and traffic conditions - in order to optimize operations, improve efficiency, and enhance the overall passenger experience, making PT more convenient, reliable, and user-friendly [31], [32]. Bus real-time data provide valuable input for developing more robust and data-driven approaches to service planning and operational management [33], [34].

Considering the growing importance of spatial data collected from GNSS-enabled systems, this review aims to provide a comprehensive overview of recent studies that investigate how spatial data and analysis have been applied to support the optimization and planning of bus transportation services. This review article categorizes and analyzes recent studies across five major application areas including bus route optimization, bus stop placement, passenger counting systems, bus operations and service reliability, and electric bus and sustainable mobility.

To date, no recent literature review has been found that focuses on the use of spatial data and spatial analysis across the most relevant applications exclusively in bus transportation services, identified through a bibliometric analysis. Reference [35] presented an overview of the GNSS technology and its applications in transportation-related fields. However, the authors focused on other modes such as aviation, marine transportation, and Autonomous Aerial Vehicles (AAVs). Reference [36] presented a study review specifically on travel mode detection based on GNSS data collected by smartphones. Reference [37] presented a review of the applications of Geographic Information System (GIS) technology, but specifically focusing on transportation network analysis. Other studies by [38] and [39] can also be cited. However, the authors focused exclusively on the application of GIS in traffic modeling and the planning of safe urban road networks.

Although these studies address the use of GNSS data and spatial analysis in transportation, a gap remains in the literature regarding the comprehensive integration of these tools for bus transportation service planning, optimization, and monitoring. Therefore, the main contributions of this

TABLE 1. List of key acronyms.

Acronyms	Description
ACO	Ant Colony Optimization
AFC	Automatic Fare Collection
AHP	Analytic Hierarchy Process
AI	Artificial Intelligence
ANN	Artificial Neural Network
AVL	Automatic Vehicle Locating (AVL)
BEB	Battery Electric Bus
BTI	Buffer Time Index
CS	Charging Station
DTM	Dwell Time Model
GA	Genetic Algorithm
GIS	Geographic Information System
GNSS	Global Navigation Satellite System
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
GTFS-RT	General Transit Feed Specification - Real Time Reference
GWR	Geographically Weighted Regression
IC	Integrated Circuit
IoT	Internet of Things
IR	Infrared
ITS	Intelligent Transportation Systems
LSTM	Long Short-Term Memory
MILP	Mixed-Integer Linear Programming
ML	Machine Learning
MLP	Multi-Layer Perceptron
NSGA-II	Non-dominated Sorting Genetic Algorithm II
PSO	Particle Swarm Optimization
PT	Public Transportation
RFID	Radio Frequency Identification

review article are:

- 1) Comprehensively investigate how spatial data and analysis have been applied to support different aspects of bus transportation service operation.
- 2) Identify and categorize key applications of spatial data and analysis in bus transportation systems.
- 3) Identify underexplored applications within the current literature.
- 4) Discuss key challenges and future perspectives in this field.
- 5) Offers a valuable reference by reviewing the state-of-the-art to support a more efficient bus transportation service.

As presented in Figure 1, the following section is organized as follows: Research methodology (Section II), including bibliographic databases, temporal analysis, and bibliometric analysis. Bus tracking systems (Section III) focusing on IoT and its integration with different technologies, such as GNSS. Section IV explores key applications in five major areas: (A) Bus route optimization, (B) Bus stop placement, (C) Passenger counting systems, (D) Bus operations and service reliability, and (E) Electric bus and sustainable mobility. Section V presents a discussion of the findings, followed by Section VI, which outlines the main challenges for future research. Finally, Section VII brings the final considerations of this review.

A list of key acronyms that appear more than once is included in Table 1.

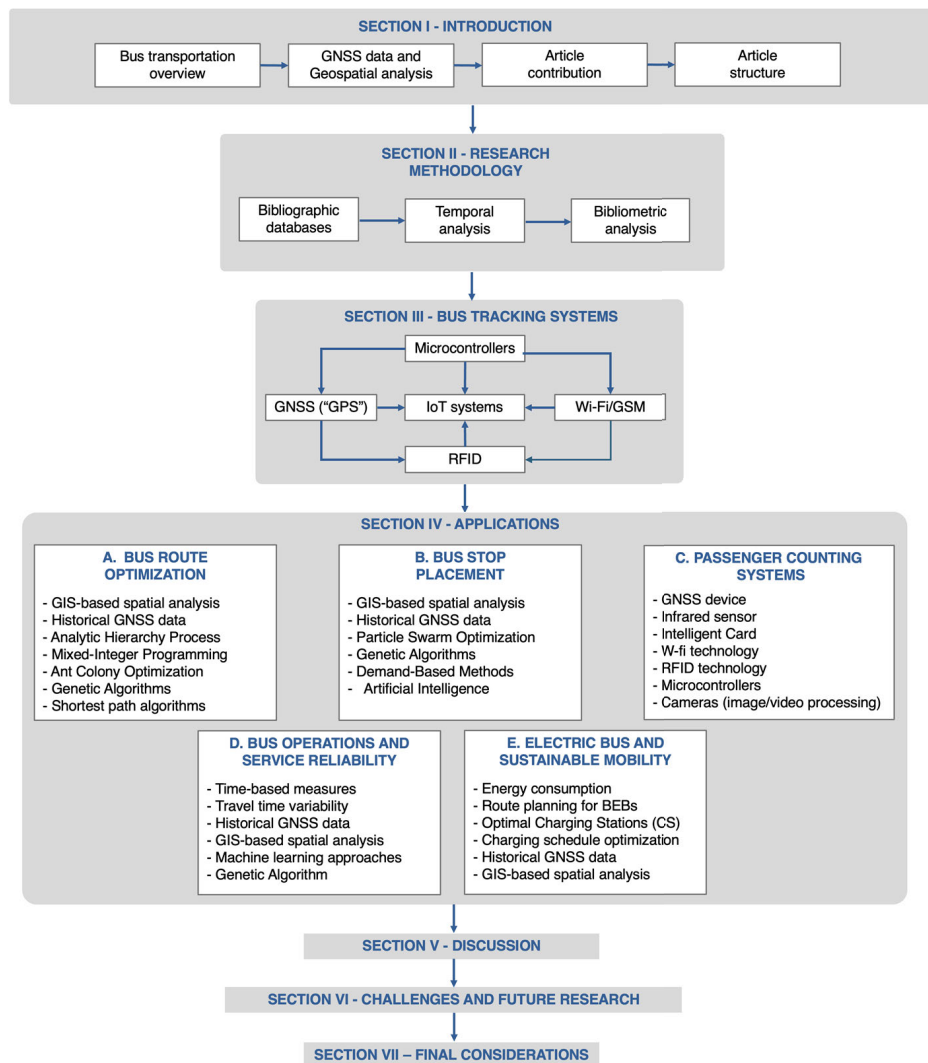


FIGURE 1. Review article structure.

II. RESEARCH METHODOLOGY

To ensure a comprehensive approach focused on recent academic research applying spatial analysis with GNSS data for optimizing and planning bus transportation service, the research methodology follows four main steps: A. Database selection, B. Search strategy, C. Inclusion and exclusion criteria, D. Thematic categorization based on bibliometric analysis.

A. DATABASE SELECTION

The bibliographic data used for this review article were obtained from four major academic databases widely recognized for their relevance in engineering, transportation, and urban studies: Scopus, Web of Science, IEEE Xplore, and Science Direct,

B. SEARCH STRATEGY

Potentially relevant articles were retrieved by performing advanced searches in each database, using the following query:

“TRANSPORT*” AND “BUS*” AND “GPS” AND “SPATIAL ANALYSIS”

The asterisk (*) was used to capture variations of root terms, such as “transportation” and “transport.” Moreover, the keyword “GPS” was adopted instead of “GNSS” because “GPS” is significantly more common in the academic literature and indexing systems. Although GNSS is technically a more accurate term, many authors still refer to satellite-based positioning as “GPS,” which ensures a more comprehensive and effective search.

Moreover, the search was limited to studies published between 2020 and 2025. This range was selected to reflect the most current trends, technologies, and practices in PT optimization using spatial analysis.

Table 2 provides a summary of the database search results, presenting (i) the number of articles found from each database using the search query, and (ii) the number of articles included after applying the inclusion and exclusion criteria.

TABLE 2. Database search results.

Academic database	Articles found	Articles included
Scopus	45	16
Web of Science	70	21
IEEE Xplore	403	51
ScienceDirect	518	39
Total number of articles included		127

C. INCLUSION AND EXCLUSION CRITERIA

Articles in the search results were screened by defining the following inclusion and exclusion criteria:

- 1) The primary inclusion criterion was that the article must focus on buses as a transportation mode. Studies related to other transportation systems - such as railways, subways, private vehicles, or air transportation - were excluded.
- 2) The article needs to be current, published from 2020 to 2025.
- 3) The full text of the article must be available and accessible online.

D. THEMATIC CATEGORIZATION BASED ON BIBLIOMETRIC ANALYSIS

Finally, to identify the most relevant and emerging themes in the literature, a bibliometric analysis was conducted using the open-source tool VOSviewer [40]. The tool enabled the construction of a co-occurrence network based on author keywords and terms occurring at least 10 times extracted from the titles and abstracts of the selected studies.

Results revealed distinct research clusters and it can be seen in Figure 2.

The bibliometric analysis from VOSviewer offers network visualizations that were used to define the structure of this review. Key major applications were identified based on the clusters and the interconnectedness of terms in the bibliometric map (See Figure 2). Section III presents an overview of the main tracking systems used for bus monitoring and responsible for collecting spatial data. Section IV introduces the key major applications identified through the bibliometric analysis, organized into the following subsections: Subsection IV-A discusses challenges of bus route optimization, which are closely connected to topics related to transportation planning, road networks, spatial analyses and GIS, as represented in the red cluster. Subsection IV-B focuses on different approaches to bus stop placement, which can be seen in the green cluster. In Subsection IV-C, studies related to passenger counting systems are presented, categorized by technologies and sensors, as seen in the blue cluster. Subsection IV-D covers research on bus operations and reliability, associated with cross-cutting themes represented in the green, blue, and purple clusters. Finally, Subsection IV-E addresses challenges related to electric bus operations, which emerge in the yellow cluster, linked

to topics such as energy consumption and environmental impact.

A total of 127 articles were analyzed in this review article, focusing on the key major applications identified through the bibliometric analysis, corresponding specifically to the studies included after the inclusion and exclusion criteria. Additional references cited throughout the manuscript provide theoretical background, methodological support, and contextual information relevant to the broader discussion.

III. BUS TRACKING SYSTEMS

According to [41] a bus tracking system based on the Internet of Things (IoT) is cost-effective and can contribute to improving the quality of PT. Over the years, IoT-based systems to track real-time locations of public buses have been developed based on a combination of different technologies, such as navigation system (GNSS), identification system (Radio Frequency Identification (RFID) technology), communication systems (e.g., Wi-fi, Bluetooth, Global System for Mobile Communications (GSM), General Packet Radio Service (GPRS), and microcontrollers [42], [43]. The integration of these systems provides real-time data on vehicle location, speed, route, and operational status along its trajectory [44].

Tracking technologies have been widely employed in IoT-based bus monitoring systems to manage and oversee different aspects of transportation. Table 3 summarizes the technologies commonly employed in bus tracking and monitoring systems.

Reference [75] proposed a GNSS-based tracking system to improve the accuracy and efficiency of bus monitoring. Bus location data are received via satellite, processed using a Kalman filter, and transmitted to web servers through cellular networks. The data are processed using the Google Maps API to display real-time locations. Reference [76] presented a real-time bus tracking and location update system based on GNSS, RFID tags, and readers placed at bus stops, with an Arduino unit managing the system and a GSM module enabling real-time tracking updates. Reference [77] also developed a system that delivers real-time bus location on Google Maps for remote users, using GNSS and GPRS technologies.

Reference [56] presented a bus tracking system based on GNSS module, RFID reader, and Wi-Fi technology to transmit the data to a remote server. A mobile application was developed to display the bus location. GNSS is used for precise bus location tracking, while RFID technology is employed for bus identification [50]. Reference [49] evaluated the performance of a GNSS module and developed a firmware for tracking buses using the a STM32 as a microcontroller. The module was configured to receive signals from multiple GNSS constellations. A low-power mode algorithm was tested, allowing the module to operate with minimal current consumption.

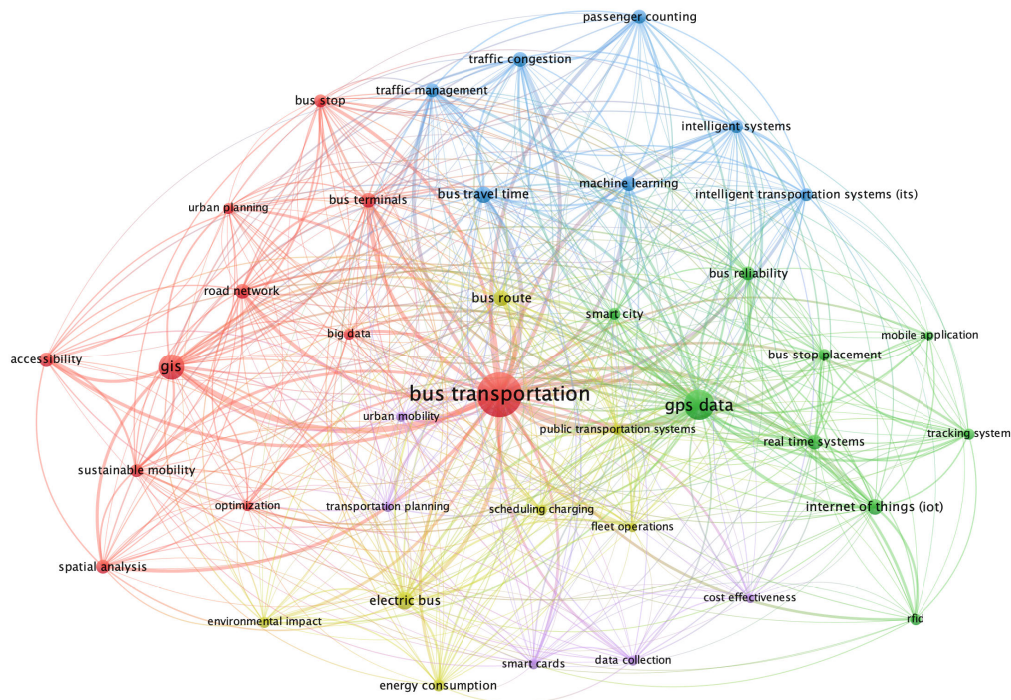


FIGURE 2. Bibliometric analysis considering 10 or more occurrences.

Reference [44] proposed a comprehensive solution for tracking school buses using Arduino as a microcontroller, along with GNSS, RFID, and GSM technologies. Additionally, an Android application was developed to monitor the bus's location. Similarly, [58] introduced a bus monitoring system incorporating an Arduino regulator with geo-location capabilities. Reference [78] also developed an Android app for real-time bus tracking locations using GNSS and RFID.

Reference [79] developed a solution responsible for collecting geo-location data from university buses using IoT technology. The system includes GSM and GNSS antennas, and Arduino Pro Mini microcontroller to provide real-time information. Reference [80] presented a GNSS-based system integrated with a fingerprint sensor and Node MCU via Arduino Uno, allowing real-time bus location updates. Furthermore, [48], [62], [69], [81], [82] presented similar systems using Arduino microcontrollers.

Reference [55] presented an IoT-based system that enables real-time bus tracking by using GNSS, GSM and RFID technologies, as well as a Raspberry Pi as a microcontroller. Reference [65] introduced a real-time monitoring smart school bus using GNSS, RFID, and an ESP32 microcontroller which was also used by [68] and [67]. A similar approach was proposed by [83], but using a PIC18F25K22 as a microcontroller.

Reference [51] applied the IoT concept to develop a tool based on GNSS devices to collect real-time location data from buses, which is transmitted via GSM to a server. This

data are then displayed on a web-based platform, offering users real-time bus locations and estimated arrival time. Reference [84] proposed an IoT bus monitoring system via mobile application. The prototype was developed by using IR sensor, GNSS and an ESP32 microcontroller which communicates using Wi-Fi. Moreover, blynk and cloud applications were used to present the data analysis on mobile apps. IoT-based systems for bus tracking and monitoring were also presented by [47], [85], and [86].

As it was outlined in this section and in the Table 3, different bus tracking systems can be found in the literature. Moreover, IoT systems integrate multiple technologies to monitor bus operations in real time. The reviewed studies commonly integrate GNSS technology with other components such as RFID, GSM, GPRS, Wi-Fi, and microcontrollers (e.g., Arduino, Raspberry Pi, ESP32, ESP 8266) to enable real-time monitoring and data collection. GNSS modules are used for the accurate location of the bus trajectory, which can be enhanced by mathematical algorithms such as Kalman filters and displayed via platforms such as Google Maps. RFID is frequently employed to identify buses at specific locations through tags installed on the buses or at the bus stops. GSM and Wi-Fi technologies are typically used to transmit location and status data to remote servers or user applications, enabling real-time communication between the tracking system and external platforms. Microcontrollers are often used to manage data collection and communication to improve efficiency and performance. Additionally, many solutions include mobile or web applications - supported by

TABLE 3. Bus tracking technologies categorized by transportation mode and year.

Source	Year	Technology used	Transportation mode	Proposed Work
[45]	2021	GNSS, Google API	Public bus	Track bus location and estimate arrival time
[46]	2022	GNSS, Android		Bus tracking and monitoring system
[47]	2022	GNSS, ESP32, IR sensors		Bus tracking system
[48]	2023	GNSS, GSM, Arduino		Track bus location and estimate arrival time
[49]	2023	GNSS, STM32		Bus tracking system for school children
[50]	2024	GNSS, Android		Provide real-time bus location information
[51]	2024	GNSS, GSM, ESP32		Real-time updates of bus coordinates
[52]	2025	GNSS, Google API		Web-based GNSS bus monitoring system
[53]	2025	GNSS, ESP8266		Provide users with a mobile app to verify bus availability and real-time location
[54]	2025	GNSS, RFID, GSM		Smart tracking system coupled with a web application
[55]	2020	GNSS, GSM, RFID, Raspberry Pi	School bus	Real-time system for bus tracking and monitoring
[43]	2021	GNSS, RFID, Android		Bus tracking and safety system for school children
[56]	2021	GNSS, RFID, Wi-Fi		Bus tracking and safety system for school children
[57]	2021	GNSS, RFID, GSM, Raspberry Pi, IR sensors		Smart school bus tracking and management system
[58]	2021	GNSS, GSM, Arduino		Bus tracking and monitoring system
[59]	2021	GNSS, QR Code		IoT-based school bus tracking system
[60]	2023	GNSS, RFID, ESP32		Bus tracking and monitoring system
[61]	2023	GNSS, RFID		Bus Tracking alert system using real-time GNSS
[62]	2023	GNSS, RFID, Arduino, IR sensor		Bus tracking, safety system and passenger counting
[63]	2023	GNSS, ESP32er		Safety system for school
[64]	2023	GNSS, Arduino, Wi-Fi		Bus tracking system with biometric authentication
[65]	2024	GNSS, RFID, ESP Cam		Real-time tracking and images of students boarding and alighting
[66]	2024	GNSS, RFID, ESP Cam		Smart school bus monitoring with a security system based on IoT
[67]	2024	GNSS, ESP32 Cam, Arduino		Smart tracking system for school bus and accident detection
[68]	2024	GNSS, RFID, ESP32, RTC Module		Bus tracking and safety system for school children
[69]	2024	GNSS, Arduino		Tracking system with smoke and distance sensors
[70]	2024	GNSS, RFID, ESP32		School transportation safety with real-time data
[71]	2024	GNSS, Google Maps API		Bus tracking and monitoring system
[72]	2025	GNSS, RFID, Wi-Fi		Safety and bus tracking platform
[73]	2021	GNSS, Android	Private bus	Bus tracking system for tourism purposes
[74]	2022	GNSS, GSM, Arduino		Track private buses

cloud services - to provide users with real-time bus location, estimated arrival times, and status updates.

GNSS tracking systems have been adopted in transportation applications and have been extensively used for gathering information on vehicle movement [87], [88], [89]. Moreover, transportation agencies and researchers are increasingly using GNSS data as an efficient and precise method for tracking bus locations in real time [90], [91], [92]. The following section presents recent studies that apply spatial analysis techniques using spatial data (GNSS data) from bus tracking systems to support the optimization and planning of bus service operations.

IV. APPLICATIONS

A. BUS ROUTE OPTIMIZATION

Bus route planning and design are essential to ensure the efficiency, reliability, and overall performance of PT services, focusing on the determination of efficient pathways through complex networks [1]. Enhancing PT through route optimization helps to reduce congestion, minimize environmental impacts, and shorten travel times, thereby encouraging increased ridership [93], [94].

Studies over the past to the recent years show a strong reliance on spatial-temporal data from sources such as

GNSS devices in buses, vehicles, and smartphones for monitoring, designing, and managing bus transportation systems [95]. Table 4 presents an overview of key studies on bus route optimization, highlighting their methods and primary applications.

Reference [111] presented a multi-objective optimization approach for PT route planning in smart cities, aiming to balance traffic patterns, cost, and environmental impact. The method employed complex optimization algorithms using real-world GNSS data and traffic reports to design more efficient routes. Multi-objective optimization models were also investigated in other studies such as in route planning scheme [112], urban bus route optimization [113], bus route design and service frequency setting [116].

Reference [106] proposed a practical approach for optimizing bus routes by integrating multi-source data, including bus GNSS and smart card transactions. By applying the Analytic Hierarchy Process (AHP) [117], [118], the authors developed a model that systematically assesses different optimization strategies. This approach enables the automatic identification of routes requiring optimization. Reference [104] also considered an AHP approach to explore the evaluation method of bus route adjustment based on multi-source data, including bus static data, bus smart card, and bus GNSS data. AHP and GIS technologies were also implemented by [103]

TABLE 4. Bus route optimization approaches.

Source	Year	Method/Model	GNSS data	Spatial Analysis	Main Application
[96]	2020	MySQL processing, GIS analysis	Yes	Yes	Route adjustments
[97]	2020	NSGA-II algorithm, Multi-Objective Optimization	Yes	No	Real-time customized bus route optimization
[98]	2020	ANN-based route recommender	Yes	Yes	Campus shuttle optimization
[99]	2020	Bilevel programming, clustering, GA	Yes	Yes	Microcirculation route layout
[100]	2021	GIS-T algorithm and Network Analyst from ArcGIS	No	Yes	Environmental impact and optimal bus route planning
[101]	2021	Routing algorithm, WebGIS interface	No	Yes	Demand-based bus routing with real-time interface
[102]	2021	Bilevel programming, MILP, ACO	Yes	Yes	Bus routing and stop selection optimization
[103]	2021	AHP and GIS analysis	No	Yes	Spatial optimization of bus routes
[104]	2021	AHP-based route evaluation	Yes	Yes	Adjustment of route design
[105]	2021	GA optimization	Yes	Yes	School bus routing
[106]	2022	AHP-based optimization strategy	Yes	Yes	Route identification for optimization
[107]	2023	Shortest path algorithms	Yes	Yes	Real-time stop and route update
[108]	2023	ACO and agent-based modeling	No	Yes	Optimal route design
[109]	2023	GIS-based path finding	No	Yes	Accessibility and route improvement
[110]	2023	Bi-level model	Yes	Yes	Smart route planning
[111]	2024	Multi-objective optimization	Yes	Yes	Smart city bus route planning
[112]	2024	Sustainable multi-objective route scheme	Yes	Yes	Eco-efficient route design
[113]	2024	Multi-objective optimization	No	Yes	Urban route optimization
[114]	2024	Bi-level programming	Yes	Yes	Dynamic route generation
[115]	2025	Heuristic approach based on the NSGA-II algorithm	Yes	Yes	Optimization problem of customized bus routes

to identify optimal bus routes through spatial analysis by analyzing urban layouts and transportation data.

Reference [109] investigated user's daily commute to work using public buses, focusing on coverage area, stop locations, and accessibility. The research identifies patterns in bus stop locations, travel distances, and commute times, combined with statistical and GIS analyses. GIS-based path finding analysis was used to assess existing routes and suggest optimized alternatives based on commuter experiences and mobility needs. Travel patterns for route optimization were also investigated by [119], [120], [121], and [122]. However, the way geographic areas are analyzed, such as how boundaries are defined or data are aggregated, may impact the findings on bus stop accessibility and route optimization.

Reference [102] presented a bilevel programming model combining Mixed-Integer Linear Programming (MILP) to simultaneously optimize bus routing, passenger route choice, and bus stop selection. The MILP model was proposed by [123] and later extended by [124] and [125]. The method integrates real-world travel data (from smartphone and GNSS device) with GIS tools, using an Ant Colony Optimization (ACO) based two-stage heuristic [126] to solve stop selection and route design. Reference [108] also implemented an ACO algorithm within an agent-based modeling framework to determine optimal routes, incorporating demand estimates at stops based on GIS-based accessibility indicators.

Reference [99] developed a bi-level programming model to design bus routes by integrating IC (Integrated Circuit) data, bus GNSS data, and taxi GNSS data. The method includes identifying areas with low conventional bus accessibility using bus operation data, followed by applying clustering analysis to determine candidate sites for bus stops. The final

bus routes and station layouts were optimized using a Genetic Algorithm (GA). A route planning strategy based on a bi-level model was also proposed by [114] and [110].

Reference [105] addressed a school bus route optimization within the smart city framework to reduce student commute time. By integrating real-time data from IoT devices, the Google Maps platform, and employing GA, the research aims to determine the most efficient paths for multiple buses. According to [127] GA can be used with other search algorithms to find optimal routes as demonstrated by [128], [129], [130], [131], [132], [133], and [134]. GA is among the popular solutions to solve the school bus routing problem [129].

Reference [98] combined GNSS-based travel data, spatial map-matching, and machine learning predictions to optimize routes with lower operational costs and environmental impacts. The method was based on an optimization tool that considers predicted travel times using Artificial Neural Network (ANN), fuel costs, and gas emissions. Reference [96] proposed a bus route optimization method using taxi GNSS data. By processing the data, the study identified pick-up and drop-off points and analyzed travel patterns. GIS tools were used to map high-demand taxi routes and detect gaps in the bus network, focusing on route adjustments to improve service coverage.

Reference [97] introduced a real-time customized bus route optimization method that addresses both static and dynamic passenger requests. The approach includes constructing a multi-objective optimization model considering a two-stage method based on the Non-dominated Sorting Genetic Algorithm II (NSGA-II): the first stage optimizes initial routes based on static requests, while the second stage updates these

routes in real-time to accommodate dynamic requests. Other studies have been focusing on customized buses using real-time data [135], [136], [137].

Reference [107] presented a method based on shortest paths algorithms to update bus stops and routes, automatically removing obsolete stops by analyzing historical GNSS data. Shortest path algorithms aim to determine the shortest possible route from a starting node to an endpoint by minimizing the total distance. The arc lengths can also be interpreted as travel times or costs [138]. References [139], [140], and [141] also proposed optimized path models for bus transit routes based on the shortest path algorithm. Moreover, [142] addressed the single-pair shortest path problem in PT by identifying the most efficient bus route between bus stations. A binary integer programming model was applied to determine the optimal path. However, future work suggested integrating GIS and using travel time as a key performance metric.

Reference [100] employed GIS application to propose optimum routes between different start and end destinations using the ArcGIS program. Moreover, the author examined the potential of bus services to reduce private car usage and CO₂ emissions. A Geographic Information Systems for Transportation (GIS-T) algorithm was applied to assess different bus route scenarios. Reference [101] also developed a bus routing algorithm in GIS framework to optimize routes for a bus service by accounting for annual demand variations. Additionally, a WebGIS-based user interface was created using GeoServer, PostgreSQL, JavaScript, and PHP to provide real-time bus location updates.

As it can be noticed in this section and in the Table 4, recent studies on bus route optimization have focused on integrating spatial analysis with multi-source data (e.g., GNSS and smart card transactions) to assess bus routes aiming to enhance PT efficiency. A key approach includes combining spatial data with decision-making frameworks to propose or redesign bus routes based on service coverage, accessibility, and passenger demand. Moreover, GIS tools are widely used for analyzing urban layouts and travel patterns, supporting the identification of efficient routes.

Optimization models frequently rely on mathematical and heuristic algorithms, such as MILP and GA to solve routing problems. Shortest path algorithms are also applied to refine existing routes by minimizing travel time, distance, and operational cost. Additionally, multi-objective optimization is often employed for bus route planning by addressing travel time, traffic patterns, cost, and environmental impacts. Moreover, some studies incorporate WebGIS platforms and real-time data streams to improve route customization, highlighting the growing role of data-driven methods in PT planning.

Across the reviewed studies, bus route optimization methods differ mainly in their data requirements, optimization strategy, and suitability for operational contexts. GIS-based approaches, such as map-matching and shortest-path models, have great advantages in spatial

assessment and the identification of coverage gaps, but offer limited adaptability to dynamic conditions. Heuristic and evolutionary algorithms such as GA, ACO, and NSGA-II provide stronger optimization capability for complex networks and multi-objective problems. However, they require higher computational effort and depend on well-structured input data. On the other hand, Bi-level and MILP-based models integrate passenger behavior and stop selection more explicitly, making them suitable for strategic route planning but less responsive in real-time conditions. Overall, GNSS-driven and multi-source frameworks enhance accuracy across methods, but choosing the proper approach depends on whether the objective is spatial assessment, real-time customization, or comprehensive route design.

B. BUS STOP PLACEMENT

Different strategies can be employed for optimizing bus stop locations, including approaches based on multi-source data, graph theory, GIS and GNSS technology, and mathematical analysis [143]. According to [144] bus stop placement depends on different factors such as topography, passenger demand, travel speed, traffic volume, intersection distance, and safety considerations. Table 5 summarizes different approaches that have been used to optimize bus stop locations.

Reference [144] evaluated the location and spacing of bus stops by using field observations, GNSS method tracking location of bus stops and GIS method produced point mapping. Reference [21] also proposed a spatial data analysis approach for network-wide consolidation of bus stop locations. The proposed method considers stop spacing and passenger boarding to eliminate undesired stops, which represent accessibility and travel time/delay. The ideal stop spacing was determined based on the ridership loss and assessment of the performance indicators.

Reference [147] employed GIS-based spatial analysis and GIS tools, including service area, location - allocation, and intersect tools, to analyze optimal locations for bus stations. The site selection considered some factors such as station spacing, passenger demands, pedestrian accommodation, land use, and existing facilities. Reference [143] proposed an integrated PT system by optimizing bus stop locations. Using GIS tools, an improved potential model was applied to create an initial bus stop layout. The NSGA-II algorithm was employed to determine optimal locations. ArcGIS was used to calculate and analyze the accessibility of residents.

Reference [152] used the AHP to systematically evaluate and prioritize potential bus stop locations. The proposed method considered factors such as travel demand, population density, land use, and accessibility for pedestrians. Reference [154] proposed a spatiotemporal integration approach for bus transit. Two data-driven algorithms were introduced: one for detecting valid bus itineraries using GNSS and stop locations, and another for clustering bus stops to define transfer areas outside physical terminals. A spatiotemporal

TABLE 5. Bus stop placement approaches.

Source	Year	Method/Model	GNSS data	Spatial Analysis	Main Application
[145]	2020	Regression analysis, GIS-based model	Yes	Yes	Optimizing bus stop locations considering passenger surveys and congestion
[146]	2020	PSO, GA, GIS-based analysis	No	Yes	Bus stop placement optimization using AI and GIS
[144]	2020	Spatial Analysis and GIS mapping	Yes	Yes	Bus stop location and spacing evaluation
[21]	2021	Spatial data analysis for stop consolidation	No	Yes	Bus stop removal based on spacing and boarding demand
[147]	2021	GIS service area and location-allocation tools	No	Yes	Optimal location of bus stations
[148]	2021	Improved k-NN (ik-NN) algorithm, data mining	Yes	Yes	Optimization of urban bus stops setting
[149]	2021	GA and PSO algorithms	No	Yes	Bus stop placement optimization
[102]	2021	Demand-based method, GIS-based travel matrices	Yes	Yes	Bus stop optimization considering walking time and demand assignment
[150]	2022	Voronoi diagrams, accessibility optimization model	No	Yes	Bus stop placement and service area optimization
[151]	2022	Weighted Voronoi diagrams	Yes	Yes	Adjusting bus stops and identifying new optimal locations
[152]	2023	AHP analysis	No	Yes	Prioritization of bus stop locations
[143]	2023	Potential model, NSGA-II algorithm, ArcGIS analysis	No	Yes	Integrated bus stop optimization
[153]	2023	PSO algorithms	No	Yes	Bus stop placement optimization
[154]	2024	GNSS data-driven algorithm and Spatiotemporal clustering algorithms	Yes	Yes	Spatiotemporal integration and transfer area definition
[155]	2024	Spatial Analysis and K-Means Clustering Method	Yes	Yes	Bus stop placement optimization
[156]	2024	Spatial Analysis and Urban Network Analysis (UNA)	No	Yes	Strategic bus stop allocation
[157]	2025	Machine learning framework and computer vision	Yes	Yes	Identify transit stop location

integration of bus transit is a strategy that allows users to change bus lines by paying a single fare [158], [159], [160].

Reference [146] proposed an approach to optimize bus stop locations based on GIS techniques and Artificial Intelligence (AI). GIS was used to generate a best-route map based on the analyzed data. Particle Swarm Optimization (PSO) and GA were applied to the geodatabase to generate optimized bus stop location maps. Bus stops were analyzed according to different conditions such as the existence of boarding bays or shelters, bus stop slope, and population served. Similar approaches considering GA and PSO were also employed by [149], [153], [161], and [162] to optimize bus stop placement along the transit route.

Reference [148] also used AI algorithms combined with data mining to present a data-driven method to optimize urban bus stop placement. By integrating GNSS and smart card data, the improved k-NN (ik-NN) algorithm is used to determine boarding stops while addressing coordinate migration issues. Reference [102] proposed a demand-based method using real-world travel demand data from smartphone devices and GIS-based travel matrices to determine optimal bus stop locations and assign passenger demand to these stops aiming to minimize total walking time. Additionally, the method considers candidate bus stop locations to integrate them into the bus route.

Reference [151] presented a method to optimize bus stop location using Voronoi diagrams [163] weighted by GNSS data, including population density and points of interest. The method identifies optimal new stop locations and adjusts

existing stops for better accessibility. Reference [150] also used Voronoi diagrams to optimize bus stop service areas. Passenger walking time was the key metric for evaluating accessibility and determining candidate stop locations. An optimization model was developed for bus stop placement and network design. Reference [145] proposed an approach that considers on-board and checkpoint passenger surveys, bus GNSS data, and a GIS-based model to identify optimal bus stop locations based on passenger volume and proximity to public facilities. Additionally, regression analysis was used to assess travel time impacts, and bus stops near congestion areas.

As it was presented in this section and in the Table 5, bus stop placement has been addressed through a variety of data-driven, spatial, and optimization-based approaches as well as multi-source data integration. Many studies use GIS and GNSS technologies to evaluate stop locations, considering factors such as passenger demand, land use, accessibility, safety, traffic conditions, and urban settings. Spatial analysis methods support decisions on stop spacing and accessibility, while optimization algorithms - including NSGA-II, GA, and PSO - are applied to refine stop positions and layouts under multiple constraints.

Artificial intelligence and data mining techniques enhance the accuracy of stop placement by integrating travel surveys, smart card data, smartphone data, and GNSS data. Additionally, decision-support models such as the AHP can be used to prioritize stop locations based on multiple criteria. Moreover, advanced spatial models, such as weighted Voronoi diagrams

TABLE 6. Technologies used in passenger counting systems.

Source	Year	Technology used	GNSS data	Spatial Analysis
[174]	2020	IR sensor	Yes	No
[175]	2020	Wi-Fi	No	No
[176]	2020	GNSS, Wi-Fi	Yes	Yes
[177]	2020	Wi-Fi	No	Yes
[178]	2020	Camera	No	No
[179]	2020	Camera & ML (Tiny-YOLOv2)	No	No
[180]	2021	Camera & ML (YOLO v3)	No	No
[181]	2021	Camera & GNSS, ML (Tiny-YOLOv3)	Yes	No
[182]	2021	GNSS, Camera	Yes	Yes
[183]	2021	GIS processed data	No	Yes
[184]	2021	Count survey & IC card	No	Yes
[185]	2022	GNSS, AI, Wi-Fi	Yes	Yes
[169]	2022	Wi-Fi	Yes	No
[186]	2022	GNSS, IR sensor, RFID	Yes	No
[187]	2022	Camera, ML (Tiny-YOLOv4)	No	No
[188]	2022	3D LiDAR video recordings	No	No
[189]	2022	GNSS, IC card	Yes	No
[190]	2023	GNSS, IC card	Yes	Yes
[191]	2023	Wi-Fi	No	No
[192]	2023	GNSS, Wi-Fi	Yes	No
[193]	2023	3D LiDAR video recordings & ML (YOLOv5)	No	No
[194]	2023	General Transit Feed Specification - Real Time Reference (GTFS-RT), Count survey & IC card	No	Yes
[168]	2024	GNSS, Wi-Fi	Yes	No
[195]	2024	GNSS, IC card	Yes	Yes
[196]	2024	IC card, smartphone	Yes	Yes
[197]	2024	GNSS, IR sensor, ML	Yes	No
[198]	2024	GNSS, Camera, ML (Tiny-YOLOv7)	Yes	Yes
[199]	2024	GIS processed data	Yes	Yes
[200]	2025	GNSS, 3D LiDAR video recordings, ESP 32 microcontroller	Yes	No
[201]	2025	GNSS, RFID	Yes	Yes

and demand-based frameworks using travel matrices or smartphone data, help align stop locations with real-world demand and minimize walking distances. Spatiotemporal clustering methods further enable better integration between routes and bus stops or transfer zones, and field data such as surveys and congestion analysis support the relocation of inefficient stops.

Overall, the studies reviewed showed some differences and similarities in how bus stop placement problems are approached. GIS-based mapping approaches, for example, consider land use, accessibility, and pedestrian conditions, but they often operate on static assumptions. In contrast, optimization techniques such as GA and PSO algorithms focus on systematically improving stop locations under multiple constraints. However, they depend on well-defined inputs and can be computationally demanding. Approaches based on location-allocation tools focus on service areas and reducing walking distances, but they are less responsive to actual passenger movement patterns. Data-driven and AI-based methods, particularly those integrating GNSS, smartphone, or smart card data, offer a realistic representation of user behavior and operational conditions. Therefore, each approach has its own strengths and limitations.

C. PASSENGER COUNTING SYSTEMS

In recent years, advances in Intelligent Transportation Systems (ITS) have significantly enhanced the automation

of passenger counting, providing greater accuracy and scalability. When integrated with technologies such as infrared sensors [164], [165], cameras [166], smart cards and GNSS [167], Wi-Fi [168], [169], and RFID [170], [171]. ITS solutions enable real-time data collection and support adaptive transportation planning, resulting in more reliable and precise operations [172]. Stop-level modeling of boarding and alighting supports both PT operations and transit-oriented development [173].

Table 6 brings an overview of the main technologies employed in bus passenger counting systems.

One of the earliest technologies for passenger counting includes data from card swiping and ticketing systems [202], [203], [204]. Reference [190] introduced a probabilistic approach to estimate alighting counts by using reverse route boarding data from smart card transactions. Reference [195] proposed a framework to estimate alighting locations using smart card data and GNSS-bus data, particularly in systems where only boarding information is available. Reference [196] proposed a method that integrates multi-source data to estimate bus passenger boarding and alighting through spatiotemporal clustering of Integrated Circuit (IC) card data - smart cards with embedded microchips for storing and processing data - combined with GNSS data. Some other studies also integrate smart card transaction with GNSS data to identify boarding and alighting at bus stops [205].

References [174] and [186] used IR sensors integrated with Raspberry Pi, GNSS and RFID technologies to collect real-time data from passenger flow and transmits it via Wi-Fi. Reference [206] also integrated IoT technologies using Wi-Fi modules for communication, GNSS to track bus location, RFID for bus identification, and two IR sensors to count passengers. A similar approach was also presented by [197]. The method also includes machine learning algorithms to estimate the number of buses needed per route. Reference [207] used RFID technology embedded in students' ID cards to detect boarding and alighting events and GNSS to track the bus location. GSM modules were also integrated to send real time notifications to parents.

Some passenger counting systems rely on devices and network-based approaches that process wireless signals, such as detecting Wi-Fi emitted by passengers' devices [175], [176], [177], [191]. Reference [192] proposed a low-cost device for real-time passenger counting using Wi-Fi probe request and a Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. The device has a Raspberry Pi and real-time GNSS positioning system, along with an external Wi-Fi adapter to capture probe requests from mobile devices. Similarly, [176] proposed a method using a Wi-Fi scanner to collect MAC addresses and GNSS data, associating them with individual passengers on the vehicle.

Passenger counting systems are also based on video cameras and image processing combined with computer vision and machine learning algorithms to detect and track people as they board or alight from vehicles [172]. Reference [180] used Thermal Imager and IP cameras for passenger detection and counting. Reference [187] used Machine Learning (ML) models such as YOLO with DeepSORT to count and identify boarding or alighting passengers. Results showed that Tiny-YOLOv4 offered high accuracy with few false predictions. Reference [179] combined YOLOv2 with an ML algorithm to propose a technology for automatically counting bus passengers. Reference [198] developed a real-time monitoring system for a transportation service using GNSS sensors and Artificial Intelligent (AI). Passenger counts were detected through thermal imaging and the YOLO-v7 algorithm. Data were integrated and visualized using Google Apps Script, AppSheet, and Google Cloud. Reference [208] proposed a method that uses tiny YOLO to detect passengers boarding and alighting from buses by training the model to recognize upper body features. Similarly, [181] presented a counting scheme to detect and count passengers entering and exiting the bus using deep learning-based object detection method and YOLOv3.

Reference [182] proposed a passenger counting system using a camera to detect and track individuals as they board or alight the bus. A neural network processes video frames to identify people. The system geo-references boarding and alighting locations by pairing each passenger detection event with the real-time GNSS coordinates of the vehicle. References [188], [193], and [200] even proposed an

approach based on LiDAR technology to create 3D models of passenger movements, allowing for highly accurate detection of individuals boarding or alighting a vehicle. However, the high cost of LiDAR systems makes them less suitable for large-scale deployments.

In addition to bus passenger counting systems, some studies focus on estimating passenger demand at bus stops by using spatial analyses. Reference [199] proposed an analytical approach based on a generalized additive model that incorporates spatial dependence. The authors employed detailed GIS-based spatial analysis to calculate variables, including land-use characteristics within areas surrounding each bus stop to estimate passenger demand. Reference [183] also applied spatial models to analyze bus stop-level ridership: the spatial autoregressive model and the spatial error model. The authors considered buffer distances from bus stops to create predictors. Moreover, land-use patterns were considered where the stop is located. The results confirmed the impact of spatial effects on bus ridership.

Reference [189] proposed a disaggregate demand model for bus ridership, addressing issues such as service frequency and spatial aggregation of data. The model accounts for overlapping stop catchments and relates cluster-level ridership to stop-level characteristics. Overlapping between disaggregate stop catchment areas was addressed using spatial tools by constructing Voronoi polygons, a method introduced by [209]. Reference [210] also modeled ridership at a stop level and investigated its relation to bus stop characteristics using spatial and spatiotemporal variables.

Reference [194] applied Geographically Weighted Negative Binomial Regression (GWNBR) to estimate bus stop-level ridership under conditions of missing boarding and alighting data. The method accounts for spatial heterogeneity and dependence by locally adjusting regression parameters based on geographic location. Previously, The same authors previously employed Ordinary Kriging (OK) as a spatial interpolation method to estimate bus stop-level ridership in an earlier study [184].

Reference [173] also compared methods for modeling bus stop-level boarding and alighting, accounting for spatial dependence and count data characteristics. The authors evaluated linear, Poisson, Geographically Weighted Regression (GWR) [211], Geographically Weighted Poisson Regression (GWPR) which is an extension of GWR [212], and Universal Kriging (UK) [213], [214] using data from a bus line.

Finally, different approaches have been proposed in the literature for forecasting passenger flow [215], [216]. References [217] and [218] trained a baseline model using historical Automatic Passenger Counting (APC) data to generate day-ahead forecast. Reference [219] proposed a method that uses a machine learning algorithm based on historical passenger travel data. Reference [220] considered regression models for development and evaluation of frameworks for real-time bus passenger occupancy prediction while [221] estimated passenger flow in a bus route using Kalman filter.

As previously described in this section and in Table 6, traditional methods based on smart card transactions have advanced through their integration with GNSS data, enabling the estimation of both boarding and alighting activities using spatiotemporal clustering and probabilistic modeling. Other approaches use infrared sensors and RFID technologies, often embedded in IoT-based systems with components such as microcontrollers, GNSS technology, and Wi-Fi or GSM transmitters to provide real-time passenger counting monitoring. Additionally, systems that detect Wi-Fi probe requests from passengers' mobile devices offer passive and cost-effective solutions for detecting passenger movements and estimating onboard occupancy.

Computer vision and deep learning methods have been explored in recent studies, using RGB, thermal, or depth cameras along with algorithms such as YOLO and DeepSORT to detect and track boarding and alighting passengers. While LiDAR-based systems provide highly accurate 3D modeling of passenger movements, their high cost poses scalability limitations. Beyond direct counting, several studies employ spatial and statistical models to estimate stop-level ridership by incorporating land use, accessibility, and spatial dependence, especially useful when data are incomplete. Forecasting passenger flow is another key area, where historical data are used with regression models, Kalman filters, or machine learning algorithms to support predictive planning and operational decision-making in PT systems.

Finally, the reviewed counting systems differ mainly in cost, accuracy, level of detail in the data, and scalability. Smart card-based approaches present a broad coverage and low operational cost, but they often lack alighting information. Infrared and RFID systems offer low-cost monitoring. However, their accuracy can be affected by installation constraints and crowded conditions. Meanwhile, Wi-Fi-based methods require inexpensive implementation. However, these systems can be sensitive to signal loss. Solutions based on images or videos present high accuracy, especially with advanced computer vision and deep learning methods, but they require higher processing capacity. Moreover, privacy and maintenance become a concern. On the other hand, spatial modeling and statistical approaches complement counting technologies by estimating ridership where direct measurements are unavailable, although their performance depends on high-quality spatial data. Therefore, integrating multiple technologies can be a powerful strategy for obtaining comprehensive passenger information.

D. BUS OPERATIONS AND SERVICE RELIABILITY

Service reliability is a key aspect of transportation in improving the quality of service operation and remains one of the most significant concerns for passengers and transportation agencies [222], [223]. Moreover, reliability is considered as one of the most important service operation factors [224], [225]. Bus reliability can be assessed by the

consistency of travel times in relation to the scheduled timetable [226]. Therefore, time-based approach is a common measure of bus service reliability [227], [228], [229]. GNSS data are often used to evaluate bus service operation [230], [231], and to obtain precise information on vehicle time, speed, and location [232]. According to [233], GNSS devices are widely used to measure running times, providing bus trajectory data that can be used to estimate travel time for passenger trips.

Table 7 provides an overview of time-based indicators and travel time variability metrics commonly used in studies to evaluate bus service reliability.

Reference [257] evaluated service reliability by using a real-time GNSS tracking mobile app on PT ridership. The method involved collecting data from users before and after using an mobile application. Other factors were also evaluated such as accessibility [258], comfort [259], [260], [261], safety and security satisfaction [262]. Results showed statistically significant improvements across all variables after using the application, highlighting the effectiveness of GNSS technology in enhancing user experience and improve service operation.

Reference [228] also evaluated bus services in terms of reliability, accessibility, and availability using GNSS data and spatial analysis. By analyzing GNSS trajectories against route data and schedules, the study identified well-performing routes, demonstrating the effectiveness of GNSS data in measuring bus service reliability. Reference [263] incorporated GNSS-based data analysis to assess service operations and reliability using headway-based indicators and data collected from an Automatic Vehicle Locating (AVL) system.

Reference [252] proposed a framework to evaluate bus performance by using GNSS data. The authors developed two algorithms: one for extracting individual bus trips and another for matching bus stop sequences, which were used to estimate travel times and dwell times at bus stops. Reference [236] also proposed algorithms based on geometric map matching [264], [265] to extract bus trips and estimate total travel time considering GNSS data. The method includes three key algorithms: geometric point-to-point map-matching, trip direction identification, and average speed determination.

Reference [245] proposed an approach to evaluate bus travel time reliability by integrating both in-stop waiting time and in-vehicle travel time as two independent stochastic processes. The reliability metric is calculated using the convolution of probability distributions for these two components, providing the probability that total bus travel time stays within an acceptable threshold. The bus location data provided by an AVL system is GNSS-based and is used to estimate travel and waiting times under different conditions, including weather, time of day, and trip distance.

Reference [229] analyzed Travel Time Variability (TTV) and reliability using GNSS-based GTFS-RT data from a

TABLE 7. Bus service reliability measures.

Source	Year	Method/Model	Technology Used	Temporal	Main Application
[234]	2020	Particle filter-based estimation framework	GNSS	1 month	Bus travel time estimation
[235]	2020	LSTM, ANN	GNSS	1 month	Arrival time prediction
[236]	2020	Geometric map-matching algorithms	GNSS	3 months	Bus trip extraction and travel time estimation
[237]	2020	Section-based travel time description	GNSS	Not mentioned	Travel time estimation
[238]	2020	LSTM	GNSS	3 months	Arrival time estimation
[239]	2020	BP neural network	GNSS	2 months	Arrival time estimation
[240]	2020	LSTM	GNSS	1 year	Arrival time prediction
[241]	2020	Machine Learning model - Deep Belief Network(DBN)	GNSS, AVL	1 month	Travel time prediction
[242]	2020	LSTM neural networks and clustering	GNSS	3 months	Arrival time prediction (departure zones focus)
[243]	2021	LSTM	GNSS, Automatic Fare Collection (AFC), and IC card	4 days	Travel Time
[244]	2021	LSTM, ANN	GNSS/AVL	1989 journeys	Travel and arrival time
[245]	2022	Convolution of waiting and in-vehicle time distributions	GNSS, AVL	1 month	Travel time reliability evaluation
[246]	2022	Multi-model prediction using LSTM	GNSS	Not mentioned	Arrival time prediction
[247]	2022	LSTM with GA optimization	GNSS and IC card	1 month	Travel and arrival time prediction
[248]	2022	Kalman filter-based RTM and DTM models	GNSS, AFC, and IC card	4 days	Travel time prediction
[249]	2022	LSTM and MLP with GA optimization	GNSS data	11 months	Travel time prediction
[250]	2022	Machine Learning models (linear and non-linear)	GNSS data	1 month	Travel time prediction
[228]	2023	GNSS-based QoS Scoring Functions	GNSS data	3 months	Bus service reliability assessment
[251]	2023	Machine learning-based estimated arrival time	GNSS and IoT in mobile applications	Not mentioned	Real-time travel time estimation
[252]	2023	Custom map matching algorithms	GNSS	5 months	Travel and dwell time estimation
[253]	2023	Deep learning and network embedding	GNSS data	6 months	Arrival time prediction
[254]	2023	LSTM	GNSS and ESP32 microcontroller	6 months	Arrival time prediction
[229]	2024	Statistical analysis of Travel Time Variability, Heatmap analysis	GNSS and GTFS-RT	1 month	Travel time reliability analysis
[255]	2024	ANN-based prediction model	GNSS data	10 months	Arrival time estimation using spatial features
[256]	2025	LSTM	GNSS	9 months	Travel time prediction

bus network. The authors considered different metrics such as Buffer Time Index (BTI), Travel Time Index (TTI), and Planning Time Index (PTI). Results were visualized through heatmaps to identify reliability patterns and congestion trends. Reference [22] also used GNSS data to assess Travel Time Reliability (TTR). The method includes statistical analysis of GNSS-derived travel times. A revised Level-of-Service (LOS) was proposed using a Weighted Delay Index (WDI) and K-means clustering.

GNSS data are often used to build models for predicting bus arrival and travel times [266]. Delivering reliable and precise bus arrival and travel time predictions is an effective strategy for improving service reliability, passenger satisfaction, and operational efficiency [256], [267]. Different machine learning (ML) algorithms are proposed to solve the problem of bus travel and arrival time prediction [268]. Reference [244] proposed and developed predictive models to estimate bus travel and arrival times based on historical GNSS data, bus route and bus stop information. Three models were developed using Linear Regression, ANN and Long Short-Term Memory network (LSTM).

Reference [251] integrated GNSS, IoT, and machine learning to improve real-time arrival time predictions by tracking and analyzing the users real time location. A deep learning model that predicts bus arrival times using only GNSS trajectory data were also proposed by [253]. The method includes a sequence embedding module to capture temporal patterns in bus trajectories, and a network embedding module to model spatial correlations between bus stops. Reference [254] also proposed a bus arrival time prediction system that relies exclusively on GNSS trajectory data and IoT technologies. It employed LSTM trained with features such as distance, traffic density, stop wait times, and road junction.

Many other studies have employed ML models such as LSTM to predict bus arrival times using GNSS data [240]. The authors developed a hybrid LSTM model to predict travel times between successive bus stops using GNSS data from public buses. Reference [246] proposed multi-model bus arrival time prediction approach using real-time online information considering four prediction models, including LSTM. Reference [235] proposed a bus arrival time prediction model based on a hybrid LSTM and ANN approach,

segmenting GNSS and spatiotemporal data by bus stops. Their results showed that the integrated LSTM-ANN model outperformed the individual models, particularly for long-distance predictions.

Similarly, [269] developed models to predict bus arrival time using GNSS trajectory data by training LSTM, which was compared against others baseline models, including ANN and historical averages. LSTM outperformed others in accuracy. Reference [255] also developed an ANN-based prediction model to predict bus arrival times using GNSS data. The model was trained on features including bus stop location and travel distance, reflecting spatial elements. Reference [242] proposed a bus arrival time prediction model using LSTM with daytime-only GNSS data from bus routes location analysis, reducing errors for buses near departure points.

Reference [250] proposed an approach based on ML aiming to predict public transit travel time to improve bus service reliability. The authors compared different ML models including linear non-linear models, based on one month of GNSS data. Input features included trip start time, day of the week, and direction of travel. Reference [241] also applied ML models to forecast travel time. The model was trained and validated using real-world GNSS-based traffic data from bus routes. Reference [249] compared two hybrid neural network models to predict real-world bus travel time: one combining LSTM with a Kalman filter, and another using a Multi-layer Perceptron (MLP) optimized by a genetic algorithm.

Reference [247] used a LSTM model to predict bus travel time based on an integration of real-time GNSS and IC card data. To enhance model accuracy and efficiency, a Genetic Algorithm (GA) was applied. Reference [249] also considered historical GNSS data from a bus line to develop two neural network models (MLP and LSTM) to predict travel times. Their models were also enhanced by a GA.

Reference [234] employed a particle filter-based vehicle model which is a probabilistic algorithm to estimate bus travel times. A real-time framework was proposed based on real-time GNSS data. Reference [248] presented a Kalman filter-based approach to predict travel times using GNSS and IC card data. The method includes a Route Travel Time Model (RTM) and the Dwell Time Model (DTM) as main components, both designed for single-bus and single-route scenarios. GNSS data were also used by [237] to determine bus displacement, speed, and motion state, which were aggregated by section to create a travel time profile on urban bus routes.

As highlighted in this section and in the Table 7, GNSS systems have become a crucial technology for assessing bus operations and service reliability. Researchers use GNSS-derived metrics such as travel time, headway, and delays to evaluate schedule adherence, service accessibility, and overall reliability. Several studies apply statistical analyses and indices such the BTI, Travel Time Index

(TTI), and Planning Time Index (PTI) to visualize travel variability. In some cases, mobile apps using real-time GNSS tracking were assessed for their impact on user satisfaction, showing improvements in perceived reliability, accessibility, and comfort.

Moreover, researchers have been focusing on using deep learning models to predict bus arrival and travel times. Techniques such as ANN, LSTM, and hybrid models combining LSTM with Kalman filters or GA have been developed. These models often rely on GNSS trajectory data and incorporate some parameters such as traffic density, time of day, and stop locations. Additional methods such as map-matching, particle filters, and Kalman filters are used to enhance the precision of travel time estimates. The integration of these approaches aims to improve operational efficiency and support more reliable PT services.

Overall, the reviewed studies showed that the use of GNSS and spatial data-based approaches offers strong reliability analyses by capturing real-time bus movements, while statistical indicators such as TTI and PTI help measure travel time variability but depend on historical data and are less responsive to real-time operational changes. On the other hand, ML and deep learning models provide higher predictive accuracy, especially for arrival-time forecasting. However, they require larger datasets and more computational processing. Therefore, choosing the most appropriate approach depends on data availability, desired accuracy, and operational needs.

E. ELECTRIC BUS AND SUSTAINABLE MOBILITY

PT systems worldwide have increasingly adopted electromobility in recent years, incorporating more electric vehicles into their fleets [270], [271], [272], [273]. Battery Electric Buses (BEBs) are recognized as a key strategy for mitigating greenhouse gas emissions. Bus Fleet Electrification (BFE) has attracted considerable attention in recent years due to its potential to contribute meaningfully to environmental sustainability efforts [274]. BEBs often present a monitoring system responsible for tracking bus location and other information [275], [276], [277]. Even when high-quality or high-frequency GNSS data are not available, some studies propose effective solutions to address this limitation.

Reference [278] proposed a method for accessing the positional error of low-cost devices based on GNSS integrity for transportation applications. The study compared two low-cost GNSS receivers - one with an embedded antenna and another with an external antenna - against a dual-frequency reference receiver through static and kinematic experiments. Data were collected and analyzed using positional error metrics to evaluate the receivers' accuracy and reliability in transportation applications.

Reference [279] proposed an unsupervised clustering method to accurately reconstruct bus trajectories from noisy, low-quality, and repetitive GNSS data. It uses a competitive density clustering algorithm where clusters attract nearby

GNSS samples and shift toward denser areas, forming a sequence of centroids that outline the route. Reference [280] presented a GNSS and spatial analysis-based method to reconstruct high-resolution speed and road slope data for BEBs using low-frequency GNSS signals and cartographic altitude data. By resampling GNSS points and applying map matching [281], the method estimates route characteristics critical for modeling energy consumption and battery State of Charge (SOC).

Table 8 provides a structured overview of research focused on electric bus operation, highlighting the use of GNSS data and its respective application within each research context.

Managing BEBs fleets presents challenges such as monitoring energy consumption, optimizing route efficiency, and minimizing operational downtime [305]. Different factors can affect vehicle performance, including route distance, frequency and characteristics [306], drive behavior and traffic condition [307], environmental conditions [308], passenger load [309], [310], trip duration, air conditioning, speed, and vehicle weight [311], [312], [313]. As a result, there has been a significant rise in the number of studies in the literature aimed at enhancing the quality of BEBs services.

Accurately estimating the energy consumption for the BEBs is essential for multiple planning and operation tasks [289]. According to [314], precise estimation of energy consumption is essential for managing BEBs operations, as it supports both the selection of the most suitable BEB model for a given route and the optimization of bus scheduling strategies. The authors investigated how the GNSS sampling interval affects the accuracy of estimating energy consumption for BEBs using a longitudinal dynamic model and real-world operational data. Some other studies took GNSS data under a specific sampling interval to model energy consumption, such as 1 to 3 sec [284], 10 to 15 sec [285], 15 sec [294], [315], and 30 sec [300]. Setting the sampling interval is still an open question for many studies [314].

Many researchers have been developing approaches to measure energy consumption of BEBs. Reference [300] used GNSS data from an entire electric bus fleet and employed spatial analysis to model energy consumption. The authors compared physics-based and supervised machine learning models which were previously introduced by [316]. Energy consumption was also analyzed by [295] from buses operating in a hilly terrain. GNSS and spatial analysis were employed in the methodology, particularly in segmenting routes by elevation changes. A linear regression model was developed to identify terrain gradient and passenger load as key factors affecting energy use. Similar studies such as [282] also investigated route slope and topography as parameters in their model to calculate energy consumption using GNSS real-world data.

Reference [301] compared the energy consumption of electric and diesel buses across different lane configurations. GNSS and spatial analysis were conducted for mapping bus

movements and associating them with lane types using spatial proximity to bus stops. The energy consumption approach was based on the framework developed by [284], which used GNSS and smart card data to calibrate the model. Reference [312] developed a deep learning model to estimate energy consumption of BEBs using GNSS coordinates of bus stops, arrival times, and trip conditions. Energy consumption between stops is calculated based on changes in battery charge levels, and spatial data are used to calculate distances and elevation differences.

Reference [302] used GNSS data to capture spatial and operational dynamics from pure BEBs to analyze their energy consumption. The authors considered Lasso regression and a Random Forest model (described by [317] as machine learning techniques. Reference [290] proposed a data-driven approach to estimate the real-time energy consumption of BEBs, using motion and GNSS data. The method incorporated a Kalman filter to reduce GNSS noise. A random forest algorithm was also used to discard unimportant features, improving prediction accuracy. Moreover, elevation data from Google API were used in this study.

Effective planning of public bus transportation systems, particularly with the integration of BEBs into fleets, requires a thorough understanding and characterization of the routes [279]. As presented in Subsection IV-A, different studies have been proposed to improve bus services through route planning. Some other studies, specifically investigate route optimization and planning for BEBs using spatial data and spatial analysis aiming to enhance battery efficiency.

Reference [286] proposed a multi-criteria methodology to prioritize the replacement of diesel buses with BEBs in urban PT. Bus routes were defined by using GNSS data and by spatial analysis to assess their suitability for electrification. Moreover, this study proposes a novel methodology that identifies bus routes of interest by priority using GIS tools. GNSS data were used to provide geolocated bus trajectories (latitude, longitude, elevation, time), which were processed using GIS tools to analyze route lengths, proximity to critical areas, and spatial coverage. Reference [287] proposed a novel deep learning-based strategy to assign suitable routes for BEBs by employing urban big data based on vehicle movement, human mobility, and geographic location. The system recommends electric bus routes that optimize energy efficiency and reduce emissions, demonstrating the potential of integrating GNSS data with advanced analytics for route planning.

Reference [303] proposed an intelligent route planning system for BEBs, aimed at minimizing passenger waiting and travel times through a multiobjective optimization model. This study used a large spatiotemporal dataset including GNSS data from buses, integrating passenger demand and operational constraints. Reference [296] presented a dynamic framework combining fixed-route transit with flexible BEBs to optimize routing, timetabling, and passenger transfer planning in real time. The authors used a Mixed-MILP model with

TABLE 8. Electric bus operation: data and applications.

Source	Year	Data used	Main Application
[282]	2020	GNSS data (longitude, latitude, and altitude), route profile, road gradient, weather condition data, vehicle characteristics	Assessing the impact of road gradient on electric vehicle energy consumption
[283]	2020	Bus route and charging station locations, from Google Maps service, bus characteristics	Charging station location and fleet sizing model
[284]	2021	GNSS data (longitude, latitude, altitude, timestamp, speed, and direction), route number, bus ID, smart card data, bus and battery characteristics	Examining influential factors on the energy consumption
[285]	2021	GNSS data (longitude, latitude, speed, altitude, acceleration), trip length and duration, temperature range, and bus characteristics.	Predict energy consumption
[286]	2021	GNSS trajectory data (longitude, latitude, and elevation), number of trips and buses	BEBs route prioritization
[287]	2021	GNSS trajectory data, bus IC card, urban road data, points of interest data, meteorological data	Route planning for BEBs
[288]	2021	Bus routes, bus stops, charging stops location, bus and battery characteristics	Electric transit route network design with delay consideration
[274]	2021	GNSS trajectory data (latitude/longitude), bus route, and smart card data, bus characteristics	Charging station location and charger configuration, charging time and vehicle flow
[289]	2022	GNSS data (longitude, latitude, speed, acceleration) bus weight, road gradient, route distance, travel time, bus and battery characteristics	Energy consumption estimation
[290]	2022	GNSS data (longitude, latitude, speed, altitude, acceleration), passenger counting data, road slope	Energy consumption estimation
[291]	2022	Geolocation data from Google Maps, charging stations location, bus and battery characteristics	Design of a multi-trip electric bus routing model
[292]	2022	GNSS data, bus characteristics, historical solar production data, average electricity price	Sustainable charging schedule under different charging types
[293]	2022	GNSS trajectory data (latitude/longitude), bus route information, bus timetable, bus and battery characteristics	Charging schedule optimization method
[294]	2023	GNSS trajectory data (longitude, latitude, timestamp, speed), temperature, bus and battery characteristics	Energy consumption estimation
[295]	2023	GNSS data (longitude, latitude, altitude, speed, and direction), dashboard images, temperature, road gradient, bus and battery characteristics	Energy consumption estimation in hilly terrain
[296]	2023	Network topology, bus routes and stops location, transfer stations	Assigning bus routing and timetabling plans
[297]	2023	GNSS data (latitude/longitude), bus route, candidate charging sites, bus characteristics	Charging station placement
[298]	2023	GNSS operational data (latitude/longitude), bus route, battery characteristics	Location planning charging stations
[299]	2023	GNSS location of bus stations, timetables, and battery characteristics	Smart and sustainable scheduling of charging events
[300]	2024	GNSS data (longitude, latitude, timestamp, and speed), route ID, bus and battery characteristics	Energy consumption estimation
[301]	2024	GNSS trajectory data (longitude, latitude, speed, and timestamp), route and bus ID, passenger flow and road elevation estimation, bus and battery characteristics	Measure energy efficiency
[302]	2024	GNSS data (longitude, latitude, speed), daily energy consumption data, bus characteristics	Predict energy consumption
[303]	2024	GNSS trajectory data, bus stop location, historical data regarding passenger arrivals and departure, traffic accident data	Route planning for BEBs
[304]	2024	GNSS trajectory data (latitude/longitude, timestamp, speed), bus ID, weather and solar irradiance data	Charging scheduling considering charging infrastructure integrated with solar photovoltaic system

embedded learning-based optimization to manage routing constraints, passenger flow, and charging limitations.

Reference [291] proposed a GA with improved features to optimize routes, charging schedules, and minimize empty trips. Geographic coordinates, Google Maps data, and k-means clustering were used to define charging stations locations. Reference [288] presented a framework for designing BEB networks and integrating route planning and charging infrastructure location using Robust Optimization and Multi-Objective Particle Swarm Optimization. The model accounts for spatial characteristics of routes and stops. Real geographic data, such as locations of bus stops and power variability zones, were considered in this study.

Another important aspect of BEBs is the optimal Charging Stations (CS) placement [318]. Selecting charging locations impacts both cost and operational performance, requiring places that ensure convenient and reliable charging [319]. CS placement is part of a transit planning process that includes other related decisions such as route design, trip timetabling and scheduling [297].

Reference [320] presented a MILP model using actual route and demand data from intercity bus networks, relying on spatial network design to optimize the locations of CS. Reference [297] also proposed a MILP model to optimize the location and size of BEBs charging stations. To validate the approach, the authors considered spatial data (latitude/longitude) from General Transit Feed Specification

Real-Time (GTFS-RT) data and candidate site coordinates. Reference [283] applied spatial analysis to develop a stochastic charging location model for BEBs and solved it through a customized Lagrangian relaxation algorithm. Charging demand was calculated using distances from Google Maps' Distance Matrix API.

Reference [274] used GNSS trajectory and smart card data from BEBs to present a novel optimization model for locating CS using nonlinear constraints and a surrogate-based optimization method [321]. Reference [322] also proposed a surrogate-based optimization method and considered spatial distribution through CS placement and demand modeling based on trip data. Reference [298] used GNSS trajectory and smart card data to estimate bus departure times, which served as inputs to the two-stage stochastic optimization model developed to determine CS optimal locations for BEBs.

Since BEBs have a limited battery capacity and driving range [323], [324], they may need to charge during their multiple trips along city routes [325]. Therefore, proper scheduling charging strategies have become a solution for fleets looking to incorporate BEBs [326]. Reference [299] presented a scheduling framework for BEBs charging based on a deep learning model that predicts energy availability. Real-world GNSS data were used to determine bus stop locations and to support route and network structure mapping. Reference [292] also considered historical GNSS data to estimate bus arrival time at the CS. The authors proposed a sustainable charging schedule for BEBs based on a rolling horizon approach [327], [328], [329].

Reference [293] proposed a charging schedule optimization method based on real-world spatial and temporal data. The authors used GNSS-based stop locations and real-time bus operation data to estimate travel distances, identify viable charging windows, and determine CS locations. Similarly, [304] proposed a MILP model to optimize daily charging schedule of BEBs. The method considered GNSS-based trajectory data to capture the spatiotemporal behavior of buses operating on different routes. GNSS data were essential for defining feasible charging windows and aligning charging schedules with bus movements and depot infrastructure.

Considering the studies presented in this section and summarized in Table 8, the integration of BEBs into PT services has driven a growing number of studies focused on monitoring, planning, and optimizing electric bus operations through the use of GNSS and spatial analysis. Several studies propose approaches to overcome limitations related to low-frequency or noisy GNSS data, such as unsupervised clustering and map-matching algorithms for reconstructing accurate trajectories and estimating road characteristics, including speed and slope. These approaches support key operational tasks such as energy consumption modeling and real-time battery monitoring. Additionally, some techniques, including physics-based models, linear regression, deep learning (e.g., LSTM), and machine learning algorithms such

as Random Forest and Lasso regression, have been used to estimate BEBs' energy consumption under different trip conditions, environmental factors, and route characteristics.

In the context of planning and optimization, GNSS and spatial data are central to defining BEBs routes, locating charging stations (CS), and managing energy-efficient scheduling. Studies employed multi-objective optimization models, genetic algorithms, and surrogate-based methods to design robust route networks and optimize CS placement based on real geospatial and operational data. Furthermore, advanced scheduling models integrate GNSS data to align charging schedules with bus operations and depot constraints. These data-driven strategies contribute significantly to reducing operational downtime, extending driving range, and enhancing the overall sustainability and efficiency of electric bus fleets.

Finally, the reviewed studies showed that spatial analysis can be a powerful tool for electric bus planning as it allows detailed reconstruction of routes, speeds, slopes, and vehicle behavior, even when data quality is low. Energy-consumption models rely on speed, route profiles, and vehicle parameters, and can be sensitive to input errors. Meanwhile, optimization models are powerful for route design, charging station placement, and scheduling, but they rely heavily on accurate spatial inputs. Therefore, each approach's effectiveness depends on the quality of the available data.

V. DISCUSSION

Figure 3 shows the distribution of studies published between 2020 and 2025, revealing a steady flow of publications over the last years, indicating a consistent research interest in bus transportation research. The peak in 2021, 2023 and 2024 aligns with the growing adoption of combination GNSS-based travel data and machine learning in PT studies, as seen in works by [171], [197], [219], [244], [251], [254], [300], and [302].

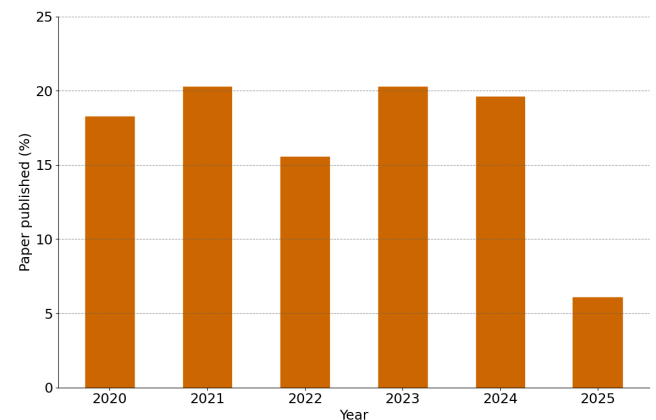


FIGURE 3. Percentage of papers published from 2020 to 2025 in bus transportation research.

The number of studies over the years reflects the persistent challenges and evolving demands in bus transportation

services. These ongoing challenges are closely associated with the rapid expansion of urban areas and continuous population growth. Moreover, the continued research attention underscores the importance of developing efficient, data-driven solutions to meet the growing expectations for urban mobility.

Figure 4 presents the distribution of bus tracking studies according to the types of bus transportation services.

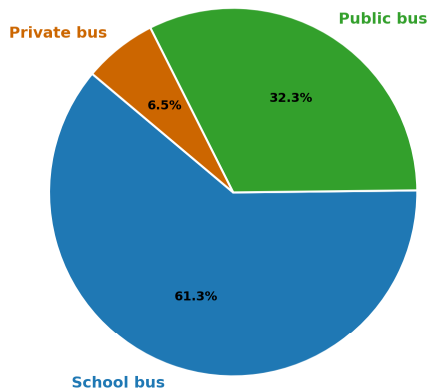


FIGURE 4. Distribution of bus tracking studies by types of bus transportation services.

The majority of recent research on bus tracking systems has focused on school buses (58.6%), reflecting a clear concern with real-time monitoring [55], [57], [58], [60] and operational management [45], [64], [69], [71], [72] in school settings. Moreover, several studies also emphasized student safety, such as security during boarding and alighting, using technologies such as QR codes, RFID, and even real-time image capture as seen in [56], [63], [65], [67], and [68]. Additionally, some recent research efforts, including [66] and [70], introduced IoT-based safety frameworks to offer real-time alerts to parents and school authorities. The high concentration of studies focusing on school bus tracking systems is partly due to the fact that school buses often operate in more controlled and predictable environments, such as specific school zones or small urban areas, which simplifies system implementation and management. The same can be noticed at university campuses where the well-defined space and routine schedules facilitate efficient monitoring and route optimization using GNSS and spatial analysis.

On the other hand, public bus monitoring (34.5%) presents significant challenges due to the many dynamic and less predictable variables, including traffic congestion, passenger demand variation, unscheduled detours, and a higher likelihood of road incidents. Studies such as [47] and [51] addressed these challenges by developing real-time GNSS-based tracking systems capable of handling larger, more complex urban environments. Additionally, service reliability studies (e.g., [229] and [255]) have emphasized the

difficulty of predicting travel times and bus arrivals due to the real-world operational complexities. Finally, for private buses (6.9%), such as tourist or corporate fleets, the research remains relatively limited. Notable examples include [73], who developed a bus tracking system for tourism, and [74], focusing on location tracking of private buses. Moreover, private buses represent a minor segment, with a significantly lower number of vehicles and users compared to PT systems as shown by [330] and [331].

Figure 5 shows the distribution of publications according to their main application within the bus transportation research field.

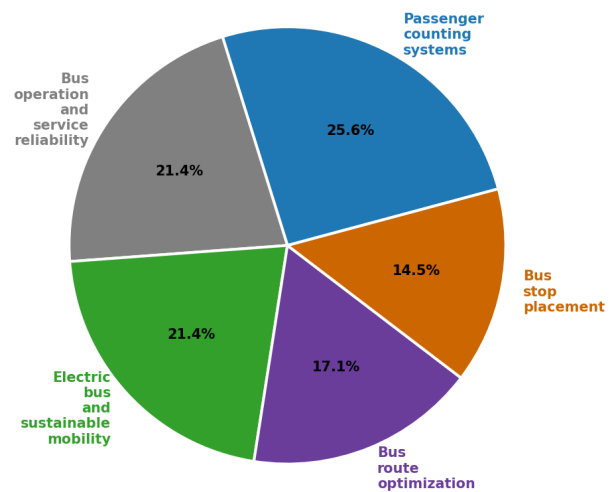


FIGURE 5. Publication trends by bus transportation application.

The studies are distributed across different applications, reflecting the challenges and opportunities in bus systems. As it can be seen, recent research efforts have been directed not only towards improving traditional operational aspects, such as passenger monitoring [242], [243], [244] and service reliability [229], [234], [246], but also towards advancing sustainable solutions.

Many studies have been focusing on electric bus operation and sustainable mobility. It reflects the global interest in decarbonizing PT systems and enhancing energy efficiency. Several studies have explored electric bus operations, focusing on energy consumption estimation [284], [295], [300] and charging infrastructure planning [274], [297], [299], as well as on smart scheduling and operational management [292], [304]. These works support the transition to greener mobility strategies and reflect the sector's efforts to adapt to the demands of sustainable urban transportation.

Figure 5 also highlights the relatively limited number of studies dedicated to bus stop placement, suggesting that this remains a research area requiring further exploration. Optimizing bus stop locations is crucial not only for enhancing accessibility, minimizing walking distances, and improving the overall efficiency of bus networks, but also

for supporting effective route planning, as well-placed stops directly influence route design and operational performance. Moreover, in the context of electric buses, bus stop location plays a strategic role in enabling opportunity charging and reducing overall energy consumption, as shown in Subsection IV-E.

This limited number of research on bus stop placement might be related to the complexity of obtaining detailed and accurate data on passenger demand, pedestrian flows, and built-environment characteristics across entire urban networks. Unlike operational metrics such as travel times or vehicle trajectories - which are increasingly accessible through GNSS and AVL systems - bus stop placement approaches often rely on data that is typically heterogeneous, fragmented, and dependent on local conditions, which includes pedestrian movement patterns, surrounding land-use dynamics, accessibility requirements, and social considerations. Therefore, it makes bus stop placement a multidisciplinary challenge that requires more frequent analytical exploration.

Figure 6 shows the distribution of publications from 2020 to 2025 based on the main applications presented in this article review.

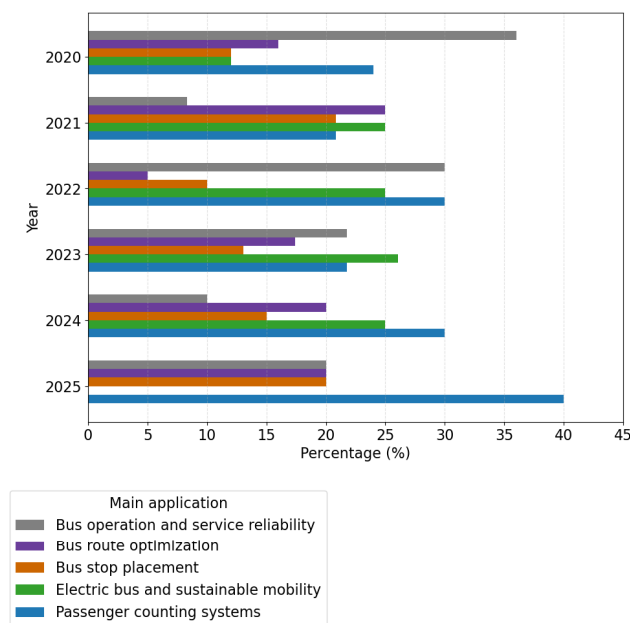


FIGURE 6. Publications by main application (2020–2025).

The development of passenger counting technologies is a topic of interest that keeps growing over the years, reflecting the importance of reliable passenger information. As it was stated by [332], such data is essential for public transport planning at different levels, such as strategic, tactical, and operational planning. Studies on electric buses and sustainable mobility also gain interest, highlighting a broader move toward cleaner and more efficient public transportation services [333], [334]. Other applications, such as bus route

optimization, bus operation and service reliability, and bus stop placement, remain present over the years but with smaller and more variable contributions. While these topics are less frequent, they remain essential, as they address operational efficiency and service quality.

Figure 7 shows the distribution of GNSS data and spatial analysis usage across the main research topics identified in this review.

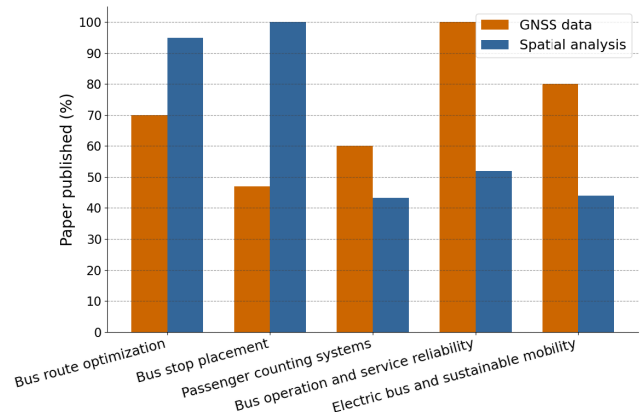


FIGURE 7. Use of GNSS data and spatial analysis by application in bus transportation.

Figure 7 highlights the predominant use of GNSS data across most research topics, particularly in studies related to bus operations and service reliability, where GNSS technology is essential for real-time tracking (see Table 3) and travel time analysis (see Table 7). The availability of spatiotemporal data enables a more accurate evaluation of service performance and support for real-time operational decisions. Additionally, GNSS plays a critical role in the context of electric buses, particularly for monitoring vehicle trajectories and optimizing route planning. By enabling precise control over travel paths and energy consumption, GNSS contributes to more sustainable operations - environmentally, by minimizing battery usage and emissions; and economically, by reducing operating costs through improved route efficiency.

In contrast, spatial analysis methods are more prominent in studies focusing on bus stop placement and bus route optimization, where spatial considerations and network design are critical elements (e.g., [21], [100], [103], [108], [109], [144], [156], [335], and [101]). Figure 7 also shows that studies on passenger counting systems make balanced use of spatial data and spatial analysis, indicating a well-integrated approach in this area.

Although this study presents several works that have GNSS-based monitoring, real-time data integration, and predictive optimization models at the core of their analyses to support the management of urban transportation systems, their contribution to equitable, accessible mobility and governance goals is not often explored. In many cases, technological advancements are primarily evaluated in terms

of accuracy, efficiency, or operational gains, while their potential social implications remain secondary. This gap highlights the need for future research to examine how data-driven optimization interacts with issues such as spatial inequality [336], [337] and service prioritization [338]. By integrating transportation policy and governance perspectives [339] into technical frameworks, future studies can help ensure that innovation not only improves system performance but also contributes to more inclusive and socially responsive mobility transitions, as well as helping to optimize how PT resources are allocated [340].

Many promising approaches still face practical adoption barriers to incorporate technological tools into daily operations due to limited institutional capacity and resource constraints [18], [341]. Concerns related to data privacy require greater attention, especially as real-time monitoring becomes more widespread [342], [343]. There is a need for stronger articulation between these technological innovations and strong public policies to ensure that advancements in transportation can assist sustainable mobility systems [344]. As cities expand electric mobility to reduce emissions and promote healthier urban environments, future research should investigate how spatial-data-driven approaches can support these goals while maintaining social equity [340]. The authors models BEBs networks with an equity-oriented objective, showing how their planning can incorporate social accessibility as a design criterion. Such studies can support transportation policies for local government and optimize the allocation of PT resources.

Beyond the technical and operational insights from this article review, the integration of spatial data and GNSS-enabled systems provides important implications for urban policy, planning, and governance. According to [345], how data are collected, analyzed, and governed can significantly influence transportation planning and policy outcomes. Moreover, data-driven frameworks enable transportation agencies and local governments to monitor service quality by precisely measuring service reliability [346]. From an urban policy perspective, GNSS-driven analytics support the transition toward data-informed decision-making in PT governance, allowing authorities to evaluate performance indicators and prioritize investments in underserved areas [347]. Such analyses are critical for ensuring that service expansion and integration align with social inclusion and sustainability goals [348]. Furthermore, linking GNSS-derived performance indicators with socio-spatial datasets can promote more equitable and resilient transportation services, ensuring that network improvements address accessibility deficits within urban spaces [159].

Finally, emerging perspectives on mobility transitions underscore the critical role of transportation systems that are predictable, socially responsive, and sustainable in the long term [349], [350]. In this context, the CalmMobility framework proposed by [351] offers a governance perspective that emphasizes gradual, coordinated, and socially

co-created mobility transformations. Future research should further explore how the CalmMobility paradigm can guide data-driven approaches to sustainable urban mobility transitions and data-driven strategies for improving PT systems. Integrating spatial data and GNSS-enabled analytics contributes to CalmMobility paradigm by enabling accurate monitoring of bus operations, improving transportation reliability, and supporting more equitable and adaptable service planning.

VI. CHALLENGES AND FUTURE RESEARCH

Despite significant advancements in IoT-based bus tracking systems, several challenges remain and need to be addressed in future research. One major issue is the reliability and stability of data transmission, especially in areas with weak or unstable GSM or Wi-Fi signals, which can affect the continuity of real-time tracking [51], [84], [278]. While many systems rely on GSM or Wi-Fi for data communication, these technologies are susceptible to signal loss, latency, and limited bandwidth, particularly densely built environments. To overcome these limitations, Low-Power Wide-Area Networks (LPWAN), such as LoRaWAN and Sigfox, provide extended coverage and reduced energy consumption, making them suitable for large-scale transportation systems with limited connectivity. Vehicular communication protocols, such as Dedicated Short-Range Communications (DSRC) and Cellular Vehicle-to-Everything (C-V2X), enable direct communication between vehicles and roadside infrastructure, reducing dependency on unstable mobile networks. Additionally, emerging technologies such as 5G and future 6G networks can offer ultra-low latency and high reliability for real-time applications. Hybrid connectivity solutions that integrate multiple communication technologies can also be implemented.

In addition, GNSS signal availability and accuracy can also be affected, especially in urban canyons, tunnels, or heavily forested areas. According to [352], it can lead to missing data, noisy data, erroneous data, conflicting data, and sparse data. While a number of studies have attempted to address this limitation [279], [280], [353], [354], it remains a challenge in the literature, especially those concerning positional accuracy in urban environments with low-cost devices. Addressing these limitations in urban environments requires the integration of complementary positioning techniques. Sensor fusion approaches, which combine GNSS with accelerometers, gyroscopes, magnetometers, and odometers, can improve positioning continuity during signal outages. Moreover, digital map-matching techniques further enhance reliability by aligning GNSS trajectories with predefined road networks, particularly in dense urban corridors.

Another challenge lies in the energy efficiency of embedded systems. Although some studies, such as [49] and [355], explored low-power mode algorithms on the GNSS, further research is needed to develop more comprehensive strategies that enable the module to operate with lower power

consumption, especially in systems operating in continuous tracking mode. Moreover, there is limited research on user experience and accessibility of mobile applications developed for tracking systems [78], [84], [356]. Future studies should include user-centered evaluations to enhance the usability and inclusiveness of such applications.

As it was presented in Subsection IV-A, recent studies have successfully employed GNSS data, GIS tools, and optimization algorithms to enhance bus route planning. However, some challenges can be noticed. References [103], [104], and [131] applied the analytic hierarchy process (AHP) to explore a practical method of bus route optimization. Although AHP provides a structured framework for decision-making, the results depend on expert input, which may not accurately reflect real-time passenger behavior, dynamic travel demand, or complex operational factors. This challenge can be addressed by employing artificial intelligence and machine learning techniques, which leverage large-scale historical and real-time data to better capture dynamic travel demand and passenger behavior. Furthermore, combining expert-based evaluations with data-driven models may offer a hybrid approach, mitigating subjectivity while maintaining operational feasibility. The integration of real-time GNSS, smart card, and passenger counting data into such models can further enhance the adaptability and reliability of bus route optimization strategies.

Many optimization models, such as those proposed by [102] and [108], consider stable travel demand, which limits their responsiveness to real-time fluctuations. Reference [102] also highlight the need to incorporate uncertain travel times and demand. Heuristic approaches, such as Ant Colony Optimization and Genetic Algorithms [105], [129], often require careful calibration, which can limit their use across different urban settings. For these cases, integrating real-time data sources and machine learning techniques can enhance adaptability to demand fluctuations and uncertain travel times. Hybrid approaches that combine heuristic algorithms with data-driven models may also reduce the dependency on extensive calibration while improving transferability across different urban contexts.

Spatial analyses conducted by [109], [119], and [120] showed that results are sensitive to how spatial boundaries are defined and how data are aggregated, suggesting the need for standardized geospatial frameworks in transportation planning. Similarly, shortest path algorithms used by [107], [140], and [142] emphasize efficiency but often overlook passenger-level preferences and multimodal transportation integration. Although dynamic real-time bus routing systems have been proposed by [97], [135], [357], [358], [359], and [360] real-world implementation remains limited due to the complexity of integrating dynamic data sources in real time to generate optimal routing decisions without compromising service reliability.

Future research should address these challenges by improving data integration methods and incorporating

real-time and uncertain variables into optimization frameworks in GIS-based approaches. In this direction, integrated dynamic models that combine heterogeneous data streams, including operational records, crowdsourced information (e.g., from Waze, OpenStreetMap), weather conditions, and traffic incidents, can provide more adaptive and resilient solutions. Predictive analytics and agent-based models [108], [361], [362] may further enhance the ability to anticipate disruptions and evaluate system-wide impacts. In addition, expanding open-source platforms [101] and assessing the operational implications of route optimization in real-world contexts can strengthen the applicability of these strategies.

Although Subsection IV-B highlights the growing application of advanced methods for bus stop placement, some challenges still remain. One major limitation lies in integrating multi-source data, which may differ in spatial detail and temporal coverage [102], [148]. Additionally, many approaches rely on static travel demand patterns, which may not fully reflect temporal variations in travel behavior or dynamic urban changes [144], [145].

While optimization algorithms such as NSGA-II, GA, and PSO have been employed [143], [146], [149], [153], [161], [162], [363], their performance is often sensitive to model calibration and parameter settings, limiting replicability across different urban settings. Similarly, GIS-based and AI-driven models may face challenges in adapting to dynamic urban environments, as they often rely on static datasets and may not reflect updated transportation conditions [146], [148]. Addressing this limitation requires frequent data updates and regular model recalibration. At the same time, AI models offer the potential to incorporate real-time and dynamic data streams into their inference processes, which may enhance responsiveness to changing travel demand and urban conditions. Future research could therefore explore hybrid approaches that combine optimization algorithms with the adaptability of AI models fed by real-time data sources.

Despite substantial progress in passenger counting systems as presented in Subsection IV-C, some challenges remain. One major issue concerns the limited availability of lighting data, particularly in systems based on smart card transactions [190], [195]. Although probabilistic and spatiotemporal clustering techniques have been proposed to address this [196], there is still a need for more robust solutions that can be applied in different urban contexts. Data integration between GNSS, smart card, and sensor networks poses another significant challenge, especially due to synchronization issues, missing data, and heterogeneous data formats [205], [364]. Furthermore, although sensor-based methods such as infrared and RFID offer low-cost alternatives [174], [207], their accuracy may decrease under crowded conditions or with environmental interference.

Future research could explore methods to estimate passenger at bus stops, complementing regular boarding and alighting data. This may include deploying sensors or smart devices at bus stops, integrating data from public surveillance

cameras, or leveraging crowdsourced information, while ensuring privacy and ethical considerations. Incorporating such data streams into existing passenger counting systems could enhance the accuracy of demand estimation and support more responsive and adaptive transit planning.

Privacy concerns also arise in Wi-Fi and Bluetooth-based systems, particularly when MAC address tracking is employed [176], [192]. Future research should focus on privacy-preserving passenger counting techniques [365], [366] that ensure data protection regulations. Computer vision and deep learning approaches, though promising in terms of accuracy [187], [208], [367], are often limited by sensitivity to lighting or occlusion and the computational resources required for real-time processing. Moreover, while LiDAR-based systems provide high-precision detection [188], [193], [200], their high cost restricts scalability in large transit networks. A key direction for future research is to develop hybrid systems that balance cost, accuracy, and scalability by combining low-cost sensors with computationally light deep learning models and cloud-based processing. Finally, future work should further explore spatiotemporal modeling and demand forecasting techniques [194], [199], [217], especially under conditions of incomplete or missing data.

Although GNSS data and machine learning techniques have been used for evaluating bus operations and service reliability, as shown in Subsection IV-D, some challenges still need to be addressed. One major issue is the generalization and scalability of predictive models. Most approaches rely on specific route configurations, limiting their applicability across different cities and transit networks [242], [244], [246]. Additionally, while models such as LSTM, ANN, and hybrid neural networks have shown improved accuracy in bus arrival and travel time predictions, there is a lack of standardization in model evaluation and feature selection [241], [255]. Despite advancements, real-time prediction systems are challenged by data latency, missing or noisy GNSS data, and the limited ability of historical data to accurately reflect dynamic traffic conditions [234], [251].

Future research should explore the integration of multi-source data such as GNSS, IC card data, and traffic sensors to enhance the accuracy of prediction models [247], [248]. There is also a need for improved interpretability of ML models to support decision-making by transit agencies. Furthermore, while tools such as real-time tracking apps have shown promise in improving passenger satisfaction [257], future studies should evaluate their long-term impact on ridership patterns and system-level performance.

Finally, considerable progress in electric mobility through BEBs has been made. As outlined in the Subsection IV-E, GNSS and spatial analysis play a critical role in supporting the integration of BEBs into PT services. However, some challenges still need to be investigated. One ongoing issue is the dependence on high-frequency and high-quality GNSS data for reliable analysis and modeling. As previously mentioned, low-resolution or incomplete GNSS datasets may

compromise the accuracy of bus trajectory reconstruction and speed estimation [279], [280]. Future research should explore robust data fusion techniques that combine GNSS with other sensors to overcome these limitations and enhance the reliability of energy consumption models under different data constraints.

Another area for future research lies in refining energy consumption estimation models. While machine learning and physics-based approaches have shown promise, current models are sensitive to factors such as GNSS sampling intervals [314], terrain variation [295], and operational variables such as passenger load and traffic conditions [308], [312]. More comprehensive, adaptive models that account for seasonal, environmental, and behavioral variables could improve accuracy. Additionally, optimal placement of charging stations remains a complex problem, especially in dense urban networks. Although many optimization methods have been proposed [274], [297], integrating real-time demand data, power grid variability, and multimodal accessibility into the decision-making process remains an open challenge. Finally, research is needed to address real-time and dynamic charging scheduling strategies that align with bus timetables, passenger flow, and energy availability, particularly under unexpected delays or route changes [292], [293], [299].

VII. FINAL CONSIDERATIONS

This review article provided a comprehensive overview of recent studies applying spatial data and analysis to optimize and plan bus transportation systems. By categorizing the literature into five key application areas, the review highlighted the increasing importance of GNSS-enabled data in improving operational efficiency, enhancing passenger experience, and supporting the development of more sustainable urban mobility solutions.

Although significant progress has been made, several challenges remain due to the growing complexity of urban environments and the increasing demand for reliable and adaptive transportation services. Persistent issues include data reliability, integration of multi-source and real-time information, energy efficiency of embedded systems, and the ability to incorporate dynamic operational variables into planning and management strategies.

The consistent interest observed in previous studies highlights that bus stop placement remains an underexplored research area that needs attentions. Despite its direct relationship with passenger accessibility, it is a critical factor that can influence user satisfaction and the overall PT system. Additionally, a growing number of recent studies have focused on supporting electric mobility and improving electric bus operations, reflecting the global trend towards cleaner and more sustainable public transport solutions. These findings offer valuable directions for future research and provide a solid foundation for the continued advancement of more intelligent, accessible, and environmentally friendly bus transportation services.

DECLARATIONS

The authors declare no conflict of interest.

REFERENCES

- [1] Y. Tang, A. Alhadlaq, A. R. Bagabaldo, and M. C. Gonzalez, "Designing transit routes based on vehicle routing behavior determined through location-based services data," *EPJ Data Sci.*, vol. 14, no. 1, pp. 1–21, Jun. 2025, doi: [10.1140/epjds/s13688-025-00559-5](https://doi.org/10.1140/epjds/s13688-025-00559-5).
- [2] L. Dell'Olio, A. Ibeas, J. de Ona, and R. de Ona, *Public Transportation Quality of Service: Factors, Models, and Applications*. Amsterdam, The Netherlands: Elsevier, 2017.
- [3] M. M. Rohani, D. C. Wijeyesekera, and A. T. A. Karim, "Bus operation, quality service and the role of bus provider and driver," *Proc. Eng.*, vol. 53, pp. 167–178, Jan. 2013, doi: [10.1016/j.proeng.2013.02.022](https://doi.org/10.1016/j.proeng.2013.02.022).
- [4] G. Mazzulla and L. Eboli, "A service quality experimental measure for public transport," *Eur. Transp.*, vol. 5, no. 34, pp. 42–53, 2006.
- [5] L. Eboli, C. Forciniti, and G. Mazzulla, "Service coverage factors affecting bus transit system availability," *Proc. Social Behav. Sci.*, vol. 111, pp. 984–993, Feb. 2014, doi: [10.1016/j.sbspro.2014.01.133](https://doi.org/10.1016/j.sbspro.2014.01.133).
- [6] N. Van Oort, *Service Reliability and Urban Public Transport Design*. Princeton, NJ, USA: Citeseer, 2011.
- [7] X. Chen, L. Yu, Y. Zhang, and J. Guo, "Analyzing urban bus service reliability at the stop, route, and network levels," *Transp. Res. A, Policy Pract.*, vol. 43, no. 8, pp. 722–734, Oct. 2009, doi: [10.1016/j.tra.2009.07.006](https://doi.org/10.1016/j.tra.2009.07.006).
- [8] T. Litman, "Evaluating accessibility for transportation planning," Victoria Transp. Policy Inst., Victoria, BC, Canada, Tech. Rep., Jan. 2008.
- [9] T. L. Lei and R. L. Church, "Mapping transit-based access: Integrating GIS, routes and schedules," *Int. J. Geographical Inf. Sci.*, vol. 24, no. 2, pp. 283–304, Feb. 2010, doi: [10.1080/13658810902835404](https://doi.org/10.1080/13658810902835404).
- [10] T. Lin, J. Xia, T. P. Robinson, K. G. Goulias, R. L. Church, D. Olaru, J. Tapin, and R. Han, "Spatial analysis of access to and accessibility surrounding train stations: A case study of accessibility for the elderly in Perth, Western Australia," *J. Transp. Geography*, vol. 39, pp. 111–120, Jul. 2014, doi: [10.1016/j.jtrangeo.2014.06.022](https://doi.org/10.1016/j.jtrangeo.2014.06.022).
- [11] A. Ceder, *Public Transit Planning and Operation: Modeling, Practice and Behavior*. Boca Raton, FL, USA: CRC Press, 2016.
- [12] E. Jenelius, "Public transport experienced service reliability: Integrating travel time and travel conditions," *Transp. Res. A, Policy Pract.*, vol. 117, pp. 275–291, Nov. 2018, doi: [10.1016/j.tra.2018.08.026](https://doi.org/10.1016/j.tra.2018.08.026).
- [13] T. Litman, "Transportation affordability," *Transports*, vol. 250, pp. 360–1560, Jan. 2016.
- [14] R. Olson and J. Austin, "A behavior-based study of bus drivers," *Prof. Saf.*, vol. 46, no. 11, pp. 20–25, 2001.
- [15] Y. Yang, J. Yan, J. Guo, Y. Kuang, M. Yin, S. Wang, and C. Ma, "Driving behavior analysis of city buses based on real-time GNSS traces and road information," *Sensors*, vol. 21, no. 3, p. 687, Jan. 2021, doi: [10.3390/s21030687](https://doi.org/10.3390/s21030687).
- [16] D. Loorbach, T. Schwanen, B. J. Doody, P. Arnfalk, O. Langeland, and E. Farstad, "Transition governance for just, sustainable urban mobility: An experimental approach from Rotterdam, The Netherlands," *J. Urban Mobility*, vol. 1, Dec. 2021, Art. no. 100009, doi: [10.1016/j.urbmob.2021.100009](https://doi.org/10.1016/j.urbmob.2021.100009).
- [17] A. Nikolaeva, P. Adey, T. Cresswell, J. Y. Lee, A. Nóvoa, and C. Temenos, "Commoning mobility: Towards a new politics of mobility transitions," *Trans. Inst. Brit. Geographers*, vol. 44, no. 2, pp. 346–360, Jun. 2019, doi: [10.1111/tran.12287](https://doi.org/10.1111/tran.12287).
- [18] I. Docherty, G. Marsden, and J. Anable, "The governance of smart mobility," *Transp. Res. A, Policy Pract.*, vol. 115, pp. 114–125, Sep. 2018, doi: [10.1016/j.tra.2017.09.012](https://doi.org/10.1016/j.tra.2017.09.012).
- [19] S. Paiva, M. Ahad, G. Tripathi, N. Feroz, and G. Casalino, "Enabling technologies for urban smart mobility: Recent trends, opportunities and challenges," *Sensors*, vol. 21, no. 6, p. 2143, Mar. 2021, doi: [10.3390/s21062143](https://doi.org/10.3390/s21062143).
- [20] G. Droj, L. Droj, and A.-C. Badea, "GIS-based survey over the public transport strategy: An instrument for economic and sustainable urban traffic planning," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 1, p. 16, Dec. 2021, doi: [10.3390/ijgi11010016](https://doi.org/10.3390/ijgi11010016).
- [21] P. K. Sahu, B. Mehran, S. P. Mahapatra, and S. Sharma, "Spatial data analysis approach for network-wide consolidation of bus stop locations," *Public Transp.*, vol. 13, no. 2, pp. 375–394, Jun. 2021, doi: [10.1007/s12469-021-00266-0](https://doi.org/10.1007/s12469-021-00266-0).
- [22] A. Kathuria, M. Parida, and R. S. Chalumuri, "Travel-time variability analysis of bus rapid transit system using GPS data," *J. Transp. Eng., A, Syst.*, vol. 146, no. 6, Jun. 2020, Art. no. 05020003, doi: [10.1061/jtepbs.0000357](https://doi.org/10.1061/jtepbs.0000357).
- [23] O. J. Ibarra-Rojas, F. Delgado, R. Giesen, and J. C. Muñoz, "Planning, operation, and control of bus transport systems: A literature review," *Transp. Res. B, Methodol.*, vol. 77, pp. 38–75, Jul. 2015, doi: [10.1016/j.trb.2015.03.002](https://doi.org/10.1016/j.trb.2015.03.002).
- [24] Y. Wang, Y. Deng, F. Ren, R. Zhu, P. Wang, T. Du, and Q. Du, "Analysing the spatial configuration of urban bus networks based on the geospatial network analysis method," *Cities*, vol. 96, Jan. 2020, Art. no. 102406, doi: [10.1016/j.cities.2019.102406](https://doi.org/10.1016/j.cities.2019.102406).
- [25] T. Shanmukhappa, I. W.-H. Ho, and C. K. Tse, "Spatial analysis of bus transport networks using network theory," *Phys. A, Stat. Mech. Appl.*, vol. 502, pp. 295–314, Jul. 2018, doi: [10.1016/j.physa.2018.02.111](https://doi.org/10.1016/j.physa.2018.02.111).
- [26] W. Li, M. Batty, and M. F. Goodchild, "Real-time GIS for smart cities," *Int. J. Geographical Inf. Sci.*, vol. 34, no. 2, pp. 311–324, Feb. 2020, doi: [10.1080/13658816.2019.1673397](https://doi.org/10.1080/13658816.2019.1673397).
- [27] D. Hernández, J. C. Muñoz, R. Giesen, and F. Delgado, "Analysis of real-time control strategies in a corridor with multiple bus services," *Transp. Res. B, Methodol.*, vol. 78, pp. 83–105, Aug. 2015, doi: [10.1016/j.trb.2015.04.011](https://doi.org/10.1016/j.trb.2015.04.011).
- [28] L. Shen and P. R. Stopher, "Review of GPS travel survey and GPS data-processing methods," *Transp. Res.*, vol. 34, no. 3, pp. 316–334, May 2014, doi: [10.1080/01441647.2014.903530](https://doi.org/10.1080/01441647.2014.903530).
- [29] National Coordination Office for Space-Based Positioning, Navigation, and Timing. (2021). *Other Global Navigation Satellite Systems (GNSS)*. Accessed: Jun. 30, 2025. [Online]. Available: <https://www.gps.gov/systems/gnss>
- [30] B. Hofmann-Wellenhof, H. Lichtenegger, and E. Wasle, *GNSS-Global Navigation Satellite Systems: GPS, GLONASS, Galileo, and More*. Cham, Switzerland: Springer, 2007.
- [31] M. Goletz and D. Ehebrect, "How can GPS/GNSS tracking data be used to improve our understanding of informal transport? A discussion based on a feasibility study from dar es salaam," *J. Transp. Geography*, vol. 88, Oct. 2020, Art. no. 102305, doi: [10.1016/j.jtrangeo.2018.08.015](https://doi.org/10.1016/j.jtrangeo.2018.08.015).
- [32] F. Maltinti, M. Coni, J. Rombi, B. Barabino, R. Ventura, and N. Rassu, "A literature review on bus comfort on-board," in *Proc. Int. Conf. Comput. Sci. Appl.*, 2024, pp. 80–97, doi: [10.1007/978-3-031-65332-2_6](https://doi.org/10.1007/978-3-031-65332-2_6).
- [33] H. Zhang, H. Cui, and B. Shi, "A data-driven analysis for operational vehicle performance of public transport network," *IEEE Access*, vol. 7, pp. 96404–96413, 2019, doi: [10.1109/ACCESS.2019.2930279](https://doi.org/10.1109/ACCESS.2019.2930279).
- [34] X. Ma and Y. Wang, "Development of a data-driven platform for transit performance measures using smart card and GPS data," *J. Transp. Eng.*, vol. 140, no. 12, Dec. 2014, Art. no. 04014063.
- [35] S. Raza, A. Al-Kaisy, R. Teixeira, and B. Meyer, "The role of GNSS-RTN in transportation applications," *Encyclopedia*, vol. 2, no. 3, pp. 1237–1249, Jul. 2022, doi: [10.3390/encyclopedia2030083](https://doi.org/10.3390/encyclopedia2030083).
- [36] L. Wu, B. Yang, and P. Jing, "Travel mode detection based on GPS raw data collected by smartphones: A systematic review of the existing methodologies," *Information*, vol. 7, no. 4, p. 67, Nov. 2016, doi: [10.3390/info7040067](https://doi.org/10.3390/info7040067).
- [37] A. D. Nagne and B. W. Gawali, "Transportation network analysis by using remote sensing and gis a review," *Int. J. Eng. Res. Appl.*, vol. 3, no. 3, pp. 70–76, 2013.
- [38] N. Singh and S. K. Katiyar, "Application of geographical information system (GIS) in reducing accident blackspots and in planning of a safer urban road network: A review," *Ecol. Informat.*, vol. 66, Dec. 2021, Art. no. 101436, doi: [10.1016/j.ecoinf.2021.101436](https://doi.org/10.1016/j.ecoinf.2021.101436).
- [39] A. Nayak and K. Goyal, "Traffic modeling and accidental data analysis using GIS: A review," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 1327, no. 1, 2024, Art. no. 012028, doi: [10.1088/1755-1315/1327/1/012028](https://doi.org/10.1088/1755-1315/1327/1/012028).
- [40] N. J. van Eck and L. Waltman, "Software survey: VOSviewer, a computer program for bibliometric mapping," *Scientometrics*, vol. 84, no. 2, pp. 523–538, Aug. 2010, doi: [10.1007/s11192-009-0146-3](https://doi.org/10.1007/s11192-009-0146-3).
- [41] W. M. Miranda, R. T. R. de Mendonça, A. A. da Silva, A. M. de Lima Curvello, F. L. D. S. de Souza, and H. J. da Silva, "BusMe: Automatic bus localization system and route registration," *Proc. Comput. Sci.*, vol. 109, pp. 1098–1103, Jan. 2017, doi: [10.1016/j.procs.2017.05.397](https://doi.org/10.1016/j.procs.2017.05.397).

- [42] M. W. Raad, M. Deriche, and T. Sheltami, "An IoT-based school bus and vehicle tracking system using RFID technology and mobile data networks," *Arabian J. Sci. Eng.*, vol. 46, no. 4, pp. 3087–3097, Apr. 2021, doi: [10.1007/s13369-020-05111-3](https://doi.org/10.1007/s13369-020-05111-3).
- [43] T. M. Hamadto, Z. A. Adam, and M. H. Elsayed, "An Android application of school bus tracker based on RFID technology," in *Proc. Int. Conf. Comput., Control, Electr., Electron. Eng. (ICCCEEE)*, Feb. 2021, pp. 1–4, doi: [10.1109/ICCCEEE49695.2021.9429629](https://doi.org/10.1109/ICCCEEE49695.2021.9429629).
- [44] S. Banu, V. Vaibhav, L. Venugopal, and P. R. Rangaswamy, "IoT based school bus monitoring system," in *Proc. 7th Int. Conf. Design Innov. 3 Cs Compute Communicate Control (ICDI3C)*, Nov. 2023, pp. 67–70, doi: [10.1109/icdi3c61568.2023.00021](https://doi.org/10.1109/icdi3c61568.2023.00021).
- [45] A. Kumar Sharma, R. Pandey, S. Tarafdar, and S. Dubey, "Towards smart mobility in cities—bus tracking and booking system," in *Proc. 9th Int. Conf. Rel., INFOCOM Technol. Optim. (Trends Future Directions) (ICRITO)*, Sep. 2021, pp. 1–5, doi: [10.1109/ICRITO51393.2021.9596492](https://doi.org/10.1109/ICRITO51393.2021.9596492).
- [46] S. Kumar, S. Moundekar, M. Rathod, and R. Parmar, "Location-based bus tracking application using Android," in *Communication, Software and Networks*. Cham, Switzerland: Springer, 2023, pp. 369–380.
- [47] F. C. Prasetyo, R. Munadi, and A. I. Irawan, "Implementasi sistem monitoring dan tracking kendaraan roda empat menggunakan global positioning system (GPS) berbasis Internet of Things," *eProceedings Eng.*, vol. 9, no. 6, pp. 1–10, 2022.
- [48] K. Mundada, S. Patti, T. Rajguru, P. Savji, and S. Shambharkar, "Smart bus real-time tracking system using GSM and GPS module," in *Proc. Int. Conf. ICT Sustain. Develop.*, 2023, pp. 509–518, doi: [10.1007/978-981-99-4932-8_46](https://doi.org/10.1007/978-981-99-4932-8_46).
- [49] I. W. Mustika, A. Bejo, A. R. Fidiyanto, and D. W. Hapsari, "Development of campus bus tracker firmware using GNSS module on the STM32 platform," in *Proc. Int. Conf. Digit. Bus. Technol. Manage. (ICONDBTM)*, Aug. 2023, pp. 1–6, doi: [10.1109/icondbtm59210.2023.10326777](https://doi.org/10.1109/icondbtm59210.2023.10326777).
- [50] S. P. Jayaram, D. Jayakumar, B. Sandhiya, M. Vanathi, and J. Karthika, "Bus tracking system using mobile GPS technology," in *Proc. Int. Conf. Inventive Comput. Technol. (ICICT)*, 2024, pp. 1811–1813, doi: [10.1109/icict60155.2024.10544750](https://doi.org/10.1109/icict60155.2024.10544750).
- [51] S. W. Saputra, D. Hidayat, R. R. Adha, R. M. Fikri, S. M. Nasution, and R. R. Septiawan, "IoT public transport monitoring system for Bandung city," in *Proc. IEEE 10th Int. Conf. Smart Instrum., Meas. Appl. (ICSIMA)*, Jul. 2024, pp. 276–280, doi: [10.1109/ICSIMA62563.2024.10675581](https://doi.org/10.1109/ICSIMA62563.2024.10675581).
- [52] S. Prabha, S. Kharod, D. C. Biswas, H. Sharma, S. K. Pandit, and G. Sangwan, "Smart transit with GPS based real-time bus tracking," *SSRN J.*, no. 5222126, pp. 1–6, Apr. 2025, doi: [10.2139/ssrn.5222126](https://doi.org/10.2139/ssrn.5222126).
- [53] S. Kavitha, S. B. Kumar, K. N. Krishnan, and R. Ashwanth, "IoT-based bus courier transit through public transport," in *Proc. 5th Int. Conf. Trends Mater. Sci. Inventive Mater. (ICTMIM)*, Apr. 2025, pp. 1318–1323, doi: [10.1109/ictmim65579.2025.10987945](https://doi.org/10.1109/ictmim65579.2025.10987945).
- [54] K. Madasanal, A. Gudnavar, S. Nagarmunoli, B. Madagouda, and K. Naregal, "Efficient smart public transit system using RFID and IoT integration with MERN web applications," in *Proc. Int. Conf. Comput., Electr. Commun. Eng. (ICCECE)*, Feb. 2025, pp. 1–8, doi: [10.1109/iccece61355.2025.10940803](https://doi.org/10.1109/iccece61355.2025.10940803).
- [55] M. Kumari, A. Kumar, and A. Khan, "IoT based intelligent real-time system for bus tracking and monitoring," in *Proc. Int. Conf. Power Electron. IoT Appl. Renew. Energy Control (PARC)*, Feb. 2020, pp. 226–230, doi: [10.1109/PARC49193.2020.246240](https://doi.org/10.1109/PARC49193.2020.246240).
- [56] S. Malathy, P. Ambarish, S. D. Kumar, and G. A. Gokul Prashanth, "Smart school bus: To ensure the safety of children," in *Proc. 7th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, vol. 1, Mar. 2021, pp. 923–927, doi: [10.1109/ICACCS51430.2021.9442044](https://doi.org/10.1109/ICACCS51430.2021.9442044).
- [57] F. M. Saleem, O. Khan, A. Fatima, A. Shahid, and I. Javed, "Smart school bus: Tracking management system," in *Proc. 6th Int. Electr. Eng. Conf. (IEEC)*, 2021, pp. 1–4.
- [58] J. Hemalatha, K. Hitaishi, V. Sharma, S. Vura, and B. Paul, "An efficient Android based school bus tracking system," *Int. J. Eng. Appl. Sci. Technol.*, vol. 6, no. 1, pp. 383–389, May 2021, doi: [10.33564/ijeast.2021.v06i01.059](https://doi.org/10.33564/ijeast.2021.v06i01.059).
- [59] H. Gull, D. Aljohar, R. Alutaibi, D. Alqahtani, M. Alarfaj, and R. Alqahtani, "Smart school bus tracking: Requirements and design of an IoT based school bus tracking system," in *Proc. 5th Int. Conf. Trends Electron. Informat. (ICOEI)*, Jun. 2021, pp. 388–394, doi: [10.1109/ICOEI51242.2021.9452818](https://doi.org/10.1109/ICOEI51242.2021.9452818).
- [60] B. Vincent, J. Sabu, C. Mathew, S. S. Nair, S. B. George, and D. Saleema, "Live college bus tracking and route mapping using Internet of Things," in *Proc. 2nd Int. Conf. Comput. Syst. Commun. (ICCCS)*, 2023, pp. 1–7, doi: [10.1109/icccs56913.2023.10143028](https://doi.org/10.1109/icccs56913.2023.10143028).
- [61] M. Sobhana, T. R. Chowdary, M. G. S. S. Venkatesh, and K. S. Devendra, "Smart campus bus tracking alert system using real-time GPS," in *Proc. 9th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, vol. 1, Mar. 2023, pp. 1777–1781, doi: [10.1109/ICACCS57279.2023.10112757](https://doi.org/10.1109/ICACCS57279.2023.10112757).
- [62] A. S. Lr, S. A. Likith, and N. Guruprasad, "Tracking and security features enhancement in a smart school bus using IoT," in *Proc. Int. Conf. IoT, Commun. Autom. Technol. (ICICAT)*, Jun. 2023, pp. 1–5, doi: [10.1109/icicat57735.2023.10263634](https://doi.org/10.1109/icicat57735.2023.10263634).
- [63] M. Selvaganesh, B. Geetha, R. Harini, and S. Jinith, "School bus monitoring and security system using IoT," in *Proc. 8th Int. Conf. Commun. Electron. Syst. (ICCES)*, Jun. 2023, pp. 432–437, doi: [10.1109/icc57224.2023.10192856](https://doi.org/10.1109/icc57224.2023.10192856).
- [64] S. S. Chandrabhan, S. V. Dagadu, S. B. Sadgir, K. A. Laxman, W. I. Santosh, and P. M. Kurhe, "Smart school bus intelligence system with biometric authentication for student attendance & safety," in *Proc. 4th Int. Conf. Comput., Autom. Knowl. Manage. (ICCAKM)*, Dec. 2023, pp. 1–4, doi: [10.1109/iccakm58659.2023.10449496](https://doi.org/10.1109/iccakm58659.2023.10449496).
- [65] M. M. Hassain, M. J. Hasan, M. A. Rahman, A. Fahim, and M. M. Uddin, "Real time student monitoring smart school bus with an IoT based security system," in *Proc. Int. Jordanian Cybersecurity Conf. (IJCC)*, 2024, pp. 93–98, doi: [10.1109/ijcc64742.2024.10847285](https://doi.org/10.1109/ijcc64742.2024.10847285).
- [66] M. R. Mahmud, A. Shakil, M. A. Uddin, M. Muhammad, and R. Ahamed, "Smart school bus monitoring with a security system based on IoT," in *Proc. 27th Int. Conf. Comput. Inf. Technol. (ICCIT)*, Dec. 2024, pp. 3629–3634, doi: [10.1109/iccit64611.2024.11022516](https://doi.org/10.1109/iccit64611.2024.11022516).
- [67] K. Bhaskar, G. Sandeep, G. S. Srinivas, D. Keerthana, and D. Tejaswi, "A smart system for school bus accident detection and GPS tracking," in *Proc. Int. Conf. Emerg. Syst. Intell. Comput. (ESIC)*, Feb. 2024, pp. 355–359, doi: [10.1109/esic60604.2024.10481563](https://doi.org/10.1109/esic60604.2024.10481563).
- [68] B. Gadade, A. Mulani, and A. Harale, "IoT based smart school bus and student monitoring system," *Naturalista Campano*, vol. 28, no. 1, pp. 730–737, 2024.
- [69] M. Mohammed, H. Aljibori, A. N. J. Al-Tamimi, M. Alifiras, V. Subramanian, A. Sharif, S. A. Jowder, Z. N. Alani, O. I. Abdullah, and A. S. Ahmad, "Toward sustainable smart cities in Bahrain: A novel IoT-based smart school bus solution for enhanced children's transportation safety and smart city integration," in *Proc. Arab ICT Conf. (AICTC)*, Aug. 2024, pp. 232–236, doi: [10.1109/AICTC58357.2024.10735035](https://doi.org/10.1109/AICTC58357.2024.10735035).
- [70] F. X. E. Deepak, P. Anitha, K. K. Kumar, and A. Kamaraja, "Revolutionizing school transportation safety with real-time data and advanced sensors," in *Proc. 2nd Int. Conf. Intell. Cyber Phys. Syst. Internet Things (ICoCI)*, Aug. 2024, pp. 499–505, doi: [10.1109/icoici62503.2024.10696511](https://doi.org/10.1109/icoici62503.2024.10696511).
- [71] S. Yeruva, A. Grandhi, S. D. Katta, S. S. C. Palutla, and G. D. Amruthaluri, "EduBusGuru: Enhancing college bus tracking and management with real-time mobile app," in *Proc. 15th Int. Conf. Comput. Commun. Netw. Technol. (ICCCNT)*, Jun. 2024, pp. 1–8, doi: [10.1109/iccncnt61001.2024.10724760](https://doi.org/10.1109/iccncnt61001.2024.10724760).
- [72] V. Jairam, B. Keerthana, and M. Johnson, "IoT-Enabled safety and tracking platform," in *Proc. Int. Conf. Comput. Commun. Technol. (ICCCNT)*, Apr. 2025, pp. 1–6, doi: [10.1109/iccncnt63501.2025.11019666](https://doi.org/10.1109/iccncnt63501.2025.11019666).
- [73] V. G. Ascueta, P. C. J. B. Bautista, J. C. Quilala, and M. A. Beninsig, "BusTap: A real-time bus tracking Android application," in *Proc. 1st Int. Conf. Inf. Comput. Res. (iCORE)*, 2021, pp. 175–180, doi: [10.1109/icore54267.2021.00050](https://doi.org/10.1109/icore54267.2021.00050).
- [74] K. Akilan, A. Chandrasekaran, A. Kumar, S. B. Jashwanth, A. Joshua, S. Rajalakshmi, and S. A. Deborah, "Customized Internet of Things-based bus tracking and management system," in *Disruptive Technologies for Big Data and Cloud Applications*. Cham, Switzerland: Springer, 2022, pp. 601–611.
- [75] S. Jain, A. Trivedi, and S. Sharma, "Application based bus tracking system," in *Proc. Int. Conf. Mach. Learn., Big Data, Cloud Parallel Comput. (COMITCon)*, Feb. 2019, pp. 152–154, doi: [10.1109/COMITCON.2019.8862254](https://doi.org/10.1109/COMITCON.2019.8862254).
- [76] A. D. Shree, J. M. S. Anusuya, and S. Malathy, "Real time bus tracking and location updation system," in *Proc. 5th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, 2019, pp. 242–245, doi: [10.1109/ICACCS.2019.8728353](https://doi.org/10.1109/ICACCS.2019.8728353).

- [77] K. I. Monica, S. Gurupriya, and S. A. Magdaline, "Bus tracking system using GPS on smartphones," *Int. J. Eng. Res. Technol.*, vol. 7, no. 11, pp. 1–4, 2019.
- [78] K. Anitha, P. Rohit, N. Chandrasekhar, and M. Satyanarayana, "RFID & GPS smart attendance system for school children tracking by using Blynk platform," *Emerg. Technol. Innov. Res.*, vol. 10, pp. 180–187, Jun. 2023.
- [79] R. Barbosa, R. Sousa, F. Oliveira, H. Oliveira, P. Luz, and L. T. Manera, "Circulino: An IoT solution applied in the university transport service," in *Proc. Brazilian Technol. Symp.* Cham, Switzerland: Springer, 2019, pp. 503–514, doi: [10.1007/978-3-030-16053-1_49](https://doi.org/10.1007/978-3-030-16053-1_49).
- [80] J. T. Raj and J. N. Sankar, "IoT based smart school bus monitoring and notification system," *Int. J. Eng. Res. Technol.*, vol. 140, pp. 89–92, 2017.
- [81] M. Vyavahare, "IoT based school bus monitoring and security system," *Int. J. Innov. Sci. Res. Technol.*, vol. 4, no. 3, pp. 244–247, 2019.
- [82] I. Korkmaz, A. Camci, C. Cengiz, D. Dirik, E. Cekci, and F. M. Akbaba, "A smart school bus tracking system," in *Proc. Int. Symp. Netw., Comput. Commun. (ISNCC)*, Jun. 2019, pp. 1–6, doi: [10.1109/ISNCC.2019.8909188](https://doi.org/10.1109/ISNCC.2019.8909188).
- [83] S. Shah and B. Singh, "RFID based school bus tracking and security system," in *Proc. Int. Conf. Commun. Signal Process. (ICCCSP)*, Apr. 2016, pp. 1481–1485, doi: [10.1109/ICCCSP.2016.7754404](https://doi.org/10.1109/ICCCSP.2016.7754404).
- [84] M. F. M. A. Hakeem, N. A. Sulaiman, M. Kassim, and N. M. Isa, "IoT bus monitoring system via mobile application," in *Proc. IEEE Int. Conf. Autom. Control Intell. Syst. (I2CACIS)*, Jun. 2022, pp. 125–130, doi: [10.1109/I2CACIS54679.2022.9815268](https://doi.org/10.1109/I2CACIS54679.2022.9815268).
- [85] N. Abdullah and S. M. M. Shazali, "Internet of Things with RFID based microcontroller for monitoring system," in *Proc. 11th Int. Conf. Robot.*, 2022, pp. 652–657, doi: [10.1007/978-981-16-8129-5_99](https://doi.org/10.1007/978-981-16-8129-5_99).
- [86] A. S. Shibghatullah, A. Jalil, M. H. A. Wahab, J. N. P. Soon, K. Subaramaniam, and T. Eldabi, "Vehicle tracking application based on real time traffic," *Int. J. Electr. Electron. Eng. Telecommun.*, vol. 11, no. 1, pp. 67–73, 2022, doi: [10.18178/ijete.11.1.67-73](https://doi.org/10.18178/ijete.11.1.67-73).
- [87] B. Furetti, P. Cintia, C. Renzo, and L. Spinsanti, "Inferring human activities from GPS tracks," in *Proc. 2nd ACM SIGKDD Int. Workshop Urban Comput.*, Aug. 2013, pp. 1–8, doi: [10.1145/2505821.2505830](https://doi.org/10.1145/2505821.2505830).
- [88] H. Gong, C. Chen, E. Bialostozky, and C. T. Lawson, "A GPS/GIS method for travel mode detection in New York City," *Comput., Environ. Urban Syst.*, vol. 36, no. 2, pp. 131–139, Mar. 2012, doi: [10.1016/j.compenurbsys.2011.05.003](https://doi.org/10.1016/j.compenurbsys.2011.05.003).
- [89] A. Sauerländer-Biebl, E. Brockfeld, D. Suske, and E. Melde, "Evaluation of a transport mode detection using fuzzy rules," *Transp. Res. Proc.*, vol. 25, pp. 591–602, Jan. 2017, doi: [10.1016/j.trpro.2017.05.444](https://doi.org/10.1016/j.trpro.2017.05.444).
- [90] J. L. Curzel, R. Lüders, K. V. O. Fonseca, and M. O. Rosa, "Temporal performance analysis of bus transportation using link streams," *Math. Problems Eng.*, vol. 2019, no. 1, Jan. 2019, Art. no. 6139379, doi: [10.1155/2019/6139379](https://doi.org/10.1155/2019/6139379).
- [91] P. Santin, F. R. Gubert, M. Fonseca, A. Munaretto, and T. H. Silva, "Characterization of public transit mobility patterns of different economic classes," *Sustainability*, vol. 12, no. 22, p. 9603, Nov. 2020, doi: [10.3390/su12229603](https://doi.org/10.3390/su12229603).
- [92] F. R. Gubert, P. Santin, M. Fonseca, A. Munaretto, and T. H. Silva, "On strategies to help reduce contamination on public transit: A multilayer network approach," *Appl. Netw. Sci.*, vol. 8, no. 1, p. 37, Jun. 2023, doi: [10.1007/s41109-023-00562-7](https://doi.org/10.1007/s41109-023-00562-7).
- [93] C. Tang, A. Ceder, Y.-E. Ge, and N. Wu, "Optimal operational strategies for multiple bus lines considering passengers' preferences," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2674, no. 5, pp. 572–586, May 2020, doi: [10.1177/0361198120917159](https://doi.org/10.1177/0361198120917159).
- [94] T. Jayasinghe, A. Munasinghe, and T. Sivakumar, "A microscopic traffic simulation framework to evaluate the performance of bus priority lanes," *J. South Asian Logistics Transp.*, vol. 3, no. 1, pp. 1–19, Mar. 2023, doi: [10.4038/jsalt.v3i1.57](https://doi.org/10.4038/jsalt.v3i1.57).
- [95] O. Hussain, E. Felemban, and F. U. Rehman, "Optimization of the mashaer shuttle-bus service in Hajj: Arafat-Muzdalifah case study," *Information*, vol. 12, no. 12, p. 496, Nov. 2021, doi: [10.3390/info12120496](https://doi.org/10.3390/info12120496).
- [96] X. Ou and C. Li, "The analysis of bus route optimization based on taxi GPS data," in *Proc. CICTP*, Aug. 2020, pp. 3282–3294, doi: [10.1061/9780784482933.283](https://doi.org/10.1061/9780784482933.283).
- [97] C. Wang, C. Ma, and X. Xu, "Multi-objective optimization of real-time customized bus routes based on two-stage method," *Phys. A, Stat. Mech. Appl.*, vol. 537, Jan. 2020, Art. no. 122774, doi: [10.1016/j.physa.2019.122774](https://doi.org/10.1016/j.physa.2019.122774).
- [98] R. M. Noor, N. B. G. Rasyidi, T. Nandy, and R. Kolandaisamy, "Campus shuttle bus route optimization using machine learning predictive analysis: A case study," *Sustainability*, vol. 13, no. 1, p. 225, Dec. 2020, doi: [10.3390/su13010225](https://doi.org/10.3390/su13010225).
- [99] R. Li and G. Li, "Research on bus route layout of regional microcirculation based on large data," in *Proc. CICTP*, Dec. 2020, pp. 2838–2850, doi: [10.1061/9780784483053.239](https://doi.org/10.1061/9780784483053.239).
- [100] R. Rendeiro Martín-Cejas, R. Suárez Vega, and P. P. Ramírez Sánchez, "GIS approach applied to tourist bus route design on lanzarote island," *Sustainability*, vol. 13, no. 19, p. 10671, Sep. 2021, doi: [10.3390/su131910671](https://doi.org/10.3390/su131910671).
- [101] P. Mohandas, M. V. Saraswathy, and A. P. Alex, "Development of an algorithm for bus routing and tracking for an educational institution: A case study," *J. Inst. Engineers (India), Ser. A*, vol. 102, no. 1, pp. 279–292, Mar. 2021, doi: [10.1007/s40030-020-00493-x](https://doi.org/10.1007/s40030-020-00493-x).
- [102] M. Wei, T. Liu, and B. Sun, "Optimal routing design of feeder transit with stop selection using aggregated cell phone data and open source GIS tool," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 4, pp. 2452–2463, Apr. 2021, doi: [10.1109/TITS.2020.3042014](https://doi.org/10.1109/TITS.2020.3042014).
- [103] S. F. Balket and N. M. Asmael, "Selecting the best route location for bus rapid transit using geographic information system (GIS): Kut city is a case study," *J. Phys., Conf. Ser.*, vol. 1895, no. 1, May 2021, Art. no. 012029, doi: [10.1088/1742-6596/1895/1/012029](https://doi.org/10.1088/1742-6596/1895/1/012029).
- [104] Q. Shi, K. Zhang, J. Weng, Y. Dong, S. Ma, and M. Zhang, "Evaluation model of bus routes optimization scheme based on multi-source bus data," *Transp. Res. Interdiscipl. Perspect.*, vol. 10, Jun. 2021, Art. no. 100342, doi: [10.1016/j.trip.2021.100342](https://doi.org/10.1016/j.trip.2021.100342).
- [105] M. Ozmen and H. Sahin, "Real-time optimization of school bus routing problem in smart cities using genetic algorithm," in *Proc. 6th Int. Conf. Inventive Comput. Technol. (ICICT)*, Jan. 2021, pp. 1152–1158, doi: [10.1109/ICICT50816.2021.9358666](https://doi.org/10.1109/ICICT50816.2021.9358666).
- [106] X. Zhou, "A dynamic diagnose on bus routes optimization using AHP based on multi-source bus data," in *Proc. 10th Int. Conf. Traffic Logistic Eng. (ICTLE)*, Aug. 2022, pp. 22–27, doi: [10.1109/ICTLE55577.2022.9902076](https://doi.org/10.1109/ICTLE55577.2022.9902076).
- [107] Y. Sun, H. Shi, D. Zhang, L. Zhang, and J. Song, "A real-time update method for bus stops and routes based on historical bus data," in *Proc. 11th Int. Conf. Inf. Technol., IoT Smart City*, 2023, pp. 211–216, doi: [10.1145/3638985.3639020](https://doi.org/10.1145/3638985.3639020).
- [108] G. Calabrò, M. L. Pira, G. Inturri, M. Ignaccolo, and A. Pluchino, "A simulation-optimization approach to solve the first and last mile of mass rapid transit via feeder services," *Transp. Res. Proc.*, vol. 69, pp. 767–774, Jan. 2023, doi: [10.1016/j.trpro.2023.02.234](https://doi.org/10.1016/j.trpro.2023.02.234).
- [109] Z. M. Yusoff, A. Shuib, S. Z. Ishak, E. S. Mokhtar, F. Othman, and N. A. Ramazan, "Mobility planning for urban employee to workplace: An analysis of bus routes network and stop locations," *IOP Conf. Ser., Earth Environ. Sci.*, vol. 1151, no. 1, Mar. 2023, Art. no. 012003, doi: [10.1088/1755-1315/1151/1/012003](https://doi.org/10.1088/1755-1315/1151/1/012003).
- [110] Y. Guan, W. Xiang, Y. Wang, X. Yan, and Y. Zhao, "Bi-level optimization for customized bus routing serving passengers with multiple-trips based on state-space-time network," *Phys. A, Stat. Mech. Appl.*, vol. 614, Mar. 2023, Art. no. 128517, doi: [10.1016/j.physa.2023.128517](https://doi.org/10.1016/j.physa.2023.128517).
- [111] M. Xiao, L. Chen, H. Feng, Z. Peng, and Q. Long, "Smart city public transportation route planning based on multi-objective optimization: A review," *Arch. Comput. Methods Eng.*, vol. 31, no. 6, pp. 3351–3375, Aug. 2024, doi: [10.1007/s11831-024-10076-9](https://doi.org/10.1007/s11831-024-10076-9).
- [112] M. Xiao, L. Chen, H. Feng, Z. Peng, and Q. Long, "Sustainable and robust route planning scheme for smart city public transport based on multi-objective optimization: Digital twin model," *Sustain. Energy Technol. Assessments*, vol. 65, May 2024, Art. no. 103787, doi: [10.1016/j.seta.2024.103787](https://doi.org/10.1016/j.seta.2024.103787).
- [113] K. Sriprateep, R. Pitakaso, S. Khonjun, T. Srichok, P. Luesak, S. Gonwirat, C. Kaewta, M. Kosacka-Olejnik, and P. Enkvetchakul, "Multi-objective optimization of resilient, sustainable, and safe urban bus routes for tourism promotion using a hybrid reinforcement learning algorithm," *Mathematics*, vol. 12, no. 14, p. 2283, Jul. 2024, doi: [10.3390/math12142283](https://doi.org/10.3390/math12142283).
- [114] Z. Yin, Y. Gu, Y. Zheng, and W. Li, "Research on online bus-hailing route planning based on a bi-level programming model," in *Proc. Int. Conf. Intell. Comput. Next Gener. Netw. (ICNGN)*, Nov. 2024, pp. 1–5, doi: [10.1109/icngn63705.2024.10871527](https://doi.org/10.1109/icngn63705.2024.10871527).
- [115] F. Gong, C. Jia, and X. Wu, "Optimization methods for customized bus routes in random environments," *J. Adv. Transp.*, vol. 2025, no. 1, Jan. 2025, Art. no. 1680317, doi: [10.1155/atr/1680317](https://doi.org/10.1155/atr/1680317).

- [116] Z. Ahern, A. Paz, and P. Corry, "Approximate multi-objective optimization for integrated bus route design and service frequency setting," *Transp. Res. B, Methodol.*, vol. 155, pp. 1–25, Jan. 2022, doi: [10.1016/j.trb.2021.10.007](https://doi.org/10.1016/j.trb.2021.10.007).
- [117] S. Moslem, M. K. Saraji, A. Mardani, A. Alkharabsheh, S. Duleba, and D. Esztergár-Kiss, "A systematic review of analytic hierarchy process applications to solve transportation problems: From 2003 to 2022," *IEEE Access*, vol. 11, pp. 11973–11990, 2023, doi: [10.1109/ACCESS.2023.3234298](https://doi.org/10.1109/ACCESS.2023.3234298).
- [118] T. L. Saaty, H. J. Zoffer, L. G. Vargas, and A. Guiora, "The analytic hierarchy process: Beyond 'getting to yes' in conflict resolution," in *Overcoming the Retributive Nature of the Israeli-Palestinian Conflict*. Cham, Switzerland: Springer, 2022, pp. 17–29, doi: [10.1007/978-3-030-83958-1_3](https://doi.org/10.1007/978-3-030-83958-1_3).
- [119] S. Pemberton, "Optimising Melbourne's bus routes for real-life travel patterns," *Case Stud. Transp. Policy*, vol. 8, no. 3, pp. 1038–1052, Sep. 2020, doi: [10.1016/j.cstp.2020.04.002](https://doi.org/10.1016/j.cstp.2020.04.002).
- [120] Z. Wang, S. Wang, and H. Lian, "A route-planning method for long-distance commuter express bus service based on OD estimation from mobile phone location data: The case of the changping corridor in Beijing," *Public Transp.*, vol. 13, no. 1, pp. 101–125, Mar. 2021, doi: [10.1007/s12469-020-00254-w](https://doi.org/10.1007/s12469-020-00254-w).
- [121] K. J. Min, A. A. Fatiah, and Z. Ponrahono, "A spatial analysis of subsidised bus route catchment feasibility in an urban area in Malaysia," *Planning Malaysia*, vol. 20, pp. 95–107, Dec. 2022, doi: [10.21837/PM.v20i24.1186](https://doi.org/10.21837/PM.v20i24.1186).
- [122] S. Tsigidinos, C. Karolemeas, M. Siti, K. Papadaki, K. Athanasopoulos, and P. G. Tzouras, "Route planning for flexible bus services in regional cities and rural areas: Combining user preferences with spatial analysis," *Future Transp.*, vol. 4, no. 4, pp. 1476–1500, Dec. 2024, doi: [10.3390/futuretransp4040071](https://doi.org/10.3390/futuretransp4040071).
- [123] G. K. Kuah and J. Perl, "Optimization of feeder bus routes and bus-stop spacing," *J. Transp. Eng.*, vol. 114, no. 3, pp. 341–354, May 1988.
- [124] S. N. Kuan, H. L. Ong, and K. M. Ng, "Applying metaheuristics to feeder bus network design problem," *Asia-Pacific J. Oper. Res.*, vol. 21, no. 4, pp. 543–560, Dec. 2004, doi: [10.1142/s0217595904000382](https://doi.org/10.1142/s0217595904000382).
- [125] A. S. Mohaymany and A. Gholami, "Multimodal feeder network design problem: Ant colony optimization approach," *J. Transp. Eng.*, vol. 136, no. 4, pp. 323–331, Apr. 2010.
- [126] S. N. Kuan, H. L. Ong, and K. M. Ng, "Solving the feeder bus network design problem by genetic algorithms and ant colony optimization," *Adv. Eng. Softw.*, vol. 37, no. 6, pp. 351–359, Jun. 2006, doi: [10.1016/j.advengsoft.2005.10.003](https://doi.org/10.1016/j.advengsoft.2005.10.003).
- [127] S. Katoch, S. S. Chauhan, and V. Kumar, "A review on genetic algorithm: Past, present, and future," *Multimedia Tools Appl.*, vol. 80, no. 5, pp. 8091–8126, Feb. 2021, doi: [10.1007/s11042-020-10139-6](https://doi.org/10.1007/s11042-020-10139-6).
- [128] A. Bolotbekova, H. Hakli, and A. Beskiri, "Trip route optimization based on bus transit using genetic algorithm with different crossover techniques: A case study in Konya/Türkiye," *Sci. Rep.*, vol. 15, no. 1, p. 2491, Jan. 2025, doi: [10.1038/s41598-025-86695-4](https://doi.org/10.1038/s41598-025-86695-4).
- [129] Z. Xue, X. Deng, B. Chen, and A. Khamis, "School bus routing using metaheuristics algorithms," in *Proc. IEEE Int. Conf. Smart Mobility (SM)*, Mar. 2023, pp. 33–38, doi: [10.1109/SM57895.2023.10112473](https://doi.org/10.1109/SM57895.2023.10112473).
- [130] N. Yoon, I. Choi, and S. W. Kang, "Analysis of route efficiency in city bus transportation vulnerable areas," in *Proc. IEEE Int. Conf. Big Data (Big Data)*, Dec. 2022, pp. 3845–3853, doi: [10.1109/BigData55660.2022.10020748](https://doi.org/10.1109/BigData55660.2022.10020748).
- [131] M. Zhao, Y. Zhao, C. Ma, and X. Wu, "Robust optimization of mixed-load school bus route based on multi-objective genetic algorithm," *Eng. Lett.*, vol. 30, no. 4, pp. 1–20, 2022.
- [132] A. R. Komijan, P. Ghasemi, K. Khalili-Damghani, and F. Hashemi Yazdi, "A new school bus routing problem considering gender separation, special students and mix loading: A genetic algorithm approach," *J. Optim. Ind. Eng.*, vol. 14, no. 2, pp. 23–39, 2021, doi: [10.22094/joie.2020.1891023.1722](https://doi.org/10.22094/joie.2020.1891023.1722).
- [133] B. Bulut, M. Gunay, K. Ozgun, and J. W. Ledet, "Optimizing bus lines using genetic algorithm for public transportation," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2021, pp. 131–136, Dec. 2021, doi: [10.5194/isprs-archives-xxvi-4-w5-2021-131-2021](https://doi.org/10.5194/isprs-archives-xxvi-4-w5-2021-131-2021).
- [134] A. Basu, B. Raja, R. Gracious, and L. Vanajakshi, "Dynamic trip planner for public transport using genetic algorithm," *Transport*, vol. 35, no. 2, pp. 156–167, Apr. 2020, doi: [10.3846/transport.2020.12477](https://doi.org/10.3846/transport.2020.12477).
- [135] C. Shen, Y. Sun, Z. Bai, and H. Cui, "Real-time customized bus routes design with optimal passenger and vehicle matching based on column generation algorithm," *Phys. A, Stat. Mech. Appl.*, vol. 571, Jun. 2021, Art. no. 125836, doi: [10.1016/j.physa.2021.125836](https://doi.org/10.1016/j.physa.2021.125836).
- [136] W. Shu and Y. Li, "A novel demand-responsive customized bus based on improved ant colony optimization and clustering algorithms," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 8, pp. 8492–8506, Aug. 2023, doi: [10.1109/TITS.2022.3145655](https://doi.org/10.1109/TITS.2022.3145655).
- [137] C. Wang and C. Ma, "Multi-objective optimization of customized bus routes based on full operation process," *Mod. Phys. Lett. B*, vol. 34, no. 25, Sep. 2020, Art. no. 2050266, doi: [10.1142/s0217984920502668](https://doi.org/10.1142/s0217984920502668).
- [138] G. Yu and J. Yang, "On the robust shortest path problem," *Comput. Oper. Res.*, vol. 25, no. 6, pp. 457–468, Jun. 1998, doi: [10.1016/s0305-0548\(97\)00085-3](https://doi.org/10.1016/s0305-0548(97)00085-3).
- [139] C. Wang, Z. Ye, and W. Wang, "A multi-objective optimization and hybrid heuristic approach for urban bus route network design," *IEEE Access*, vol. 8, pp. 12154–12167, 2020, doi: [10.1109/ACCESS.2020.2966008](https://doi.org/10.1109/ACCESS.2020.2966008).
- [140] B. Y. Chen, X.-W. Chen, H.-P. Chen, and W. H. K. Lam, "A fast algorithm for finding k shortest paths using generalized spur path reuse technique," *Trans. GIS*, vol. 25, no. 1, pp. 516–533, Feb. 2021, doi: [10.1111/tgis.12699](https://doi.org/10.1111/tgis.12699).
- [141] S. M. Faradunbeh, H. Li, M. Kang, and C. Iim, "The optimized path for the public transportation of incheon in South Korea," 2023, *arXiv:2309.10006*.
- [142] E. Özceylan, "Offering a new bus route between campus and bus terminal using shortest path algorithm," in *Logistics and Supply Chain Management*, Z. Molamohamadi, E. B. Tirkolaee, A. Mirzazadeh, and G.-W. Weber, Eds., Cham, Switzerland: Springer, 2021, pp. 271–276, doi: [10.1007/978-3-030-89743-7_15](https://doi.org/10.1007/978-3-030-89743-7_15).
- [143] S. Wang, Z. Zuo, and Y. Liu, "Study on location of bus stop in subway service area based on Residents' travel accessibility," *Sustainability*, vol. 15, no. 5, p. 4517, Mar. 2023, doi: [10.3390/su15054517](https://doi.org/10.3390/su15054517).
- [144] G. Bahiru, B. Teklu, and T. Tilahun, "Evaluation of public bus stops location and spacing: The case of selected routes of Anbessa city bus service enterprise," *Zede J.*, vol. 42, no. 1, pp. 1–13, 2024, doi: [10.20372/zede.v42i.10178](https://doi.org/10.20372/zede.v42i.10178).
- [145] T. H. Aung, K. K. Lwin, and Y. Sekimoto, "Analysis of bus operation at peak hours using bus GPS data: A case study of YBS-36," *J. Disaster Res.*, vol. 15, no. 3, pp. 426–436, Mar. 2020, doi: [10.20965/jdr.2020.p0426](https://doi.org/10.20965/jdr.2020.p0426).
- [146] N. Shatnawi, A. A. Al-Omari, and H. Al-Qudah, "Optimization of bus stops locations using GIS techniques and artificial intelligence," *Proc. Manuf.*, vol. 44, pp. 52–59, Jan. 2020, doi: [10.1016/j.promfg.2020.02.204](https://doi.org/10.1016/j.promfg.2020.02.204).
- [147] S. F. Balket and N. M. Asmael, "The use of geographic information system to analyze stations locating for proposed rapid bus route in AL KUT city," *J. Phys., Conf. Ser.*, vol. 1973, no. 1, Aug. 2021, Art. no. 012243, doi: [10.1088/1742-6596/1973/1/012243](https://doi.org/10.1088/1742-6596/1973/1/012243).
- [148] G. Duan, X. Ma, J. Wang, Z. Wang, and Y. Wang, "Optimization of urban bus stops setting based on data mining," *Int. J. Pattern Recognit. Artif. Intell.*, vol. 35, no. 8, Jun. 2021, Art. no. 2159028, doi: [10.1142/s021800142159028x](https://doi.org/10.1142/s021800142159028x).
- [149] M. Ghasedi, M. Ghorbanzadeh, and I. Bargegol, "Robust optimization of bus stop placement based on dynamic demand using meta heuristic approaches: A case study in a developing country," *Transp. Telecommun. J.*, vol. 22, no. 1, pp. 39–52, Feb. 2021, doi: [10.2478/tjt-2021-0004](https://doi.org/10.2478/tjt-2021-0004).
- [150] F. Wang, M. Ye, H. Zhu, and D. Gu, "Optimization method for conventional bus stop placement and the bus line network based on the Voronoi diagram," *Sustainability*, vol. 14, no. 13, p. 7918, Jun. 2022, doi: [10.3390/su14137918](https://doi.org/10.3390/su14137918).
- [151] T. Stadler, S. Hofmeister, and J. Dünnweber, "A method for the optimized placement of bus stops based on Voronoi diagrams," in *Proc. 55th Hawaii Int. Conf. Syst. Sci.*, 2022, pp. 1–9.
- [152] M. M. R. Bhuiya, R. Rafiq, K. N. M. Morshed, and M. I. Rahman, "Suitable bus stop locations for a proposed bus rapid transit corridor in a developing country city: An analytical hierarchy process approach," *Transp. Developing Economies*, vol. 9, no. 1, p. 10, Apr. 2023, doi: [10.1007/s40890-023-00179-6](https://doi.org/10.1007/s40890-023-00179-6).
- [153] C. Li, R. Ge, X. Wu, and A. Khamis, "Optimal placement of bus stops using particle swarm optimization," in *Proc. IEEE Int. Conf. Smart Mobility (SM)*, Mar. 2023, pp. 15–20, doi: [10.1109/SM57895.2023.10112283](https://doi.org/10.1109/SM57895.2023.10112283).
- [154] J. Borges, A. M. Peixoto, T. H. Silva, A. Munaretto, and R. Luders, "Towards spatiotemporal integration of bus transit with data-driven approaches," 2024, *arXiv:2402.17866*.
- [155] J. Yu, Y. Zhu, C. Wu, and G. Song, "Optimization of bus stop location based on spatial analysis and k-means clustering method—A case study in Hohhot city," *Res. Square*, vol. 1, pp. 1–17, Feb. 2024, doi: [10.21203/rs.3.rs-3913979/v1](https://doi.org/10.21203/rs.3.rs-3913979/v1).

- [156] L. Ashour and D. Tarawneh, "Strategic bus stop allocation for enhanced accessibility in the amman bus rapid transit," in *Proc. 3rd Int. Conf. Electr., Comput., Commun. Mechatronics Eng. (ICECCME)*, Jul. 2023, pp. 1–10, doi: [10.1109/iceccme57830.2023.10252452](https://doi.org/10.1109/iceccme57830.2023.10252452).
- [157] J. Wang, K. Wang, and Y. Zhao, "Identifying potential upgradable bus stop locations with on-demand shuttle ridership with VIA data in Jersey City," *Transp. Res. A, Policy Pract.*, vol. 196, Jun. 2025, Art. no. 104480, doi: [10.1016/j.tra.2025.104480](https://doi.org/10.1016/j.tra.2025.104480).
- [158] S. Wang, Y. Sun, C. Musco, and Z. Bao, "Public transport planning: When transit network connectivity meets commuting demand," in *Proc. Int. Conf. Manage. Data*, Jun. 2021, pp. 1906–1919, doi: [10.1145/3448016.3457247](https://doi.org/10.1145/3448016.3457247).
- [159] R. Żochowska, M. J. Klos, P. Soczówka, and M. Pilch, "Assessment of accessibility of public transport by using temporal and spatial analysis," *Sustainability*, vol. 14, no. 23, p. 16127, Dec. 2022, doi: [10.3390/su142316127](https://doi.org/10.3390/su142316127).
- [160] T. Liu, W. Ji, K. Gkiotsalitis, and O. Cats, "Optimizing public transport transfers by integrating timetable coordination and vehicle scheduling," *Comput. Ind. Eng.*, vol. 184, Oct. 2023, Art. no. 109577, doi: [10.1016/j.cie.2023.109577](https://doi.org/10.1016/j.cie.2023.109577).
- [161] J. H. Taplin and Y. Sun, "Optimizing bus stop locations for walking access: Stops-first design of a feeder route to enhance a residential plan," *Environ. Planning B, Urban Anal. City Sci.*, vol. 47, no. 7, pp. 1237–1259, Sep. 2020, doi: [10.1177/2399808318824108](https://doi.org/10.1177/2399808318824108).
- [162] A. Taghavi, R. Ghanbari, K. Ghorbani-Moghadam, A. Davoodi, and A. Emrouznejad, "A genetic algorithm for solving bus terminal location problem using data envelopment analysis with multi-objective programming," *Ann. Oper. Res.*, vol. 309, no. 1, pp. 259–276, Feb. 2022, doi: [10.1007/s10479-021-04244-4](https://doi.org/10.1007/s10479-021-04244-4).
- [163] B. She, X. Zhu, X. Ye, W. Guo, K. Su, and J. Lee, "Weighted network Voronoi diagrams for local spatial analysis," *Comput., Environ. Urban Syst.*, vol. 52, pp. 70–80, Jul. 2015, doi: [10.1016/j.compenurbysys.2015.03.005](https://doi.org/10.1016/j.compenurbysys.2015.03.005).
- [164] A. Shokrollahi, J. A. Persson, R. Malekian, A. Sarkheyli-Hägele, and F. Karlsson, "Passive infrared sensor-based occupancy monitoring in smart buildings: A review of methodologies and machine learning approaches," *Sensors*, vol. 24, no. 5, p. 1533, Feb. 2024, doi: [10.3390/s24051533](https://doi.org/10.3390/s24051533).
- [165] R. L. dos Santos, H. C. de Oliveira, M. C. de Almeida, D. F. Vieira, E. P. L. Junior, and T. Ji, "A low-cost bidirectional people counter device for assisting social distancing monitoring for COVID-19," *J. Control. Autom. Electr. Syst.*, vol. 33, no. 4, pp. 1148–1160, Aug. 2022, doi: [10.1007/s40313-022-00916-z](https://doi.org/10.1007/s40313-022-00916-z).
- [166] C. Pronello and X. R. Garzón Ruiz, "Evaluating the performance of video-based automated passenger counting systems in real-world conditions: A comparative study," *Sensors*, vol. 23, no. 18, p. 7719, Sep. 2023, doi: [10.3390/s23187719](https://doi.org/10.3390/s23187719).
- [167] C.-H. Cheng, M.-C. Tsai, and Y.-C. Cheng, "An intelligent time-series model for forecasting bus passengers based on smartcard data," *Appl. Sci.*, vol. 12, no. 9, p. 4763, May 2022, doi: [10.3390/app12094763](https://doi.org/10.3390/app12094763).
- [168] M. Alatiyyah, "Smartphone-based Wi-Fi analysis for bus passenger counting," *Comput., Mater. Continuum*, vol. 79, no. 1, pp. 875–907, 2024, doi: [10.32604/cmc.2024.047790](https://doi.org/10.32604/cmc.2024.047790).
- [169] V. M. Cheme, M. H. Martinez, R. Durango, and V. S. Padilla, "Occupancy estimation at bus stops through Wi-Fi connectivity assessment—A study in Guayaquil city," *Transp. Telecommun. J.*, vol. 23, no. 1, pp. 93–101, Feb. 2022, doi: [10.2478/tj-2022-0009](https://doi.org/10.2478/tj-2022-0009).
- [170] D. T. Cardoso, D. Manfroí, and E. P. de Freitas, "Improvement in the detection of passengers in public transport systems by using UHF RFID," *Int. J. Wireless Inf. Netw.*, vol. 27, no. 1, pp. 116–132, Mar. 2020, doi: [10.1007/s10776-019-00457-4](https://doi.org/10.1007/s10776-019-00457-4).
- [171] V. K. Balla, H. Nandibhatla, G. Santhoshi, A. V. Krishna, M. Mohan, and P. Purushotham, "Smart bus ticketing system with RFID using IoT," in *Proc. IEEE 6th Int. Conf. Cybern., Cognition Mach. Learn. Appl. (ICCCMLA)*, Oct. 2024, pp. 297–301, doi: [10.1109/ICCCMLA63077.2024.10871743](https://doi.org/10.1109/ICCCMLA63077.2024.10871743).
- [172] A. Radovan, L. Mršić, G. Đambić, and B. Mihaljević, "A review of passenger counting in public transport concepts with solution proposal based on image processing and machine learning," *Eng.*, vol. 5, no. 4, pp. 3284–3315, Dec. 2024, doi: [10.3390/eng5040172](https://doi.org/10.3390/eng5040172).
- [173] S. D. F. Marques and C. S. Pitombo, "Transit ridership modeling at the bus stop level: Comparison of approaches focusing on count and spatially dependent data," *Appl. Spatial Anal. Policy*, vol. 16, no. 1, pp. 277–313, Mar. 2023, doi: [10.1007/s12061-022-09482-y](https://doi.org/10.1007/s12061-022-09482-y).
- [174] A. Rahmatulloh, F. M. S. Nursuwars, I. Darmawan, and G. Febrizki, "Applied Internet of Things (IoT): The prototype bus passenger monitoring system using PIR sensor," in *Proc. 8th Int. Conf. Inf. Commun. Technol. (ICoICT)*, Jun. 2020, pp. 1–6, doi: [10.1109/ICoICT49345.2020.9166420](https://doi.org/10.1109/ICoICT49345.2020.9166420).
- [175] S. Ryu, B. B. Park, and S. El-Tawab, "Wi-Fi sensing system for monitoring public transportation ridership: A case study," *KSCE J. Civil Eng.*, vol. 24, no. 10, pp. 3092–3104, Oct. 2020, doi: [10.1007/s12205-020-0316-7](https://doi.org/10.1007/s12205-020-0316-7).
- [176] A. Hidayat, S. Terabe, and H. Yaginuma, "Estimating bus passenger volume based on a Wi-Fi scanner survey," *Transp. Res. Interdiscipl. Perspect.*, vol. 6, Jul. 2020, Art. no. 100142, doi: [10.1016/j.trip.2020.100142](https://doi.org/10.1016/j.trip.2020.100142).
- [177] M. Ribeiro, B. Galvão, C. Prandi, and N. Nunes, "Passive Wi-Fi monitoring in public transport: A case study in the Madeira Island," 2020, *arXiv:2006.16083*.
- [178] Y. Zhang, W. Tu, K. Chen, C. H. Wu, L. Li, W. H. Ip, and C. Y. Chan, "Bus passenger flow statistics algorithm based on deep learning," *Multimedia Tools Appl.*, vol. 79, nos. 39–40, pp. 28785–28806, Oct. 2020, doi: [10.1007/s11042-020-09487-0](https://doi.org/10.1007/s11042-020-09487-0).
- [179] L. Liu, J. He, Y. Hou, and C. Zhang, "A technology for automatically counting bus passenger based on YOLOv2 and MIL algorithm," in *Proc. IEEE 5th Int. Conf. Image*, vol. 14, Aug. 2020, pp. 166–170, doi: [10.1109/ICIVC50857.2020.9177434](https://doi.org/10.1109/ICIVC50857.2020.9177434).
- [180] P. Kuchár, R. Pirník, T. Tichý, K. Rástočný, M. Skuba, and T. Tettamanti, "Noninvasive passenger detection comparison using thermal imager and IP cameras," *Sustainability*, vol. 13, no. 22, p. 12928, Nov. 2021, doi: [10.3390/su132212928](https://doi.org/10.3390/su132212928).
- [181] J. Kniess, J. C. Rutke, and W. A. C. Castañeda, "An IoT transport architecture for passenger counting: A real implementation," in *Proc. IFIP/IEEE Int. Symp. Integr. Netw. Manage. (IM)*, Jul. 2021, pp. 613–617.
- [182] G. Recalde, D. Gregor, D. Britez, and M. Arzamendia, "Low-cost in-bus people counting system for the reordering of urban passenger traffic based on artificial vision and neural networks," in *Proc. IEEE CHILEAN Conf. Electr., Electron. Eng., Inf. Commun. Technol. (CHILECON)*, Dec. 2021, pp. 1–4, doi: [10.1109/CHILECON54041.2021.9703018](https://doi.org/10.1109/CHILECON54041.2021.9703018).
- [183] M. Rahman, S. Yasmin, A. Faghih-Imani, and N. Eluru, "Examining the bus ridership demand: Application of spatio-temporal panel models," *J. Adv. Transp.*, vol. 2021, pp. 1–10, May 2021, doi: [10.1155/2021/8844743](https://doi.org/10.1155/2021/8844743).
- [184] S. D. F. Marques and C. S. Pitombo, "Ridership estimation along bus transit lines based on kriging: Comparative analysis between network and Euclidean distances," *J. Geovisualization Spatial Anal.*, vol. 5, no. 1, p. 7, Jun. 2021, doi: [10.1007/s41651-021-00075-w](https://doi.org/10.1007/s41651-021-00075-w).
- [185] M. P. R. Junior, R. M. A. Medrano, and C. F. Almeida, "UAI-FI: Using artificial intelligence for automatic passenger counting through Wi-Fi and GPS data UAI-FI: Utilização de inteligência artificial para contagem automática de passageiros através de Wi-Fi e dados GPS," *Transportes*, vol. 30, no. 2, p. 2555, Aug. 2022, doi: [10.14295/transportes.v30i2.2555](https://doi.org/10.14295/transportes.v30i2.2555).
- [186] S. Desai, R. Suthar, V. Yadav, V. Ankar, and V. Gupta, "Smart bus fleet management system using IoT," in *Proc. 4th Int. Conf. Emerg. Res. Electron., Comput. Sci. Technol. (ICERECT)*, Dec. 2022, pp. 01–06, doi: [10.1109/ICERECT56837.2022.10059646](https://doi.org/10.1109/ICERECT56837.2022.10059646).
- [187] I. J. C. Valencia, M. Guillermo, E. P. Dadios, A. M. Fillone, E. Sybingco, and R. K. C. Billones, "Overhead view bus passenger detection and counter using DeepSORT and tiny-YOLO V4," in *Proc. IEEE 14th Int. Conf. Humanoid*, vol. 1, Sep. 2022, pp. 1–6, doi: [10.1109/HNICEM57413.2022.10109539](https://doi.org/10.1109/HNICEM57413.2022.10109539).
- [188] R. Seidel, N. Jahn, S. Seo, T. Goerttler, and K. Obermayer, "NAPC: A neural algorithm for automated passenger counting in public transport on a privacy-friendly dataset," *IEEE Open J. Intell. Transp. Syst.*, vol. 3, pp. 33–44, 2022, doi: [10.1109/OJITS.2021.3139393](https://doi.org/10.1109/OJITS.2021.3139393).
- [189] L. Deepa, A. R. Pinjari, S. K. Nirmale, K. K. Srinivasan, and T. Rambha, "A direct demand model for bus transit ridership in Bengaluru, India," *Transp. Res. A, Policy Pract.*, vol. 163, pp. 126–147, Sep. 2022, doi: [10.1016/j.tra.2022.07.004](https://doi.org/10.1016/j.tra.2022.07.004).
- [190] K. Özgün, M. Günay, D. Başaran, and J. Ledet, "Estimation of alighting counts for public transportation vehicle occupancy levels using reverse direction boarding," *J. Public Transp.*, vol. 25, Mar. 2023, Art. no. 100070, doi: [10.1016/j.jpuptr.2023.100070](https://doi.org/10.1016/j.jpuptr.2023.100070).
- [191] H. Jiang, S. Chen, Z. Xiao, J. Hu, J. Liu, and S. Dustdar, "Pa-count: Passenger counting in vehicles using Wi-Fi signals," *IEEE Trans. Mobile Comput.*, vol. 23, no. 4, pp. 2684–2697, Apr. 2024, doi: [10.1109/TMC.2023.3263229](https://doi.org/10.1109/TMC.2023.3263229).

- [192] G. Delzanno, L. Caputo, D. D'Agostino, D. Grosso, A. H. Mustajab, L. Bixio, and M. Rulli, "Automatic passenger counting on the edge via unsupervised clustering," *Sensors*, vol. 23, no. 11, p. 5210, May 2023, doi: [10.3390/s23115210](https://doi.org/10.3390/s23115210).
- [193] Y. Liu, Q. Lyu, J. Zhang, X. Suo, C. Hong, L. Lin, and D. Wang, "APC: Automatic passenger counting and seat occupancy detection for sightseeing tram based on 3D LiDAR and camera," in *Proc. IEEE Int. Conf. Cybern. Intell. Syst. (CIS) IEEE Conf. Robot.*, Jul. 2023, pp. 44–49, doi: [10.1109/CIS-RAM55796.2023.10370009](https://doi.org/10.1109/CIS-RAM55796.2023.10370009).
- [194] S. D. F. Marques and C. S. Pitombo, "Local modeling as a solution to the lack of stop-level ridership data," *J. Transp. Geography*, vol. 112, Oct. 2023, Art. no. 103682, doi: [10.1016/j.jtrangeo.2023.103682](https://doi.org/10.1016/j.jtrangeo.2023.103682).
- [195] P. Mohanraj, J. Barros, H. Lin, R. Murcio, and H. Bei, "Alighting location estimation from public transit data: A case study of Shenzhen," *Transp. Planning Technol.*, vol. 48, no. 5, pp. 937–952, Jul. 2025, doi: [10.1080/03081060.2024.2382247](https://doi.org/10.1080/03081060.2024.2382247).
- [196] L. Li, S. Xue, and Y. Xiao, "An efficient approach for identifying potential bus passenger demand based on multisource data," *J. Adv. Transp.*, vol. 2024, no. 1, Jan. 2024, Art. no. 5368577, doi: [10.1155/2024/5368577](https://doi.org/10.1155/2024/5368577).
- [197] P. Mohanraj, T. Rajasekar, V. Jeeva, J. Jerrin, and P. Nithinkrishna, "Smart public transport monitoring and demand analysis with machine learning," in *Proc. 5th Int. Conf. Electron. Sustain. Commun. Syst. (ICESC)*, 2024, pp. 1674–1679.
- [198] J. M. B. Besmonte, J. I. C. D. Cueva, J. A. A. Jiménez, D. D. D. Valenzuela, M. L. C. Guico, and J. K. A. Galicia, "Public vehicle passenger counting system using infrared imaging with data analytics," in *Proc. IEEE Region 10 Conf. (TENCON)*, Aug. 2024, pp. 409–413, doi: [10.1109/TENCON61640.2024.10902683](https://doi.org/10.1109/TENCON61640.2024.10902683).
- [199] Y. Montero-Lamas, R. Fernández-Casal, F.-A. Varela-García, A. Orro, and M. Novales, "A spatial statistical approach to estimate bus stop demand using GIS-processed data," *J. Transp. Geography*, vol. 118, Jun. 2024, Art. no. 103906, doi: [10.1016/j.jtrangeo.2024.103906](https://doi.org/10.1016/j.jtrangeo.2024.103906).
- [200] M. Pown, R. Dhillip, and A. Ganesh, "Autonomous passenger counting and tracking system," in *Proc. Int. Conf. Data Sci., Agents Artif. Intell. (ICDSAAI)*, Mar. 2025, pp. 1–6, doi: [10.1109/icd-saaai65575.2025.11011735](https://doi.org/10.1109/icd-saaai65575.2025.11011735).
- [201] T. G. Rao, K. Tejaswi, B. L. Sree, D. P. Hyma, and D. S. Rani, "School bus boarding, debarking, and location notification system based on GPS and the Internet of Things," *Int. J. Eng. Res. Sci. Technol.*, vol. 21, no. 2, pp. 180–186, 2025.
- [202] J. Y. Park, D.-J. Kim, and Y. Lim, "Use of smart card data to define public transit use in Seoul, South Korea," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2063, no. 1, pp. 3–9, Jan. 2008.
- [203] T. Mohammed and T. Fujiyama, "Investigating paper ticket usage on London underground's network," in *Proc. 14th Conf. Adv. Syst. Public Transp. (CASPT)*, 2018, pp. 23–25.
- [204] P. Chowdhury, P. Bala, D. Addy, S. Giri, and A. R. Chaudhuri, "RFID and Android based smart ticketing and destination announcement system," in *Proc. Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, Sep. 2016, pp. 2587–2591, doi: [10.1109/ICACCI.2016.7732447](https://doi.org/10.1109/ICACCI.2016.7732447).
- [205] A. Fidanoglu and I. Gökaşar, "A method for public transit OD estimation using transit smart card data," *J. Transp. Logistics*, vol. 7, no. 2, pp. 479–488, Dec. 2022, doi: [10.26650/jtl.2022.1192128](https://doi.org/10.26650/jtl.2022.1192128).
- [206] M. Kassim, A. S. Salleh, S. Shahbudin, M. Yusoff, and N. A. Kamaluddin, "IoT bus tracking system localization via GPS-RFID," in *Proc. IEEE Int. Conf. Power Eng. Appl. (ICPEA)*, Mar. 2022, pp. 1–6, doi: [10.1109/ICPEA53519.2022.9744710](https://doi.org/10.1109/ICPEA53519.2022.9744710).
- [207] A. Sawant, P. Narayanan, O. Mourya, S. Adhangale, and G. Kedare, "GPS/GSM based school bus boarding-deboarding and location notifying system," in *Proc. 4th Int. Conf. Electron. Sustain. Commun. Syst. (ICESC)*, Jul. 2023, pp. 534–539, doi: [10.1109/icesc57686.2023.10193226](https://doi.org/10.1109/icesc57686.2023.10193226).
- [208] S. Zhang, Y. Wu, C. Men, and X. Li, "Tiny YOLO optimization oriented bus passenger object detection," *Chin. J. Electron.*, vol. 29, no. 1, pp. 132–138, Jan. 2020, doi: [10.1049/cje.2019.11.002](https://doi.org/10.1049/cje.2019.11.002).
- [209] M. W. Horner, I. Casas, and J. A. Downs, "Exploration of a polygon decomposition technique based on the ordinary Voronoi diagram," *Ann. GIS*, vol. 16, no. 4, pp. 223–236, Dec. 2010, doi: [10.1080/19475683.2010.525796](https://doi.org/10.1080/19475683.2010.525796).
- [210] S. J. Berrebi, S. Joshi, and K. E. Watkins, "On bus ridership and frequency," *Transp. Res. A, Policy Pract.*, vol. 148, pp. 140–154, Jun. 2021, doi: [10.1016/j.tra.2021.03.005](https://doi.org/10.1016/j.tra.2021.03.005).
- [211] C. Brunsdon, A. S. Fotheringham, and M. E. Charlton, "Geographically weighted regression: A method for exploring spatial nonstationarity," *Geographical Anal.*, vol. 28, no. 4, pp. 281–298, Oct. 1996, doi: [10.1111/j.1538-4632.1996.tb00936.x](https://doi.org/10.1111/j.1538-4632.1996.tb00936.x).
- [212] G. Al-Hasani, M. Asaduzzaman, and A.-H. Soliman, "Geographically weighted Poisson regression models with different kernels: Application to road traffic accident data," *Commun. Statist., Case Stud., Data Anal. Appl.*, vol. 7, no. 2, pp. 166–181, Apr. 2021, doi: [10.1080/23737484.2020.1869628](https://doi.org/10.1080/23737484.2020.1869628).
- [213] N. Cressie, *Statistics for Spatial Data*. Hoboken, NJ, USA: Wiley, 1993.
- [214] G. Matheron, "The theory of regionalised variables and its applications," *Les Cahiers du Centre de Morphologie Mathématique*, vol. 5, p. 212, Jan. 1971.
- [215] N. Banerjee, A. Morton, and K. Akartunali, "Passenger demand forecasting in scheduled transportation," *Eur. J. Oper. Res.*, vol. 286, no. 3, pp. 797–810, Nov. 2020, doi: [10.1016/j.ejor.2019.10.032](https://doi.org/10.1016/j.ejor.2019.10.032).
- [216] M. Nar and S. Arslankaya, "Passenger demand forecasting for railway systems," *Open Chem.*, vol. 20, no. 1, pp. 105–119, Mar. 2022, doi: [10.1515/chem-2022-0124](https://doi.org/10.1515/chem-2022-0124).
- [217] J. Hoppe, F. Schwinger, H. Haeger, J. Wernz, and M. Jarke, "Improving the prediction of passenger numbers in public transit networks by combining short-term forecasts with real-time occupancy data," *IEEE Open J. Intell. Transp. Syst.*, vol. 4, pp. 153–174, 2023, doi: [10.1109/OJITS.2023.3251564](https://doi.org/10.1109/OJITS.2023.3251564).
- [218] P. Wang, X. Chen, J. Chen, M. Hua, and Z. Pu, "A two-stage method for bus passenger load prediction using automatic passenger counting data," *IET Intell. Transp. Syst.*, vol. 15, no. 2, pp. 248–260, Feb. 2021, doi: [10.1049/itr2.12018](https://doi.org/10.1049/itr2.12018).
- [219] A. Altıntaş, L. Davidson, G. Kostaras, and M. Isaac, "The day-ahead forecasting of the passenger occupancy in public transportation by using machine learning," in *Proc. Int. Conf. Intell. Transp. Syst.*, 2022, pp. 3–12.
- [220] J. Wood, Z. Yu, and V. V. Gayah, "Development and evaluation of frameworks for real-time bus passenger occupancy prediction," *Int. J. Transp. Sci. Technol.*, vol. 12, no. 2, pp. 399–413, Jun. 2023, doi: [10.1016/j.ijst.2022.03.005](https://doi.org/10.1016/j.ijst.2022.03.005).
- [221] G. S. Vidya, V. S. Hari, and S. Shivasagaran, "Estimation of passenger flow in a bus route using Kalman filter," in *Proc. 6th Int. Conf. Adv. Comput. Commun. Syst. (ICACCS)*, Mar. 2020, pp. 1248–1251, doi: [10.1109/ICACCS48705.2020.9074363](https://doi.org/10.1109/ICACCS48705.2020.9074363).
- [222] L. Eboli and G. Mazzulla, "A new customer satisfaction index for evaluating transit service quality," *J. Public Transp.*, vol. 12, no. 3, pp. 21–37, Sep. 2009, doi: [10.5038/2375-0901.12.3.2](https://doi.org/10.5038/2375-0901.12.3.2).
- [223] R. F. Abenoza, O. Cats, and Y. O. Susilo, "Travel satisfaction with public transport: Determinants, user classes, regional disparities and their evolution," *Transp. Res. A, Policy Pract.*, vol. 95, pp. 64–84, Jan. 2017, doi: [10.1016/j.tra.2016.11.011](https://doi.org/10.1016/j.tra.2016.11.011).
- [224] M. L. Outwater, G. Spitz, J. Lobb, M. Campbell, R. M. Pendyala, B. Sana, and W. Woodford, "TCRP H-37 characteristics of premium transit services that affect mode choice: Summary of phase 1," Transit Cooperat. Res. Program, USA, Tech. Rep. 37, 2010. Accessed: Jun. 25, 2025. [Online]. Available: <https://rosap.nhtl.bts.gov/view/dot/34320>
- [225] A. Carrel, A. Halvorsen, and J. L. Walker, "Passengers' perception of and behavioral adaptation to unreliability in public transportation," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2351, no. 1, pp. 153–162, Jan. 2013, doi: [10.3141/2351-17](https://doi.org/10.3141/2351-17).
- [226] D. van Lierop, M. G. Badami, and A. M. El-Geneidy, "What influences satisfaction and loyalty in public transport? A review of the literature," *Transp. Rev.*, vol. 38, no. 1, pp. 52–72, Jan. 2018, doi: [10.1080/01441647.2017.1298683](https://doi.org/10.1080/01441647.2017.1298683).
- [227] M. M. Harsha, R. H. Mulangi, and H. D. D. Kumar, "Analysis of bus travel time variability using automatic vehicle location data," *Transp. Res. Proc.*, vol. 48, pp. 3283–3298, Jan. 2020, doi: [10.1016/j.trpro.2020.08.123](https://doi.org/10.1016/j.trpro.2020.08.123).
- [228] R. Chawuthai, A. Sumalee, and T. Threepak, "GPS data analytics for the assessment of public city bus transportation service quality in Bangkok," *Sustainability*, vol. 15, no. 7, p. 5618, Mar. 2023, doi: [10.3390/su15075618](https://doi.org/10.3390/su15075618).
- [229] R. Fedujwar and A. Agarwal, "Travel time variability of bus routes in Delhi using real-time GTFS data," in *Proc. 16th Int. Conf. Commun. Syst. Netw. (COMSNETS)*, Jan. 2024, pp. 210–215, doi: [10.1109/comsnets59351.2024.10427234](https://doi.org/10.1109/comsnets59351.2024.10427234).

- [230] E. Mazloumi, G. Currie, and G. Rose, "Using GPS data to gain insight into public transport travel time variability," *J. Transp. Eng.*, vol. 136, no. 7, pp. 623–631, Jul. 2010.
- [231] R. Chawuthai, K. Pruekwangkha, and T. Threepak, "Spatial-temporal traffic speed prediction on Thailand roads," in *Proc. 7th Int. Conf. Eng., Appl. Sci. Technol. (ICEAST)*, Apr. 2021, pp. 58–62, doi: [10.1109/ICEAST52143.2021.9426257](https://doi.org/10.1109/ICEAST52143.2021.9426257).
- [232] N. Schuessler and K. W. Axhausen, "Processing raw data from global positioning systems without additional information," *Transp. Res. Rec.*, *J. Transp. Res. Board*, vol. 2105, no. 1, pp. 28–36, Jan. 2009, doi: [10.3141/2105-04](https://doi.org/10.3141/2105-04).
- [233] A. Danaher, J. Wensley, A. Dunham, T. Orosz, R. Avery, K. Cobb, K. Watkins, C. Queen, S. Berrebi, and M. Connor, *Minutes Matter: A Bus Transit Service Reliability Guidebook*. Washington, DC, USA: National Academy Press, 2020, doi: [10.17226/25727](https://doi.org/10.17226/25727).
- [234] T. Elliott and T. Lumley, "Modelling the travel time of transit vehicles in real-time through a GTFS-based road network using GPS vehicle locations," *Austral. New Zealand J. Statist.*, vol. 62, no. 2, pp. 153–167, Jun. 2020, doi: [10.1111/anzs.12294](https://doi.org/10.1111/anzs.12294).
- [235] H. Liu, H. Xu, Y. Yan, Z. Cai, T. Sun, and W. Li, "Bus arrival time prediction based on LSTM and spatial-temporal feature vector," *IEEE Access*, vol. 8, pp. 11917–11929, 2020, doi: [10.1109/ACCESS.2020.2965094](https://doi.org/10.1109/ACCESS.2020.2965094).
- [236] J. Munasinghe, D. Dhananjaya, W. Fernando, A. Kumarage, and A. Perera, "Bus route travel time estimation using GPS data: A case-based study for the Kandy city," in *Proc. Nat. Eng. Res. Symp. (NERS)*, vol. 1, 2020, pp. 95–102.
- [237] Z. Czapla and S. Krawiec, "Travel time description for urban routes operated with electric buses," in *Proc. Smart City Symp. Prague (SCSP)*, Jun. 2020, pp. 1–5, doi: [10.1109/SCSP49987.2020.9133867](https://doi.org/10.1109/SCSP49987.2020.9133867).
- [238] K. Panyo, J. Bootkrajang, P. Inkeaw, and J. Chaijaruwanich, "Bus arrival time estimation for public transportation system using LSTM," in *Proc. 5th Int. Conf. Inf. Technol. (InCIT)*, 2020, pp. 134–138, doi: [10.1109/incit50588.2020.9310940](https://doi.org/10.1109/incit50588.2020.9310940).
- [239] X. Wu, S. Huang, J. Zou, J. S. Shen, W. Cai, and J. Zhao, "Bus arrival time estimation based on GPS data by the artificial bee colony optimization BP neural network," in *Proc. 5th Int. Conf. Smart Grid Electr. Autom. (ICS-GEA)*, vol. 6, 2020, pp. 264–267, doi: [10.1109/icsgea51094.2020.00063](https://doi.org/10.1109/icsgea51094.2020.00063).
- [240] Q. Han, K. Liu, L. Zeng, G. He, L. Ye, and F. Li, "A bus arrival time prediction method based on position calibration and LSTM," *IEEE Access*, vol. 8, pp. 42372–42383, 2020, doi: [10.1109/ACCESS.2020.2976574](https://doi.org/10.1109/ACCESS.2020.2976574).
- [241] C. Chen, H. Wang, F. Yuan, H. Jia, and B. Yao, "Bus travel time prediction based on deep belief network with back-propagation," *Neural Comput. Appl.*, vol. 32, no. 14, pp. 10435–10449, Jul. 2020, doi: [10.1007/s00521-019-04579-x](https://doi.org/10.1007/s00521-019-04579-x).
- [242] Y. Seitbekova, B. Assilbekov, I. Beisembetov, and A. Kuljabekov, "The bus arrival time prediction using LSTM neural network and location analysis," *J. Adv. Technol. Eng. Res.*, vol. 6, no. 2, pp. 46–57, 2020.
- [243] A. Baimbetova, K. Konyrova, A. Zhumabayeva, and Y. Seitbekova, "Bus arrival time prediction: A case study for almaty," in *Proc. IEEE Int. Conf. Smart Inf. Syst. Technol. (SIST)*, Apr. 2021, pp. 1–6, doi: [10.1109/SIST50301.2021.9465963](https://doi.org/10.1109/SIST50301.2021.9465963).
- [244] A. Tapparita and M. Brady, "Bus journey and arrival time prediction based on archived AVL/GPS data using machine learning," in *Proc. 7th Int. Conf. Models Technol. Intell. Transp. Syst. (MT-ITS)*, Jun. 2021, pp. 1–6, doi: [10.1109/MT-ITS49943.2021.9529328](https://doi.org/10.1109/MT-ITS49943.2021.9529328).
- [245] Z. Zhuang, Z. Cheng, J. Yao, J. Wang, and S. An, "Bus travel time reliability incorporating stop waiting time and in-vehicle travel time with AVL data," *Int. J. Coal Sci. Technol.*, vol. 9, no. 1, p. 71, Dec. 2022, doi: [10.1007/s40789-022-00544-7](https://doi.org/10.1007/s40789-022-00544-7).
- [246] Z. Yang, C.-T. Lam, and B. K. Ng, "Multi-model bus arrival time prediction using real-time online information," in *Proc. IEEE 22nd Int. Conf. Commun. Technol. (ICCT)*, Nov. 2022, pp. 1918–1922, doi: [10.1109/ICCT56141.2022.10072901](https://doi.org/10.1109/ICCT56141.2022.10072901).
- [247] R. Jiang, D. Hu, S. I.-J. Chien, Q. Sun, and X. Wu, "Predicting bus travel time with hybrid incomplete data—A deep learning approach," *Promet-Traffic Transp.*, vol. 34, no. 5, pp. 673–685, Sep. 2022, doi: [10.7307/ptt.v34i5.4052](https://doi.org/10.7307/ptt.v34i5.4052).
- [248] X. Zhang, L. Lauber, H. Liu, J. Shi, M. Xie, and Y. Pan, "Travel time prediction of urban public transportation based on detection of single routes," *PLoS ONE*, vol. 17, no. 1, Jan. 2022, Art. no. e0262535, doi: [10.1371/journal.pone.0262535](https://doi.org/10.1371/journal.pone.0262535).
- [249] T. Nimpanomprasert, L. Xie, and N. Klierer, "Comparing two hybrid neural network models to predict real-world bus travel time," *Transp. Res. Proc.*, vol. 62, pp. 393–400, Jan. 2022, doi: [10.1016/j.trpro.2022.02.049](https://doi.org/10.1016/j.trpro.2022.02.049).
- [250] B. P. Ashwini, R. Sumathi, and H. S. Sudhira, "Bus travel time prediction: A comparative study of linear and non-linear machine learning models," *J. Phys., Conf. Ser.*, vol. 2161, no. 1, Jan. 2022, Art. no. 012053, doi: [10.1088/1742-6596/2161/1/012053](https://doi.org/10.1088/1742-6596/2161/1/012053).
- [251] M. Suresh, S. Saranya, A. Punitha, and R. Kowsalya, "Estimation of bus arrival by tracking and analyzing the users real time location," in *Proc. Int. Conf. Syst., Comput., Autom. Netw. (ICSCAN)*, Nov. 2023, pp. 1–6, doi: [10.1109/icscan58655.2023.10395687](https://doi.org/10.1109/icscan58655.2023.10395687).
- [252] S. Ratneswaran and U. Thayasivam, "Extracting potential travel time information from raw GPS data and evaluating the performance of public transit—A case study in Kandy, Sri Lanka," in *Proc. 3rd Int. Conf. Intell. Commun. Comput. Techn. (ICCT)*, Jan. 2023, pp. 1–7, doi: [10.1109/ICCT56969.2023.10075789](https://doi.org/10.1109/ICCT56969.2023.10075789).
- [253] C. Li, S. Lin, H. Zhang, H. Zhao, L. Liu, and N. Jia, "A sequence and network embedding method for bus arrival time prediction using GPS trajectory data only," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 5024–5038, May 2023, doi: [10.1109/TITS.2023.3237320](https://doi.org/10.1109/TITS.2023.3237320).
- [254] N. Shanthi, R. C. Suganth, V. Manimaran, P. M. Kumar, A. K. Bharathi, and E. Mythili, "GPS-based arrival time prediction for bus schedules: Enhancing commuter experience," in *Proc. 12th Int. Conf. Adv. Comput. (ICoAC)*, Aug. 2023, pp. 1–6, doi: [10.1109/icoac59537.2023.10249289](https://doi.org/10.1109/icoac59537.2023.10249289).
- [255] Y. T. Lim, S. L. Keoh, Y.-W. Chong, N. F. Ibrahim, and S. K. A. Rahim, "An ANN-based prediction model for public bus journey time," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIC)*, Feb. 2024, pp. 578–583, doi: [10.1109/icaic60209.2024.10463440](https://doi.org/10.1109/icaic60209.2024.10463440).
- [256] A. Mansurova, A. Mansurova, and A. Sakhipov, "Bus arrival time prediction via hybrid LSTM using GPS-derived run and dwell times," in *Proc. 11th Int. Conf. Vehicle Technol. Intell. Transp. Syst.*, 2025, pp. 514–519, doi: [10.5220/0013349200003941](https://doi.org/10.5220/0013349200003941).
- [257] W. C. Chan, W. H. W. Ibrahim, M. C. Lo, M. K. Suaidi, and S. T. Ha, "Sustainability of public transportation: An examination of user behavior to real-time GPS tracking application," *Sustainability*, vol. 12, no. 22, p. 9541, Nov. 2020, doi: [10.3390/su12229541](https://doi.org/10.3390/su12229541).
- [258] L. dell'Olio, A. Ibeas, and P. Cecin, "The quality of service desired by public transport users," *Transp. Policy*, vol. 18, no. 1, pp. 217–227, Jan. 2011, doi: [10.1016/j.tranpol.2010.08.005](https://doi.org/10.1016/j.tranpol.2010.08.005).
- [259] B. Hussain, M. M. Zefreh, and A. Torok, "Designing the appropriate data collection method for public transport passenger satisfaction analysis," *Int. J. Traffic Transp. Eng.*, vol. 8, no. 2, pp. 177–183, 2018, doi: [10.7708/ijtte.2018.8\(2\).03](https://doi.org/10.7708/ijtte.2018.8(2).03).
- [260] A. Christopher, "A comfort index for public transportation riders," Ph.D. dissertation, Building, Civil Environ. Eng., Concordia Univ., Montreal, QC, Canada, 2016.
- [261] R. G. Mugion, M. Toni, H. Raharjo, L. Di Pietro, and S. P. Sebathu, "Does the service quality of urban public transport enhance sustainable mobility?" *J. Cleaner Prod.*, vol. 174, pp. 1566–1587, Feb. 2018.
- [262] T. Litman, "A new transit safety narrative," *J. Public Transp.*, vol. 17, no. 4, pp. 114–135, Dec. 2014, doi: [10.5038/2375-0901.17.4.7](https://doi.org/10.5038/2375-0901.17.4.7).
- [263] R. Ishaq and O. Cats, "Designing bus rapid transit systems: Lessons on service reliability and operations," *Case Stud. Transp. Policy*, vol. 8, no. 3, pp. 946–953, Sep. 2020, doi: [10.1016/j.cstp.2020.05.001](https://doi.org/10.1016/j.cstp.2020.05.001).
- [264] J. Weng, C. Wang, H. Huang, Y. Wang, and L. Zhang, "Real-time bus travel speed estimation model based on bus GPS data," *Adv. Mech. Eng.*, vol. 8, no. 11, Nov. 2016, Art. no. 1687814016678162, doi: [10.1177/1687814016678162](https://doi.org/10.1177/1687814016678162).
- [265] S. Nadeeshan and A. S. Perera, "A map matching algorithm for noisy, low frequent public transportation GPS data," in *Proc. 7th Int. Conf. Control, Decis. Inf. Technol. (CoDIT)*, vol. 1, Jun. 2020, pp. 1081–1086, doi: [10.1109/CoDIT49905.2020.9263797](https://doi.org/10.1109/CoDIT49905.2020.9263797).
- [266] E. Mazloumi, G. Rose, G. Currie, and S. Moridpour, "Prediction intervals to account for uncertainties in neural network predictions: Methodology and application in bus travel time prediction," *Eng. Appl. Artif. Intell.*, vol. 24, no. 3, pp. 534–542, Apr. 2011, doi: [10.1016/j.engappai.2010.11.004](https://doi.org/10.1016/j.engappai.2010.11.004).
- [267] S. I.-J. Chien, Y. Ding, and C. Wei, "Dynamic bus arrival time prediction with artificial neural networks," *J. Transp. Eng.*, vol. 128, no. 5, pp. 429–438, Sep. 2002, doi: [10.1061/\(asce\)0733-947x\(2002\)128:5\(429\)](https://doi.org/10.1061/(asce)0733-947x(2002)128:5(429)).
- [268] P. Chinmayee, A. H. Patil, V. Bhavika, B. Ashwini, and R. Sumathi, "A comprehensive survey on parametric and non-parametric machine learning approaches for bus arrival time prediction," in *Proc. 4th Int. Conf. Adv. Comput., Commun. Control Netw. (ICAC3N)*, Dec. 2022, pp. 816–822, doi: [10.1109/ICAC3N56670.2022.10074390](https://doi.org/10.1109/ICAC3N56670.2022.10074390).

- [269] O. Alam, A. Kush, A. Emami, and P. Pouladzadeh, "Predicting irregularities in arrival times for transit buses with recurrent neural networks using GPS coordinates and weather data," *J. Ambient Intell. Humanized Comput.*, vol. 12, no. 7, pp. 7813–7826, Jul. 2021, doi: [10.1007/s12652-020-02507-9](#).
- [270] J.-Q. Li, "Battery-electric transit bus developments and operations: A review," *Int. J. Sustain. Transp.*, vol. 10, no. 3, pp. 157–169, Mar. 2016, doi: [10.1080/15568318.2013.872737](#).
- [271] A. Ajanovic, R. Haas, and M. Schrödl, "On the historical development and future prospects of various types of electric mobility," *Energies*, vol. 14, no. 4, p. 1070, Feb. 2021, doi: [10.3390/en14041070](#).
- [272] Y. Cao, J. Cui, S. Liu, X. Li, Q. Zhou, C. Hu, Y. Zhuang, and Z. Liu, "A holistic review on E-mobility service optimization: Challenges, recent progress, and future directions," *IEEE Trans. Transp. Electrification*, vol. 10, no. 2, pp. 3712–3741, Jun. 2024, doi: [10.1109/TTE.2023.3311410](#).
- [273] J. Husár and P. Trojanowski, "Integration of electromobility into public transport systems: A case study," in *Proc. Int. Conf. Future Access Enablers Ubiquitous Intell. Infrastructures*, 2024, pp. 104–118, doi: [10.1007/978-3-031-72393-3_9](#).
- [274] X. Liu, X. Qu, and X. Ma, "Optimizing electric bus charging infrastructure considering power matching and seasonality," *Transp. Res. D, Transp. Environ.*, vol. 100, Nov. 2021, Art. no. 103057, doi: [10.1016/j.trd.2021.103057](#).
- [275] L. C. da Silva, M. G. Villalva, M. C. de Almeida, J. L. Brittes, J. Yasuoka, J. G. Cyprino, D. Dotta, J. T. V. Pereira, M. B. Salles, G. B. Archilli, and J. G. Campos, "Sustainable campus model at the University of Campinas—Brazil: An integrated living lab for renewable generation, electric mobility, energy efficiency, monitoring and energy demand management," in *Towards Green Campus Operations*. Cham, Switzerland: Springer, 2018, pp. 457–472, doi: [10.1007/978-3-319-76885-4_30](#).
- [276] J. J. Astrain, F. Falcone, A. J. Lopez-Martin, P. Sanchis, J. Villadangos, and I. R. Matias, "Monitoring of electric buses within an urban smart city environment," *IEEE Sensors J.*, vol. 22, no. 12, pp. 11364–11372, Jun. 2022, doi: [10.1109/JSEN.2021.3077846](#).
- [277] Y. Yi, J. A. Jang, and I. Lee, "Data design for driving monitoring of hydrogen electric bus," in *Proc. Int. Conf. Inf. Commun. Technol. Converg. (ICTC)*, 2021, pp. 697–700, doi: [10.1109/ictc52510.2021.9621036](#).
- [278] L. A. Barbosa, H. C. D. Oliveira, W. C. Machado, and D. C. Costa, "A new method for evaluation of the positional error of low-cost devices based on GNSS integrity for transportation applications," *Boletim de Ciências Geodésicas*, vol. 28, no. 1, 2022, Art. no. 2022001, doi: [10.1590/s1982-21702022000100001](#).
- [279] F. T. Mariotto, N. B. Yoma, and M. C. D. Almeida, "Unsupervised competitive learning clustering and visual method to obtain accurate trajectories from noisy repetitive GPS data," *IEEE Trans. Intell. Transp. Syst.*, vol. 26, no. 2, pp. 1562–1572, Feb. 2025, doi: [10.1109/TITS.2024.3520393](#).
- [280] A. Hitz, A. Konzept, B. Reick, and K. Rheinberger, "Efficient GPS route matching method for battery electric bus fleets," SAE Tech. Paper 2024-24-0026, 2024, doi: [10.4271/2024-24-0026](#).
- [281] O. Pink and B. Hummel, "A statistical approach to map matching using road network geometry, topology and vehicular motion constraints," in *Proc. 11th Int. IEEE Conf. Intell. Transp. Syst.*, Oct. 2008, pp. 862–867, doi: [10.1109/ITSC.2008.4732697](#).
- [282] I. Sagaama, A. Kchiche, W. Trojet, and F. Kamoun, "Impact of road gradient on electric vehicle energy consumption in real-world driving," in *Proc. Int. Conf. Adv. Inf. Netw. Appl.* Cham, Switzerland: Springer, 2020, pp. 393–404, doi: [10.1007/978-3-030-44041-1_36](#).
- [283] K. An, "Battery electric bus infrastructure planning under demand uncertainty," *Transp. Res. C, Emerg. Technol.*, vol. 111, pp. 572–587, Feb. 2020, doi: [10.1016/j.trc.2020.01.009](#).
- [284] X. Ma, R. Miao, X. Wu, and X. Liu, "Examining influential factors on the energy consumption of electric and diesel buses: A data-driven analysis of large-scale public transit network in Beijing," *Energy*, vol. 216, Feb. 2021, Art. no. 119196, doi: [10.1016/j.energy.2020.119196](#).
- [285] O. A. Hjelkrem, K. Y. Lervåg, S. Babri, C. Lu, and C.-J. Södersten, "A battery electric bus energy consumption model for strategic purposes: Validation of a proposed model structure with data from bus fleets in China and Norway," *Transp. Res. D, Transp. Environ.*, vol. 94, May 2021, Art. no. 102804, doi: [10.1016/j.trd.2021.102804](#).
- [286] K.-P. Wenz, X. Serrano-Guerrero, A. Barragán-Escandón, L. G. González, and J.-M. Clairand, "Route prioritization of urban public transportation from conventional to electric buses: A new methodology and a study of case in an intermediate city of Ecuador," *Renew. Sustain. Energy Rev.*, vol. 148, Sep. 2021, Art. no. 111215, doi: [10.1016/j.rser.2021.111215](#).
- [287] X. Lu, K. Ota, M. Dong, C. Yu, and H. Jin, "Clairvoyance: Intelligent route planning for electric buses based on urban big data," 2021, *arXiv:2112.04682*.
- [288] C. Iliopoulou and K. Kepaptsoglou, "Robust electric transit route network design problem (RE-TRNDP) with delay considerations: Model and application," *Transp. Res. C, Emerg. Technol.*, vol. 129, Aug. 2021, Art. no. 103255, doi: [10.1016/j.trc.2021.103255](#).
- [289] J. Ji, Y. Bie, Z. Zeng, and L. Wang, "Trip energy consumption estimation for electric buses," *Commun. Transp. Res.*, vol. 2, Dec. 2022, Art. no. 100069, doi: [10.1016/j.commtr.2022.100069](#).
- [290] Y. Liu and H. Liang, "A data-driven approach for electric bus energy consumption estimation," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 10, pp. 17027–17038, Oct. 2022, doi: [10.1109/TITS.2022.3152679](#).
- [291] J. Zhang and X. Zhang, "A multi-trip electric bus routing model considering equity during short-notice evacuations," *Transp. Res. D, Transp. Environ.*, vol. 110, Sep. 2022, Art. no. 103397, doi: [10.1016/j.trd.2022.103397](#).
- [292] L. A. L. Zaneti, N. B. Arias, M. C. de Almeida, and M. J. Rider, "Sustainable charging schedule of electric buses in a university campus: A rolling horizon approach," *Renew. Sustain. Energy Rev.*, vol. 161, Jun. 2022, Art. no. 112276, doi: [10.1016/j.rser.2022.112276](#).
- [293] Y. Tomizawa, Y. Ihara, Y. Kodama, Y. Iino, Y. Hayashi, O. Ikeda, and J. Yoshinaga, "Multipurpose charging schedule optimization method for electric buses: Evaluation using real city data," *IEEE Access*, vol. 10, pp. 56067–56080, 2022, doi: [10.1109/ACCESS.2022.3177618](#).
- [294] J. Jiang, Y. Yu, H. Min, Q. Cao, W. Sun, Z. Zhang, and C. Luo, "Trip-level energy consumption prediction model for electric bus combining Markov-based speed profile generation and Gaussian processing regression," *Energy*, vol. 263, Jan. 2023, Art. no. 125866, doi: [10.1016/j.energy.2022.125866](#).
- [295] A. Sharma, S. Das, A. Nasar, P. Gairola, S. A. Gandhi, and N. Nezamuddin, "Evaluation of electric bus operations in hilly Terrain—A case study of leh," in *Proc. Conf. Transp. Res. Group India*. Cham, Switzerland: Springer, 2025, pp. 17–34, doi: [10.1007/978-981-97-9943-5_2](#).
- [296] X. Tang, J. Yang, X. Lin, F. He, and J. Si, "Dynamic operations of an integrated mobility service system of fixed-route transits and flexible electric buses," *Transp. Res. E, Logistics Transp. Rev.*, vol. 173, May 2023, Art. no. 103081, doi: [10.1016/j.trce.2023.103081](#).
- [297] D. McCabe and X. Ban, "Optimal locations and sizes of layover charging stations for electric buses," *Transp. Res. C, Emerg. Technol.*, vol. 152, Jul. 2023, Art. no. 104157, doi: [10.1016/j.trc.2023.104157](#).
- [298] X. Liu, X. Liu, X. Zhang, Y. Zhou, J. Chen, and X. Ma, "Optimal location planning of electric bus charging stations with integrated photovoltaic and energy storage system," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 38, no. 11, pp. 1424–1446, Jul. 2023, doi: [10.1111/mice.12935](#).
- [299] P. Jarvis, L. Climent, and A. Arbelaez, "Smart and sustainable scheduling of charging events for electric buses," in *EU Cohesion Policy Implementation—Evaluation Challenges Opportunities*. Cham, Switzerland: Springer, 2023, p. 121, doi: [10.1007/978-3-031-18161-0_8](#).
- [300] F. Basso, F. Feijoo, R. Pezosa, M. Varas, and B. Vidal, "The impact of electromobility in public transport: An estimation of energy consumption using disaggregated data in Santiago, Chile," *Energy*, vol. 286, Jan. 2024, Art. no. 129550, doi: [10.1016/j.energy.2023.129550](#).
- [301] S. Sun, S. F. A. Batista, M. Menéndez, Y. Wang, and S. Zhang, "Powering up urban mobility: A comparative study of energy efficiency in electric and diesel buses across various lane configurations," *Sustain. Cities Soc.*, vol. 101, Feb. 2024, Art. no. 105086, doi: [10.1016/j.scs.2023.105086](#).
- [302] W. Wang, X. Chen, and X. Liu, "Comparative analysis of energy consumption prediction of pure electric bus based on multiple influencing factors," in *Proc. CICTP*, Dec. 2024, pp. 2976–2985, doi: [10.1061/9780784485484.281](#).
- [303] C. Zhou, P. Dai, X. Huang, and F. Wang, "Intelligent route planning recommendation for electric bus transport," *Int. J. Intell. Syst.*, vol. 2024, no. 1, Jan. 2024, Art. no. 5947433, doi: [10.1155/2024/5947433](#).
- [304] X. Liu, S. Yeh, P. Plötz, W. Ma, F. Li, and X. Ma, "Electric bus charging scheduling problem considering charging infrastructure integrated with solar photovoltaic and energy storage systems," *Transp. Res. E, Logistics Transp. Rev.*, vol. 187, Jul. 2024, Art. no. 103572, doi: [10.1016/j.trce.2024.103572](#).
- [305] P. Nisha, "Fleet management system for electric buses using ESP32 microcontroller," in *Proc. 9th Int. Conf. Commun. Electron. Syst. (ICCES)*, Dec. 2024, pp. 100–105, doi: [10.1109/icc63552.2024.10859383](#).

- [306] P. Á. Szilassy and D. Földes, "Consumption estimation method for battery-electric buses using general line characteristics and temperature," *Energy*, vol. 261, Dec. 2022, Art. no. 125080, doi: [10.1016/j.energy.2022.125080](https://doi.org/10.1016/j.energy.2022.125080).
- [307] Y. Al-Wreikat, C. Serrano, and J. R. Sodr , "Driving behaviour and trip condition effects on the energy consumption of an electric vehicle under real-world driving," *Appl. Energy*, vol. 297, Sep. 2021, Art. no. 117096, doi: [10.1016/j.apenergy.2021.117096](https://doi.org/10.1016/j.apenergy.2021.117096).
- [308] P. Li, Y. Zhang, Y. Zhang, K. Zhang, and M. Jiang, "The effects of dynamic traffic conditions, route characteristics and environmental conditions on trip-based electricity consumption prediction of electric bus," *Energy*, vol. 218, Mar. 2021, Art. no. 119437, doi: [10.1016/j.energy.2020.119437](https://doi.org/10.1016/j.energy.2020.119437).
- [309] X. Yang and L. Liu, "Analysis of the influence of passenger load on bus energy consumption a vehicle-engine combined model-based simulation framework," *Sci. Rep.*, vol. 12, no. 1, p. 14535, Aug. 2022, doi: [10.1038/s41598-022-18866-6](https://doi.org/10.1038/s41598-022-18866-6).
- [310] D. Fan, Q. Feng, A. Zhang, M. Liu, Y. Ren, and Y. Wang, "Optimization of scheduling and timetabling for multiple electric bus lines considering nonlinear energy consumption model," *IEEE Trans. Intell. Transp. Syst.*, vol. 25, no. 6, pp. 5342–5355, Jun. 2024, doi: [10.1109/TITS.2023.3336326](https://doi.org/10.1109/TITS.2023.3336326).
- [311] S. C. Onar, B. Oztaysi,  . Otay, and C. Kahraman, "Multi-expert wind energy technology selection using interval-valued intuitionistic fuzzy sets," *Energy*, vol. 90, pp. 274–285, Oct. 2015, doi: [10.1016/j.energy.2015.06.086](https://doi.org/10.1016/j.energy.2015.06.086).
- [312] T. Pamula and W. Pamula, "Estimation of the energy consumption of battery electric buses for public transport networks using real-world data and deep learning," *Energies*, vol. 13, no. 9, p. 2340, May 2020, doi: [10.3390/en13092340](https://doi.org/10.3390/en13092340).
- [313] H. Ibrahim, A. M. Ali, and T. Attia, "Optimizing electric bus performance via predictive maintenance: A combined experimental and modeling approach," *Frontiers Future Transp.*, vol. 5, Jan. 2025, Art. no. 1506866, doi: [10.3389/ffutr.2024.1506866](https://doi.org/10.3389/ffutr.2024.1506866).
- [314] J. Ji, L. Wang, and Y. Bie, "Determination of GPS sampling interval for trip energy consumption estimation of electric buses: Analysis of real-world data," *Proc. Inst. Mech. Eng., D, J. Automobile Eng.*, vol. 239, no. 7, pp. 2753–2765, Jun. 2025, doi: [10.1177/09544070241239384](https://doi.org/10.1177/09544070241239384).
- [315] X. Li, T. Wang, J. Li, Y. Tian, and J. Tian, "Energy consumption estimation for electric buses based on a physical and data-driven fusion model," *Energies*, vol. 15, no. 11, p. 4160, Jun. 2022, doi: [10.3390/en15114160](https://doi.org/10.3390/en15114160).
- [316] W. Zeng, T. Miwa, and T. Morikawa, "Exploring trip fuel consumption by machine learning from gps and can bus data," *J. Eastern Asia Soc. Transp. Stud.*, vol. 11, pp. 906–921, Jan. 2015, doi: [10.11175/easts.11.906](https://doi.org/10.11175/easts.11.906).
- [317] M. Parzinger, L. Hanfstaengl, F. Sigg, U. Spindler, U. Wellisch, and M. Wirnsberger, "Comparison of different training data sets from simulation and experimental measurement with artificial users for occupancy detection—Using machine learning methods random forest and LASSO," *Building Environ.*, vol. 223, Sep. 2022, Art. no. 109313, doi: [10.1016/j.buildenv.2022.109313](https://doi.org/10.1016/j.buildenv.2022.109313).
- [318] M. Momenitabar, Z. D. Ebrahimi, and K. Bengtson, "Optimal placement of battery electric bus charging stations considering energy storage technology: Queuing modeling approach," *Transp. Res. Rec., J. Transp. Res. Board*, vol. 2677, no. 6, pp. 663–672, Jun. 2023, doi: [10.1177/03611981221149729](https://doi.org/10.1177/03611981221149729).
- [319] Y.-T. Hsu, S. Yan, and P. Huang, "The depot and charging facility location problem for electrifying urban bus services," *Transp. Res. D, Transp. Environ.*, vol. 100, Nov. 2021, Art. no. 103053, doi: [10.1016/j.trd.2021.103053](https://doi.org/10.1016/j.trd.2021.103053).
- [320] T. Uslu and O. Kaya, "Location and capacity decisions for electric bus charging stations considering waiting times," *Transp. Res. D, Transp. Environ.*, vol. 90, Jan. 2021, Art. no. 102645, doi: [10.1016/j.trd.2020.102645](https://doi.org/10.1016/j.trd.2020.102645).
- [321] Z. Luo, F. He, X. Lin, J. Wu, and M. Li, "Joint deployment of charging stations and photovoltaic power plants for electric vehicles," *Transp. Res. D, Transp. Environ.*, vol. 79, Feb. 2020, Art. no. 102247, doi: [10.1016/j.trd.2020.102247](https://doi.org/10.1016/j.trd.2020.102247).
- [322] T.-Y. Ma and S. Xie, "Optimal fast charging station locations for electric ridesharing with vehicle-charging station assignment," *Transp. Res. D, Transp. Environ.*, vol. 90, Jan. 2021, Art. no. 102682, doi: [10.1016/j.trd.2020.102682](https://doi.org/10.1016/j.trd.2020.102682).
- [323] Z. Gao, Z. Lin, T. J. LaClair, C. Liu, J.-M. Li, A. K. Birky, and J. Ward, "Battery capacity and recharging needs for electric buses in city transit service," *Energy*, vol. 122, pp. 588–600, Mar. 2017, doi: [10.1016/j.energy.2017.01.101](https://doi.org/10.1016/j.energy.2017.01.101).
- [324] P. Li, Y. Zhang, Y. Zhang, Y. Zhang, and K. Zhang, "Prediction of electric bus energy consumption with stochastic speed profile generation modelling and data driven method based on real-world big data," *Appl. Energy*, vol. 298, Sep. 2021, Art. no. 117204, doi: [10.1016/j.apenergy.2021.117204](https://doi.org/10.1016/j.apenergy.2021.117204).
- [325] T. Paul and H. Yamada, "Operation and charging scheduling of electric buses in a city bus route network," in *Proc. 17th Int. IEEE Conf. Intell. Transp. Syst. (ITSC)*, Oct. 2014, pp. 2780–2786, doi: [10.1109/ITSC.2014.6958135](https://doi.org/10.1109/ITSC.2014.6958135).
- [326] M. Jiang, Y. Zhang, and Y. Zhang, "Electric bus scheduling considering limited charging facility capacity for large-scale operation," *J. Transp. Eng., A, Syst.*, vol. 148, no. 9, Sep. 2022, Art. no. 04022058, doi: [10.1061/jteps.0000706](https://doi.org/10.1061/jteps.0000706).
- [327] J. A. Pinzon, P. P. Vergara, L. C. P. da Silva, and M. J. Rider, "Optimal management of energy consumption and comfort for smart buildings operating in a microgrid," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 3236–3247, May 2019, doi: [10.1109/TSG.2018.2822276](https://doi.org/10.1109/TSG.2018.2822276).
- [328] C. Sabillon, J. F. Franco, M. J. Rider, and R. Romero, "Joint optimal operation of photovoltaic units and electric vehicles in residential networks with storage systems: A dynamic scheduling method," *Int. J. Electr. Power Energy Syst.*, vol. 103, pp. 136–145, Dec. 2018, doi: [10.1016/j.ijepes.2018.05.015](https://doi.org/10.1016/j.ijepes.2018.05.015).
- [329] Q. Yang and C. Ji, "Intraday rolling optimization strategy of PV-energy storage-integrated charging station serving multiple electrical bus lines," in *Proc. 5th Asia Conf. Power Electr. Eng. (ACPEE)*, Jun. 2020, pp. 1143–1149, doi: [10.1109/ACPEE48638.2020.9136480](https://doi.org/10.1109/ACPEE48638.2020.9136480).
- [330] C. Musili and D. Salon, "Do private transport services complement or compete against public transit? Evidence from the commuter vans in eastern Queens, New York," *Urban Sci.*, vol. 3, no. 1, p. 24, Feb. 2019, doi: [10.3390/urbansci3010024](https://doi.org/10.3390/urbansci3010024).
- [331] M. Altieri, C. Silva, and S. Terabe, "Give public transit a chance: A comparative analysis of competitive travel time in public transit modal share," *J. Transp. Geography*, vol. 87, Jul. 2020, Art. no. 102817, doi: [10.1016/j.jtrangeo.2020.102817](https://doi.org/10.1016/j.jtrangeo.2020.102817).
- [332] C. Iliopoulou and K. Kepaptsoglou, "Combining ITS and optimization in public transportation planning: State of the art and future research paths," *Eur. Transp. Res. Rev.*, vol. 11, no. 1, pp. 1–15, Dec. 2019, doi: [10.1186/s12544-019-0365-5](https://doi.org/10.1186/s12544-019-0365-5).
- [333] R. C. G. Lobo, D. Denes-Santos, and F. J. Ricardo, "Electric bus adoption for sustainable urban mobility: Public policies, technological maturity of cities, and small business innovation," *Res. Transp. Bus. Manage.*, vol. 62, Oct. 2025, Art. no. 101463, doi: [10.1016/j.rtbm.2025.101463](https://doi.org/10.1016/j.rtbm.2025.101463).
- [334] L. C. da Silva and A. J. R. Mello, "Desafios e impactos dos  nibus el tricos para o alcance da mobilidade sustent vel: Uma revis o sistem tica da literatura," *Transporte y Territorio*, vol. 10, no. 30, pp. 96–113, 2024, doi: [10.34096/rtt.i30.12929](https://doi.org/10.34096/rtt.i30.12929).
- [335] C. Yu, H. Lin, Y. Chen, C. Yang, A. Yin, and Q. Yuan, "Creating most needed customized bus services: A collaborative analysis of user-route dynamics," *Transp. Res. D, Transp. Environ.*, vol. 133, Aug. 2024, Art. no. 104312, doi: [10.1016/j.trd.2024.104312](https://doi.org/10.1016/j.trd.2024.104312).
- [336] G. Y. Purifoye, "Transporting urban inequality through public transit designs & systems," *City Community*, vol. 16, no. 4, pp. 364–368, Dec. 2017, doi: [10.1111/cico.12266](https://doi.org/10.1111/cico.12266).
- [337] N. S. Iyer, "Measuring spatial inequalities in public transportation systems: Three key perspectives," Ph.D. dissertation, Dept. Comput. Sci., Univ. of Exeter, U.K., 2024.
- [338] M. Kalifa, A.  zdemir, A.  zkan, and M. Banar, "Application of multi-criteria decision analysis including sustainable indicators for prioritization of public transport system," *Integr. Environ. Assessment Manage.*, vol. 18, no. 1, pp. 25–38, Jul. 2021, doi: [10.1002/ieam.4486](https://doi.org/10.1002/ieam.4486).
- [339] G. Marsden and L. Reardon, "Questions of governance: Rethinking the study of transportation policy," *Transp. Res. A, Policy Pract.*, vol. 101, pp. 238–251, Jul. 2017, doi: [10.1016/j.tra.2017.05.008](https://doi.org/10.1016/j.tra.2017.05.008).
- [340] Y. Yan, W. Du, P. Tong, and J. Li, "Equity-considered design method for battery electric bus networks," *Sustainability*, vol. 17, no. 22, p. 10149, Nov. 2025, doi: [10.3390/su172210149](https://doi.org/10.3390/su172210149).
- [341] T. Rye, J. Monios, R. Hrelja, and K. Isaksson, "The relationship between formal and informal institutions for governance of public transport," *J. Transp. Geography*, vol. 69, pp. 196–206, May 2018, doi: [10.1016/j.jtrangeo.2018.04.025](https://doi.org/10.1016/j.jtrangeo.2018.04.025).

- [342] Y. Miao, Y. Yang, X. Li, K.-K.-R. Choo, X. Meng, and R. H. Deng, "Comprehensive survey on privacy-preserving spatial data query in transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 13603–13616, Dec. 2023, doi: [10.1109/ITITS.2023.3295798](https://doi.org/10.1109/ITITS.2023.3295798).
- [343] U. Madububambachu, R. Fatima, A. Sherif, and K. Khalil, "Security and privacy solutions in intelligent transportation systems: A survey," *Internet Things*, vol. 34, Nov. 2025, Art. no. 101812, doi: [10.1016/j.iot.2025.101812](https://doi.org/10.1016/j.iot.2025.101812).
- [344] F. Creutzig, M. Franzen, R. Moeckel, D. Heinrichs, K. Nagel, S. Nieland, and H. Weisz, "Leveraging digitalization for sustainability in urban transport," *Global Sustainability*, vol. 2, p. 14, Jan. 2019, doi: [10.1017/sus.2019.11](https://doi.org/10.1017/sus.2019.11).
- [345] J. M. Klopp, F. Delattre, and A. Chevre, "Open data for inclusive urban public transport globally," French Develop. Agency, France, Tech. Rep. 1, 2019.
- [346] G. Padrón, T. Cristóbal, F. Alayón, A. Quesada-Arencibia, and C. García, "System proposal for mass transit service quality control based on GPS data," *Sensors*, vol. 17, no. 6, p. 1412, Jun. 2017, doi: [10.3390/s17061412](https://doi.org/10.3390/s17061412).
- [347] S. Pensa, E. Masala, M. Arnone, and A. Rosa, "Planning local public transport: A visual support to decision-making," *Proc.-Social Behav. Sci.*, vol. 111, pp. 596–603, Feb. 2014, doi: [10.1016/j.sbspro.2014.01.093](https://doi.org/10.1016/j.sbspro.2014.01.093).
- [348] Y. He, B. Liu, C. Xu, and D. Wu, "Literature review on public transport and land use: Based on CiteSpace statistical analysis," *Land*, vol. 14, no. 5, p. 1096, May 2025, doi: [10.3390/land14051096](https://doi.org/10.3390/land14051096).
- [349] F. Canitez, "Pathways to sustainable urban mobility in developing megacities: A socio-technical transition perspective," *Technological Forecasting Social Change*, vol. 141, pp. 319–329, Apr. 2019, doi: [10.1016/j.techfore.2019.01.008](https://doi.org/10.1016/j.techfore.2019.01.008).
- [350] S.-P. Duarte, J. P. de Sousa, and J. F. de Sousa, "Designing urban mobility policies in a socio-technical transition context," *Transp. Res. Proc.*, vol. 62, pp. 17–24, Jan. 2022, doi: [10.1016/j.trpro.2022.02.003](https://doi.org/10.1016/j.trpro.2022.02.003).
- [351] K. Turon, "Sustainable urban mobility transitions—From policy uncertainty to the CalmMobility paradigm," *Smart Cities*, vol. 8, no. 5, p. 164, Oct. 2025, doi: [10.3390/smartcities8050164](https://doi.org/10.3390/smartcities8050164).
- [352] S. Si, W. Xiong, and X. Che, "Data quality analysis and improvement: A case study of a bus transportation system," *Appl. Sci.*, vol. 13, no. 19, p. 11020, Oct. 2023, doi: [10.3390/app131911020](https://doi.org/10.3390/app131911020).
- [353] A. Z. Rakhman, L. Edi Nugroho, W. Widyawan, and N. K. Utami, "From GPS to maps: Path reconstruction techniques for understanding people mobility with low-sampling-rate data," in *Proc. 3rd Int. Conf. Intell. Cybern. Technol. Appl. (ICICyTA)*, Dec. 2023, pp. 142–147, doi: [10.1109/icicyta60173.2023.10428754](https://doi.org/10.1109/icicyta60173.2023.10428754).
- [354] L. Jiang, C.-X. Chen, and C. Chen, "L2MM: Learning to map matching with deep models for low-quality GPS trajectory data," *ACM Trans. Knowl. Discovery Data*, vol. 17, no. 3, pp. 1–25, Jun. 2023, doi: [10.1145/3550486](https://doi.org/10.1145/3550486).
- [355] A. Botla and C. Paidimarry, "Development of low power GNSS correlator in Zynq SoC for GPS and GLONASS," *J. VLSI circuits Syst.*, vol. 6, no. 2, pp. 14–22, 2024, doi: [10.31838/jvcs/06.02.02](https://doi.org/10.31838/jvcs/06.02.02).
- [356] N. Khamis and K. K. Li, "User experience evaluation for a bus tracking apps in smart campus initiative," *Bull. Electr. Eng. Informat.*, vol. 10, no. 4, pp. 2254–2262, Aug. 2021, doi: [10.11591/eei.v10i4.3112](https://doi.org/10.11591/eei.v10i4.3112).
- [357] K. Huang, L. Xu, Y. Chen, Q. Cheng, and K. An, "Customized bus route optimization with the real-time data," *J. Adv. Transp.*, vol. 2020, pp. 1–9, Aug. 2020, doi: [10.1155/2020/8838994](https://doi.org/10.1155/2020/8838994).
- [358] B. Zhang, Z. Zhong, X. Zhou, Y. Qu, and F. Li, "Optimization model and solution algorithm for rural customized bus route operation under multiple constraints," *Sustainability*, vol. 15, no. 5, p. 3883, Feb. 2023, doi: [10.3390/su15053883](https://doi.org/10.3390/su15053883).
- [359] Y. Guan, Y. Wang, X. Yan, H. Guo, and Y. Zhao, "The one e-ticket customized bus service mode for passengers with multiple trips and the routing problem," *Sustainability*, vol. 14, no. 4, p. 2124, Feb. 2022, doi: [10.3390/su14042124](https://doi.org/10.3390/su14042124).
- [360] L. Zhen, X. He, S. Wang, J. Wu, and K. Liu, "Vehicle routing for customized on-demand bus services," *IJSE Trans.*, vol. 55, no. 12, pp. 1277–1294, Dec. 2023, doi: [10.1080/24725854.2023.2179139](https://doi.org/10.1080/24725854.2023.2179139).
- [361] G. Calabrò, M. Le Pira, N. Giuffrida, G. Inturri, M. Ignaccolo, and G. H. D. A. Correia, "Fixed-route vs. demand-responsive transport feeder services: An exploratory study using an agent-based model," *J. Adv. Transp.*, vol. 2022, pp. 1–20, Jan. 2022, doi: [10.1155/2022/8382754](https://doi.org/10.1155/2022/8382754).
- [362] S. I. Rasca, B. Hu, B. Biesinger, and M. Prandtstetter, "Agent-based decision-support model for bus route redesign in networks of small cities and towns: Case study of Agder, Norway," *Public Transp.*, vol. 16, no. 2, pp. 583–618, Jun. 2024, doi: [10.1007/s12469-024-00358-7](https://doi.org/10.1007/s12469-024-00358-7).
- [363] I. Bargegol, M. Ghorbanzadeh, M. Ghasedi, and M. Rastbod, "Evaluation of effective factors on travel time in optimization of bus stops placement using genetic algorithm," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 245, pp. 1–13, Mar. 2017, Art. no. 042002, doi: [10.1088/1757-899x/245/4/042002](https://doi.org/10.1088/1757-899x/245/4/042002).
- [364] Z. Chen and W. Fan, "Extracting bus transit boarding stop information using smart card transaction data," *J. Mod. Transp.*, vol. 26, no. 3, pp. 209–219, Sep. 2018, doi: [10.1007/s40534-018-0165-y](https://doi.org/10.1007/s40534-018-0165-y).
- [365] P. Ghaziamin, K. Bajaj, N. Bouguila, and Z. Patterson, "A privacy-preserving edge computing solution for real-time passenger counting at bus stops using overhead fisheye camera," in *Proc. IEEE 18th Int. Conf. Semantic Comput. (ICSC)*, Feb. 2024, pp. 25–32, doi: [10.1109/ICSC59802.2024.00011](https://doi.org/10.1109/ICSC59802.2024.00011).
- [366] N. Shafaeipour, V.-D. Stanciu, M. van Steen, and M. Wang, "Understanding the protection of privacy when counting subway travelers through anonymization," *Comput., Environ. Urban Syst.*, vol. 110, Jun. 2024, Art. no. 102091, doi: [10.1016/j.compenvurbsys.2024.102091](https://doi.org/10.1016/j.compenvurbsys.2024.102091).
- [367] W. Liu, X. Du, Q. Geng, J. Li, H. Li, and L. Liu, "Metro passenger flow statistics based on YOLOv3," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 688, no. 4, Nov. 2019, Art. no. 044025, doi: [10.1088/1757-899x/688/4/044025](https://doi.org/10.1088/1757-899x/688/4/044025).



RAFAEL L. SANTOS received the Technical degree in land surveying and the B.S. degree in land surveying and cartographic engineering from IFSULDEMINAS, Inconfidentes Campus, Inconfidentes, Brazil, in 2010 and 2017, respectively, and the M.Sc. degree in civil engineering, with a major in transportation from the State University of Campinas (UNICAMP), Campinas, Brazil, in 2021, where he is currently pursuing the Ph.D. degree in civil engineering with a major in

transportation.

He completed a Complementary Education Program in geographic information systems with Niagara College, Welland, ON, Canada, through the Science Without Borders Program, from 2014 to 2015. He was a Specialist in teaching in professional and technological education with IFSULDEMINAS, in 2020. In 2022, he was a Research Intern with the Department of Civil Engineering, Purdue University, West Lafayette, IN, USA. He is a member of the Digital Cartography Laboratory (LCD), Department of Infrastructure and Environment (InfrA), School of Civil Engineering, Architecture, and Urban Design (FECFAU), UNICAMP. During his master's research, he was part of the Living Laboratory for Electric Mobility in Public Transportation Project at UNICAMP, where he focused on the application of geospatial data for routing techniques, passenger volume estimation at bus stops, delay modeling considering bus occupancy, dwell time estimation, and timetable optimization. His major field of study is geomatics applied to civil engineering and transportation. His research interests include mobility data analysis, transport planning, and geospatial techniques applied to intelligent transportation systems.



HENRIQUE C. OLIVEIRA received the Technical degree in geomatics from the State University of Campinas (UNICAMP), Campinas, Brazil, in 2005, and the B.S. degree in cartographic engineering and the M.Sc. and Ph.D. degrees in cartographic sciences from São Paulo State University (UNESP), Presidente Prudente, Brazil, in 2010, 2013, and 2016, respectively.

His research internship with the Department of Civil Engineering, Purdue University, West Lafayette, IN, USA, from 2014 to 2015. From 2016 to 2017, he was a Professor with the Federal University of Uberlândia, Brazil. From 2017 to 2022, he was an Assistant Professor (a Professor Doutor) with UNICAMP. From 2022 to 2023, he was an Assistant Professor with the Department of Surveying Engineering, The Pennsylvania State University, State College, PA, USA. Since 2024, he has been an Associate Professor with the Department of Infrastructure and Environment (Infra), School of Civil Engineering, Architecture, and Urban Design (FECFAU), UNICAMP, where he coordinates the Digital Cartography Laboratory (LCD). He is currently the Coordinator of the Graduate Program in Civil Engineering (PPGEC) at UNICAMP, where he teaches courses and advises master's and doctoral students. His major field of study is geomatics applied to civil engineering and transportation. His research interests include transportation planning, mobility data analysis, and geospatial techniques applied to intelligent transportation solutions.

Dr. Oliveira has been a Permanent Faculty Member of PPGEC–UNICAMP, since 2017. He is a member of Brazilian Society of Cartography, Geodesy, Photogrammetry, and Remote Sensing (SBC). He received the title of an Associate Professor (Livre-Docente) in Surveying and Geodesy at UNICAMP, in 2022.



DÉBORA P. SIMÕES received the B.S. degree in land surveying and cartographic engineering and the Specialization degrees in environmental management and in teaching (professional and technological education) from IFSULDEMINAS, Inconfidentes Campus, Inconfidentes, Brazil, in 2016, 2018, and 2020, respectively, and the M.Sc. degree in civil engineering, with a major in transportation from the State University of Campinas (UNICAMP), Campinas, Brazil,

in 2022, where she is currently pursuing the Ph.D. degree in civil engineering, with a major in transportation, where she conducts research on the planning of safe routes for AAVs (drones).

Since 2018, she has been a Lecturer with the Surveying and Cartography Department, IFSULDEMINAS. She is a member of the Digital Cartography Laboratory (LCD), Department of Infrastructure and Environment (Infra), School of Civil Engineering, Architecture, and Urban Design (FECFAU), UNICAMP. Her research interests include photogrammetry, artificial intelligence, and transportation.

• • •